

Manipulation-Robust Mark Design for Perpetual Futures

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June 22, 2026

Abstract

A perpetual futures contract's *mark price* is built from a reference an adversary can manipulate to shift funding and trigger liquidations. How responsive should the mark be? The robustness literature answers “less,” treating manipulability as exogenous. We show it is endogenous: the reference's depth comes from traders who lose to the manipulator, so manipulability is a fixed point of the mark's reliance. A mark calibrated to the normal-times order book over-relies on a fragile reference, thinning it until it collapses; what looks deeper is trusted more, hence more dangerous. We develop *reflexive mark design*, with an estimable trigger and a fragility law.

Keywords: perpetual futures, market manipulation, mechanism design, reference prices, oracles, liquidations, funding rate, robustness, market microstructure.

JEL: G14, G18, G23, D47, D82, G13.

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1 Introduction

Perpetual futures are the dominant venue for leveraged exposure in cryptocurrency markets (Makarov and Schoar, 2020) and, following the CFTC’s 2026 policy statement and related order, are now being listed on regulated U.S. exchanges (CFTC, 2026). Unlike a dated future, a perpetual never converges to spot by expiry. A *mark price* tethers it instead: a reference the exchange builds from spot prints and oracle feeds. The mark has two uses. First, it determines the periodic funding payment from one side of the contract to the other: the mechanism by which no-arbitrage pricing tethers a perpetual to spot (Shiller, 1993; Akerer et al., 2025; He et al., 2024). Second, it marks open positions to decide which are liquidated. The mark is a public signal that allocates large transfers and forces large position changes.

That signal is manipulable (Allen and Gale, 1992; Kumar and Seppi, 1992), and reference-price manipulation is empirically documented in cryptocurrency markets (Griffin and Shams, 2020). An adversary can move the reference at a cost, by wicking a thin venue, pushing an oracle, or pumping and dumping the spot. Doing so moves the mark, the funding transfer, and the liquidation boundary, all without taking the honest other side of a trade. The harm is not a transfer captured from a counterparty in the usual sense. It is destroyed information about fair value, mistimed liquidations, and the resources spent attacking and defending.

A concrete case fixes the mechanism. In October 2022 an attacker took a large position in the MNGO perpetual on Mango Markets, then bought the token across the three thin spot venues feeding the oracle from which the mark was built, driving the oracle-reported price more than thirteen-fold in about thirty minutes. The inflated mark, counted as collateral, let him withdraw roughly \$110 million and drain the protocol. No honest counterparty stood on the other side (the loss was the corrupted reference, the collateral it mis-valued, and the resources the attack consumed), and it worked because the mark listened, fully and at once, to a reference too thin to defend. The Commodity Futures Trading Commission charged the episode as an “oracle manipulation” case, alleging that more than \$110 million in digital assets was misappropriated (CFTC, 2023); it was one of dozens of oracle- and reference-price attacks that drained hundreds of millions from such venues that year. The lesson is not that manipulation should be policed but that the mark was built to trust a reference it could not afford to: how much a mark listens to its reference is a design choice with first-order consequences.

This paper studies the optimal construction and use of the mark when the reference is manipulable. We specify an economy with a continuum of leveraged positions, a fair value that follows a mean-reverting diffusion, and an adversary who injects a bias into the reference at a convex cost. A designer commits to how responsive the mark is to the reference, how long to average it, and at which price to liquidate; a privately informed adversary then acts. The welfare objective is not assumed but derived from the two channels the mark feeds.

The natural way to treat a manipulable reference is as a signal-extraction problem (Kyle, 1985; Glosten and Milgrom, 1985): regard the contamination as noise, filter it, trust a deeper or more broadly aggregated reference more, and take the depth of the reference market as given, as the manipulation-robust benchmark literature does (Duffie and Stein, 2015; Duffie et al., 2017; Duffie and Dworczak, 2021). That instinct is half right (the optimal mark does discount the reference), but in several specific, finance-relevant ways it has the sign backwards. Whether a manipulation of one mark spreads to correlated others or instead hedges them; whether the reference market’s depth can be treated as a primitive at all; and even whether the mark should under- or over-weight its reference: on each, the price-taking, signal-extraction view gives the wrong answer. What reverses it is a force specific to a *designed* reference. The reference is corrupted because it is used (it allocates transfers and forces liquidations, so moving it pays, and it pays more the more the

mark relies on it), and the liquidity that keeps it honest is supplied by traders who themselves lose to the manipulator. Reliance and manipulability are therefore jointly determined, by the mark’s design and by the market that supports it; a measure used as a target decays as a measure, and the design question is how to build the target so the decay is slowest.

The analysis yields three economic conclusions, each an answer to a question an exchange faces. The first is how much to trust the reference. The optimal mark under-weights it, but the reason is sharper than the truism that manipulable signals are noisy: what forces the discount is the designer’s *uncertainty* about the manipulation, not its size. A bias the designer can anticipate is undone at no cost by the mark’s intercept; only the unpredictable part survives to distort funding and to mistime liquidations. It is the variance of the attacker’s incentive, not its mean, that the mark must guard against, so a venue facing a few large but well-understood manipulators may rationally trust its reference more than one facing many small, idiosyncratic ones.

The second is what makes the reference trustworthy in the first place. Averaging the mark over a longer window raises the cost of moving it (a transient push enters the reference only through its time average, so a longer window forces the attacker to hold the dislocation longer against arbitrage), but the same averaging makes the mark stale, so it tracks a drifting fair value with error. We derive the staleness from the price process itself, so both blades of the trade-off come from a single object, the averaging window. The optimal window is interior: it exists when the robustness bought at the margin exceeds the staleness it costs, lengthens, under slow mean reversion, when manipulation is more profitable, and shortens when the underlying is more volatile, comparative statics an exchange can read off its own market.

The third is when the reference is safe at all, and how a single manipulation propagates once it is not. A reference deep enough relative to the strongest sabotage incentive lets the mark track fair value fully; below that depth the designer is forced to discount, and welfare falls *discontinuously* at the threshold. This is the regulators’ “deep, active, continuous” spot-market requirement, here a derived cutoff rather than a slogan, with a longer averaging window a substitute for raw depth. And the harm does not stay local: liquidating every position at a single common mark spreads one manipulation across the entire cross-section, whereas liquidating each position at its own execution price contains it: the perpetual-futures counterpart of the choice between uniform-price and pay-as-bid allocation, with a sharp severity threshold separating the regimes.

These conclusions are not signal extraction in disguise. A merely noisy reference would be shrunk by ordinary filtering, with the direction of the adjustment set by the value’s dynamics and reversible under a trending reference. What is special here is that the corruption *responds* to the weight, and that feedback, not the linear-quadratic scaffolding, is what produces the quartic manipulation penalty, the discontinuous sabotage threshold, and the averaging window’s role as an instrument that raises the price of corruption rather than discounting the corrupted signal. Isolating the strategic feedback from the mechanical shrinkage shows the optimal mark can even *over-weight* a trending, weakly-manipulable reference, so the headline results are predictions about the world rather than restatements of the model’s premises.

Relative to the no-arbitrage pricing of perpetuals (Shiller, 1993; Akerer et al., 2025; He et al., 2024), which takes the reference as given and asks what funding tethers the price, we take the reference as manipulable and ask how the mark should be built and used. The robustness of the mark is itself an object of design, chosen through the averaging window against a staleness cost derived from the price dynamics; the same reference governs two channels, liquidation and funding, that respond differently to the available instruments; and the welfare criterion is microfounded from a cross-section of positions that also delivers the localization result. We go further than treating the reference market as a primitive: its depth is supplied by traders exposed to the manipulator, so reliance and manipulability are jointly determined, and the reference can tip from a deep, honest

regime into a thin, manipulated one: a reflexive feedback that the price-taking view of manipulation cannot represent (Section 9). The closest precedents (the theory of cash-settlement manipulation and the design of manipulation-resistant benchmarks and oracles) model the manipulation, or price a single instrument against it; none chooses the mark’s responsiveness, window, and liquidation convention together from welfare, and none lets the reference’s depth respond to its own use. Taking that feedback as the organizing principle rather than a correction yields the paper’s framework, *reflexive mark design*: the reference’s manipulability is a fixed point of the reliance placed on it, indexed by a single reflexivity gain that is zero for the price-taking literature and divergent at the fold. Its sharpest implication is a result the price-taking view cannot produce: a mark optimized against the order book an exchange measures in normal times can, on a fragile underlying, choose a reliance that destroys the very book it needs, collapsing the reference it was built to read (Section 10). In dynamic form that threshold becomes a fragility law in which the expected time to a collapse scales as the exponential of a universal power of the distance to the fold.

The paper contributes to four literatures. The first is the theory of settlement and reference-price manipulation (Kumar and Seppi, 1992; Allen and Gale, 1992; Jarrow, 1992; Kyle and Viswanathan, 2008; Bénabou and Laroque, 1992), of which the mark of a perpetual (a cash-settlement price consulted continuously rather than once at expiry) is a natural object; to it the paper adds the joint, welfare-based design of the reference’s responsiveness, averaging window, and liquidation convention against a strategic adversary, with the competition-among-manipulators logic of Kumar and Seppi (1992) reappearing as a discount governed by the *spread* rather than the level of the attack. The second is the design of manipulation-robust benchmarks (Duffie and Stein, 2015; Duffie et al., 2017; Duffie and Dworczak, 2021), which models a fixing as an estimator built from the trades of agents whose profits depend on it; the listen-less mark is such a fixing, and its affine form rests on the worst-case robustness foundation of Carroll (2015). The third is the feedback-effects literature, in which trading on a price corrupts the information it carries (Goldstein and Guembel, 2008; Bond et al., 2012); reflexive mark design is its benchmark-design analogue, the force by which a reference decays because it is used. The fourth is the theory of runs and liquidity crashes (Diamond and Dybvig, 1983; Goldstein and Pauzner, 2005; Brunnermeier and Pedersen, 2009; Cespa and Foucault, 2014; Goldstein et al., 2025), to which the paper contributes a fragility whose object is a designed reference price and whose severity the designer controls through how much the mark is built to listen. The departure common to all four is endogenous manipulability: where each takes the cost of moving the reference as a primitive, here that cost is supplied by traders exposed to the manipulator and so responds to the reliance the mark places on it. The closest single precedent is Kumar and Seppi (1992), and the point of departure is its price-taking treatment of liquidity, which reflexive mark design overturns; Section 2 develops these connections.

The results bear directly on how perpetuals are regulated and run. The condition for a fully responsive mark to be sabotage-safe is a derived version of the “deep, active, continuous” spot-market requirement regulators impose before admitting an onshore perpetual: a computable threshold in the ratio of the strongest sabotage incentive to the reference’s depth, with a longer mark window a quantified substitute for raw depth that lets a thinner underlying be admitted at a priced staleness cost. The choice between liquidating at a common mark and at own execution prices is the exchange’s analogue of uniform-price versus pay-as-bid, optimal one way for deep references and the other under severe manipulation, with a severity threshold the designer can locate. And because that depth is itself endogenous (heavy reliance can hollow out the very liquidity that keeps the reference honest), an exchange’s reliance on its own reference is a risk-management variable rather than a fixed parameter: over-reliance can extinguish that liquidity, with hysteresis, so the object to monitor is the feedback gain that measures how close the reference sits to its fold.

The paper draws on two traditions, and the first is asset pricing and market microstructure,

which supplies its primitives. The cost of moving the reference is a price-impact relation (Kyle, 1985; Glosten and Milgrom, 1985): the robustness κ (the depth of the reference market, the inverse of a Kyle-type impact coefficient) governs that cost, and the mark is a signal-extraction problem (the optimal filter of a contaminated price), so “listen less” is the shrinkage of a noisy signal whose noise the designer partly controls. The no-arbitrage pricing of perpetuums, in which funding tethers the contract to spot and pins the basis (Akerer et al., 2025; He et al., 2024), is the frictionless benchmark this paper perturbs by letting the reference be moved. The object of study, a widely-used number that is at once a public good and an attack surface, places the work in the lineage of benchmark manipulation and design that the interest-rate and foreign-exchange fixings made salient (Duffie and Stein, 2015; Duffie and Dworczak, 2021). What it adds to that lineage is endogenous informativeness: the benchmark decays because it is used, a force the price-taking treatment of manipulation cannot represent and the benchmark analogue of the feedback by which trading on a price corrupts the information it carries (Goldstein and Guembel, 2008; Bond et al., 2012). The optimal design thus trades the value of the signal against the corruption its own use invites.

The second tradition is the theory of financial intermediaries, which supplies the agent. The exchange is a central counterparty: it posts margin, forces liquidations, and moves funding, and the mark is the public signal on which all three turn. The designer is an intermediary that produces a benchmark and internalizes the welfare of the cross-section that relies on it: a delegated monitor whose monitoring technology is itself a manipulable price. The choice between liquidating at a common mark and at own execution prices is a clearinghouse design choice, the perpetual analogue of how a central counterparty mutualizes or contains a defaulting member. The localization result states which convention spreads a shock across the cross-section and which absorbs it. And the endogenous-depth result is an intermediation fragility of a familiar shape. The liquidity that keeps the reference honest is supplied by traders who lose to the manipulator, so heavy reliance can dry it up and the reference can run from a deep regime to a thin one. It is a coordination failure of the Diamond and Dybvig (1983) kind and a liquidity spiral in the sense of Brunnermeier and Pedersen (2009), relocated from the liability side of a bank to the information infrastructure of an exchange. The same global-game coordination logic governs runs on the liability side of crypto intermediaries (on stablecoin redemptions (Gill and John, 2026a,b) and on restaking capacity (Gill and John, 2026c)) where the run is on the redeemable claim; here it is on the reference that prices and liquidates it.

Section 3 specifies the environment and Section 4 derives the welfare loss from the liquidation and funding channels. Section 5 solves the adversary’s problem and records the staleness moments. Section 6 characterizes the robust mark and Section 7 the optimal window; Section 8 treats sabotage and admissibility. Section 9 endogenizes the reference’s depth, and Section 10, the core of the paper, develops reflexive mark design: the conservatism wedge, the over-trust catastrophe, the dynamic fragility law, and its microfoundation. Section 11 contrasts common- and own-price liquidation, Section 12 shows the fragility condition is estimable, Section 14 examines the robustness of the modeling choices, and Section 16 discusses the scope of the contribution and concludes. Longer proofs are in Appendix A, and a companion working paper develops the supporting extensions: the optimal nonlinear mark, the maintenance-margin instrument, the multi-venue robust index, the funding clamp, and the window’s dynamics. The environment and welfare loss are built from primitives rather than sketched: the manipulation cost from the same averaging operator that produces the staleness (Section 3.4), the cross-section from a population of optimizing leveraged traders (Section 4.1), the adversary’s type from its book (Section 5.1), and the static criterion from the stationary flow loss (Section 4.7), with equilibrium and well-posedness proved in Appendix A.4.

2 Related literature

Perpetual-futures pricing. The pricing strand takes the reference as given. [Shiller \(1993\)](#) proposed perpetual claims settled against a cash index and already flagged the index’s vulnerability to mismeasurement; modern treatments derive no-arbitrage perpetual prices and the funding that tethers them to spot. [Ackerer et al. \(2025\)](#) price linear, inverse, and quanto perpetuals as risk-neutral expectations of the spot sampled at a funding-driven random time and identify funding specifications under which the basis is zero, and [He et al. \(2024\)](#) derive no-arbitrage prices and trading-cost bounds and document the empirical basis deviations the clamp produces. This literature asks what funding tethers the price to an honest reference; we take the reference as manipulable and ask how the mark that reads it should be built and used, the design half these papers leave to the venue.

Manipulation of settlement and reference prices. The closest manipulation precedent is the theory of cash-settlement manipulation. [Kumar and Seppi \(1992\)](#) show, in a [Kyle \(1985\)](#) market, that a trader holding a cash-settled position profits by moving the settlement price, and that as manipulators multiply their profits compete away while their liquidity footprint does not. The mark of a perpetual is a cash-settlement price consulted continuously rather than once at expiry, so the paper sits in this lineage and refines it: the competition logic of [Kumar and Seppi \(1992\)](#) reappears as the dispersed-attacker result of Section 8, and the settlement-window of the “banging-the-close” literature becomes the optimized averaging window of Section 7. The broader manipulation taxonomy of [Allen and Gale \(1992\)](#) and [Jarrow \(1992\)](#), the welfare-based definition of harmful manipulation of [Kyle and Viswanathan \(2008\)](#), and the analysis of a manipulable public signal’s credibility in [Bénabou and Laroque \(1992\)](#) frame what counts as manipulation and why a designer should resist it. That reference-price manipulation is empirically documented in crypto ([Griffin and Shams, 2020](#)), in markets whose fragmentation and limited cross-venue arbitrage ([Makarov and Schoar, 2020](#)) leave many references thin, makes the attack surface concrete rather than hypothetical; what the paper adds is the design side they hold fixed: responsiveness, window, and liquidation convention chosen jointly from a microfounded objective.

Benchmark design. The design precedent is [Duffie and Dworczak \(2021\)](#), who model a benchmark fixing as an estimator of fair value built from the trades of agents whose profits depend on the fixing, and characterize the optimal and collusion-robust linear fixing; [Duffie et al. \(2017\)](#) study benchmarks in search markets, and [Duffie and Stein \(2015\)](#) set out the policy problem the LIBOR and foreign-exchange scandals exposed. The mark is a fixing in this sense, and the listen-less rule is a manipulation-robust estimator whose affine form has a robustness foundation beyond the Gaussian case, established directly in Theorem 14.2. Against an adversary whose type law is known only through its variance, the affine mark is minimax-optimal, the mark’s analogue of the linear-rule selection [Carroll \(2015\)](#) obtains for unknown adversary actions. Where the paper departs is reflexivity: [Duffie and Dworczak \(2021\)](#) and the oracle literature treat the cost of manipulation (the liquidity an attacker must move) as a primitive, whereas here that liquidity is supplied by traders exposed to the manipulator and so responds to the reliance the fixing places on it. The fixing’s manipulability is endogenous to its own use, which is the source of every result the price-taking design cannot reach.

Price impact and the resilient reference book. The model’s primitives are standard market microstructure. The cost of moving the reference is a [Kyle \(1985\)](#) price-impact relation, with the

robustness κ the inverse impact coefficient and the adverse selection that thins the book the [Glosten and Milgrom \(1985\)](#) channel; the directional manipulator’s continuous-trading problem is of the [Back \(1992\)](#) form. The reference book is *resilient*: depth consumed by an attack is replenished by liquidity providers, at a finite speed, exactly the supply/demand dynamics of [Obizhaeva and Wang \(2013\)](#) and the equilibrium limit-order-book resilience of [Foucault et al. \(2005\)](#). The averaging window works because resilience makes a sustained dislocation costly, and the endogenous-depth section microfounds the supply of that resilience from the providers’ entry and exit.

Feedback effects and the reflexivity of prices. The paper’s organizing idea has a precise home in the feedback-effects literature. [Goldstein and Guembel \(2008\)](#) show that when a price guides real decisions, the feedback creates an incentive to trade so as to corrupt it, and that the corruption lowers the price’s own informativeness; [Bond et al. \(2012\)](#) survey this real-effects channel. Reflexive mark design is the benchmark-design analogue: the mark guides liquidations and funding, so reliance on it changes the liquidity that keeps it honest, and the optimal price-taking mark can corrupt the reference it reads. The over-trust catastrophe is the design-side counterpart of the informativeness loss [Goldstein and Guembel \(2008\)](#) identify: here it is the designer, not a trader, whose optimal action degrades the signal, and the degradation is in the reference’s depth rather than its information content.

Liquidity spirals and runs. The fragility results connect to the literature on destabilizing liquidity feedback and runs. [Brunnermeier and Pedersen \(2009\)](#) show that margins can be destabilizing and that market and funding liquidity reinforce each other into spirals in which liquidity suddenly dries up; the manipulation trap of Section 9 is a spiral of this kind in the reference’s depth, and the reference run of Section 10 is its realization. The run is a coordination failure with a tipping point of the [Diamond and Dybvig \(1983\)](#) kind, and the fragility law (a unique threshold below which a run is exponentially rare and above which it is immediate) is the benchmark-design analogue of the unique run probability that [Goldstein and Pauzner \(2005\)](#) derive for demand deposits and of the amplification mechanisms surveyed by [Goldstein et al. \(2025\)](#). The self-reinforcing loop between the reference’s depth and its informativeness is, in particular, the manipulation-design counterpart of the liquidity crash of [Cespa and Foucault \(2014\)](#), in which a small fall in liquidity feeds back, through cross-asset learning, into a large one. The contribution relative to all of these is that the fragile object is a designed *reference price*, and its fragility is a controllable function of how much the mark is built to listen to it.

Contribution. Against this background there is, to our knowledge, no prior formal theory of optimal, manipulation-robust *mark-price design* for perpetuals (how responsive the mark should be, how long to average it, and at which price to liquidate), derived together from welfare against a strategic adversary whose effectiveness the design itself determines. The specific contributions are five. First, the manipulation discount governed by the *spread* rather than the level of the manipulation (Theorem 6.2, Proposition 6.4), sharpening the competition result of [Kumar and Seppi \(1992\)](#) into a statement about which moment of the attack distribution binds. Second, the optimal averaging window as the unique solution to a manipulation-versus-staleness trade-off with signed comparative statics (Lemma 5.5, Theorem 7.2, Proposition 7.3), turning the settlement-window and oracle instruments into an optimized object. Third, the admissibility threshold that rationalizes the deep-spot-market rule and prices the window as a substitute for depth (Proposition 8.1, Corollary 8.2). Fourth, and central, reflexive mark design (Section 9, Section 10): manipulability is a fixed point of reliance, indexed by a reflexivity gain that nests the price-taking benchmark-design

literature at zero, with the over-trust catastrophe (Theorem 10.4) (an optimal price-taking mark that destroys its own reference) characterized by a single estimable condition that holds for any depth-supply microfoundation, and a dynamic fragility law (Theorem 10.7) for the expected time to a reference run. Fifth, the common-mark versus own-price localization of liquidation manipulation, the perpetual analogue of uniform-price versus pay-as-bid, with a unique severity threshold (Theorem 11.5).

3 The environment

The environment has three actors and one contested number. An exchange must publish a mark that simultaneously prices funding and triggers liquidations; a population of leveraged traders is exposed to that mark; and an adversary can bend the reference from which the mark is built, at a cost. The fair value the mark should report is unobservable and moving, and the exchange's only window onto it is a stream of prints the adversary can contaminate. The design problem is to turn that contaminated stream into a number that does the least damage. We specify each primitive in turn, motivating it before formalizing it, and keep the notation light: a deviation v of fair value from a slow anchor, a window τ over which prints are averaged, a responsiveness a with which the mark loads on the averaged reference, and a depth κ that measures how costly the reference is to move.

3.1 Probability space and the fair value

Fix a complete filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in \mathbb{R}}, \mathbb{P})$ carrying a standard Brownian motion B . The *fair value* of the underlying, measured as a deviation from a slow anchor (a long-window reference, or the previous settlement), is a stationary Ornstein–Uhlenbeck process

$$dv_t = -\phi v_t dt + \varsigma dB_t, \quad \phi > 0, \varsigma > 0, \quad (3.1)$$

in its stationary law, so that $v_t \sim \mathcal{N}(0, \sigma^2)$ with

$$\sigma^2 = \frac{\varsigma^2}{2\phi}, \quad \text{Cov}(v_t, v_{t-h}) = \sigma^2 e^{-\phi|h|}. \quad (3.2)$$

Here ϕ is the rate at which the fair value reverts to its anchor, ς the instantaneous volatility, and σ^2 the stationary variance of the deviation the mark must track. The anchor is normalized to zero. We work at a representative date and write $v := v_t$ for the current deviation.

3.2 The contract and the two roles of the mark

A perpetual future on the underlying is held by a unit continuum of traders $i \in [0, 1]$, described in Section 4. The contract has no maturity, and in place of the convergence to spot that disciplines a dated future, an exchange publishes a *mark* M and an *index* I , both functions of observed data, and puts the mark to two uses. It liquidates: each position carries a maintenance boundary, and a position is closed when the mark crosses it; the efficient rule would liquidate when *fair value* crosses the boundary, so a mark that departs from fair value closes solvent positions and spares insolvent ones, both costly. And it sets funding: every interval a transfer proportional to a clamped function of the mark–index gap passes between the two sides of the contract, so a mark away from fair value misprices that transfer and pushes positions away from where they belong. The mark is thus a single public number that decides at once who is liquidated and who pays whom, which is what

makes moving it valuable to an adversary, and its design a genuine economic problem rather than a filtering exercise. The index I is the exchange's best estimate of contemporaneous fair value; we take $I = v$ for the honest index and treat the mark M as the designed object, since it is the mark, not the index, that is smoothed and that governs liquidation.¹

3.3 The reference: averaging and manipulation

The exchange does not observe v . It observes a *reference* formed by averaging recent prints and then contaminated by an adversary. Averaging over a window of length $\tau \geq 0$ produces the time average

$$\bar{v}_\tau := \frac{1}{\tau} \int_{t-\tau}^t v_s ds \quad (\tau > 0), \quad \bar{v}_0 := v, \quad (3.3)$$

which lags the current level by the *staleness gap* $\varepsilon := v - \bar{v}_\tau$. An adversary injects an additive *bias* $\delta \in \mathbb{R}$ into the reference, so the exchange observes

$$b = \bar{v}_\tau + \delta = v - \varepsilon + \delta. \quad (3.4)$$

Moving the reference is costly. Producing a bias δ costs the adversary $\frac{\kappa(\tau)}{2} \delta^2$, where $\kappa(\tau) > 0$ is the *reference robustness*: the marginal cost of moving the inputs to the mark. Robustness is a property of construction. A reference averaged over a longer window, or aggregated across more and deeper venues, is costlier to move and to hold moved, so κ is increasing in τ ; Section 7 adopts a parametric form. We write $\kappa := \kappa(\tau)$ when τ is fixed.

3.4 The reference technology and the robustness function

The cost $\frac{\kappa(\tau)}{2} \delta^2$ and the monotonicity of κ were stated as primitives. We derive both from the averaging operator that already produces the staleness of Section 5.4, so that the two sides of the central trade-off (a robustness that rises with the window and a staleness that also rises with it) rest on a single object: the window over which prints are averaged.

The exchange averages a stream of prints. Write the contaminated print at instant s as

$$p_s = v_s + y_s, \quad (3.5)$$

the sum of fair value and an attacker displacement $y_s \geq 0$ (a one-directional push over the window, without loss for a directional attack on the average). The reference is the window average of the print, so the bias injected into it is the time average of the displacement,

$$\delta = \frac{1}{\tau} \int_{t-\tau}^t p_s ds - \bar{v}_\tau = \frac{1}{\tau} \int_{t-\tau}^t y_s ds =: \langle y \rangle_\tau. \quad (3.6)$$

The same operator that lags the clean signal by $\varepsilon = v - \bar{v}_\tau$ thus dilutes the attacker's displacement: because only the window average reaches the mark, a longer window forces a longer-held push to achieve a given bias.

Moving a print is costly through two standard microstructure channels. *Establishing* a displacement requires trading against the book: to hold the print displaced by an amount, the attacker carries an inventory proportional to that amount against a book with Kyle depth λ , at a temporary-impact cost quadratic in the displacement, $\frac{\kappa_0}{2} (\cdot)^2$ with $\kappa_0 \propto 1/\lambda$. *Sustaining* the displacement

¹In practice the index is a volume-weighted average across many spot venues and the mark a smoothed, median-filtered composite; we collapse the index to fair value and locate all design choices in the mark, which is where averaging and responsiveness operate.

bleeds a carry: holding the print away from value over time costs adverse selection against arbitrageurs at a flow rate quadratic in the instantaneous displacement, $\frac{1}{2}\theta(r)y_s^2$ at elapsed holding time r , with carry-rate profile $\theta(\cdot) > 0$.

Assumption 1 (Reference cost technology). The cost of a displacement path $\{y_s\}_{s \in [t-\tau, t]}$, $y_s \geq 0$, is

$$C_\tau[y] = \frac{\kappa_0}{2} \left(\max_s y_s \right)^2 + \frac{1}{2} \int_{t-\tau}^t \theta(t-s) y_s^2 ds, \quad \kappa_0 > 0, \theta > 0,$$

an establishment cost on the displacement level and a carry on its time profile.

Lemma 3.1 (Robustness from the averaging operator). *Under Assumption 1 with constant carry rate $\theta \equiv \gamma$, the cheapest displacement path achieving bias δ is the constant push $y_s \equiv \delta$, and the minimized cost is $\frac{\kappa(\tau)}{2}\delta^2$ with*

$$\kappa(\tau) = \kappa_0 + \gamma\tau.$$

Robustness is therefore strictly increasing and affine in the window, with instantaneous level $\kappa(0) = \kappa_0 = 1/\lambda$ fixed by spot depth and slope γ fixed by the carry of holding the print displaced.

Proof. Any admissible path has $\max_s y_s \geq \langle y \rangle_\tau = \delta$, so the establishment term is at least $\frac{\kappa_0}{2}\delta^2$. By the Cauchy–Schwarz inequality applied to $y_s \cdot 1$,

$$\left(\int_{t-\tau}^t y_s ds \right)^2 \leq \tau \int_{t-\tau}^t y_s^2 ds, \quad \text{i.e.} \quad \int_{t-\tau}^t y_s^2 ds \geq \tau \delta^2,$$

so the carry term is at least $\frac{\gamma}{2}\tau\delta^2$. Both bounds bind at the constant path $y_s \equiv \delta$, which is admissible; summing gives $C_\tau^{\min} = \frac{1}{2}(\kappa_0 + \gamma\tau)\delta^2$. The identification $\kappa_0 = 1/\lambda$ records that establishing a unit displacement requires inventory $1/\lambda$ at linear impact, the constant of proportionality absorbed into units. $\square$

The affine law $\kappa(\tau) = \kappa_0 + \gamma\tau$ is the constant-carry case. Letting the carry rate depend on how long the manipulation has run delivers the iso-elastic technology of Section 7 and gives its exponent an economic reading.

Proposition 3.2 (Iso-elastic robustness from a power-law carry). *Adopt the constant-displacement convention $y_s \equiv \delta$ of Lemma 3.1 and let the carry rate at elapsed holding time r be the power law*

$$\theta(r) = \frac{\kappa_0\beta}{\tau_0} \left(1 + \frac{r}{\tau_0} \right)^{\beta-1}, \quad \beta > 0, \tau_0 > 0.$$

Then robustness is exactly the iso-elastic form of Assumption 4,

$$\kappa(\tau) = \kappa_0 + \int_0^\tau \theta(r) dr = \kappa_0 \left(1 + \frac{\tau}{\tau_0} \right)^\beta,$$

and the manipulability index $\rho(\tau) = \text{Var}(w)/(\kappa(\tau)^2\sigma^2) = \rho_0(1 + \tau/\tau_0)^{-2\beta}$ is strictly decreasing and strictly convex. More generally, every technology in the class of Assumption 1 with $\theta > 0$ has $\kappa(0) = \kappa_0 > 0$, $\kappa' > 0$, and $\rho' < 0$, with ρ convex whenever $3\kappa'^2 \geq \kappa\kappa''$. The exponent β is the elasticity of the marginal cost of duration: $\beta = 1$ is the constant-carry baseline of Lemma 3.1, $\beta > 1$ a carry intensifying as the manipulation persists (arbitrage and venue defenses thicken), and $\beta < 1$ a carry that relents.

Proof. The holding cost of the constant path under rate $\theta(\cdot)$ is $\frac{1}{2}\delta^2 \int_0^\tau \theta(r) dr$, so $\kappa(\tau) = \kappa_0 + \int_0^\tau \theta$. For the stated θ , $\int_0^\tau \theta = \kappa_0[(1 + \tau/\tau_0)^\beta - 1]$, giving $\kappa(\tau) = \kappa_0(1 + \tau/\tau_0)^\beta$ and $\rho \propto \kappa^{-2} = \rho_0(1 + \tau/\tau_0)^{-2\beta}$. Differentiating, $(\kappa^{-2})' = -2\kappa'\kappa^{-3} < 0$ and $(\kappa^{-2})'' = 2\kappa^{-4}(3\kappa'^2 - \kappa\kappa'')$, positive under $3\kappa'^2 \geq \kappa\kappa''$; for the iso-elastic κ , $3\kappa'^2 - \kappa\kappa'' = \kappa_0^2\beta(2\beta + 1)\tau_0^{-2}(1 + \tau/\tau_0)^{2\beta-2} > 0$. The general signs are immediate from $\theta > 0$. $\square$

Remark 3.3 (One operator, both sides of the trade-off). Lemma 5.5 and Lemma 3.1 read the same window in opposite directions. Lengthening τ averages over a longer stretch of the mean-reverting path, raising the staleness variance $\text{Var}(\varepsilon) = \sigma^2(1 - 2C + V)$ toward σ^2 , and simultaneously forces the attacker to hold a displacement longer, raising the carry and hence $\kappa(\tau)$. The manipulation–staleness trade-off priced throughout the paper is therefore not an assumed cost set against a derived one, but two consequences of a single averaging choice.

Remark 3.4 (Single-interval timing and the scope of window deterrence). The deterrence in $\delta^* = wa/\kappa(\tau) \rightarrow 0$ as τ grows rests on a timing convention worth stating, because it is the one a microstructure reader will press. The carry in $\kappa(\tau)$ (Assumption 1) is *window-integrated*: it runs over the whole averaging window $[t - \tau, t]$. The adversary’s benefit (the funding transfer at the next settlement, the liquidations triggered at the marking instant, the sabotage payoff) is realized *once* per attack, at a single marking instant. This is a separation of timescales, and the empirically relevant one: a mark window is minutes and a funding interval hours, so a displacement held across one mark window collects a one-shot benefit against a window-length carry, and lengthening τ raises the carry without raising the benefit. The scope of the window-as-substitute result (Theorem 7.2, Corollary 8.2) is accordingly the single-interval sweep: the form the documented attacks take (Section 1). A *sustained* attacker who holds the displacement across many funding intervals and extracts each one rescales benefit and carry together: the program $\max_\delta [wa\delta \cdot (\text{duration}) - \frac{\kappa(\tau)}{2}\delta^2]$ gives $\delta^* = wa\tau/\kappa(\tau)$, which under an affine carry does *not* vanish in τ , and the window stops buying robustness. Against that adversary depth, not the window, is the instrument; the two are complements, and the substitution result is the statement for the single-interval attack the carry is built to price.

3.5 The mark map and the designer’s instruments

The exchange (the *designer*) publishes a mark that is an affine function of the reference,

$$M(b) = a b + c, \quad a \in [0, 1], \quad c \in \mathbb{R}, \quad (3.7)$$

with *responsiveness* a and *intercept* c . Responsiveness $a = 1$ tracks the reference one for one; $a < 1$ anchors the mark toward the slow reference and is the sense in which the mark “listens less.” No exchange literally publishes a fractional multiple of its index, and a should not be read as a dial an operator turns. It is the model’s stand-in for a concrete and measurable object (the *effective pass-through* of a contaminated reference to the published mark) which the real robustness toolkit of multi-venue volume weighting, outlier trimming, a median filter, and a clamp moves below one without ever writing a down. The two readings coincide for the design problem, and a may be read either way. The restriction to affine rules is the natural linear-quadratic benchmark and, under the Gaussian primitives, is without loss within the class of rules that are optimal against a linear adversary; we discuss the non-Gaussian extension in Section 16. The designer’s instruments are the triple

$$(a, c, \tau) \in [0, 1] \times \mathbb{R} \times [0, \infty) \quad \text{and the liquidation convention of Section 11.}$$

The affine restriction is more than a benchmark: under the maintained Gaussian primitives it is exact against the directional adversary. The unconstrained problem, choose any measurable $M : \mathbb{R} \rightarrow \mathbb{R}$, has an affine optimum if and only if the relevant pair is jointly Gaussian, the standard linear-projection result.

Proposition 3.5 (Affine optimality). *Suppose (v, ε, w) is jointly Gaussian, equivalently, since the type w is independent of the price path, (v, ε) is jointly Gaussian and the type w is Gaussian, and the adversary is directional (Section 5.2). Then the loss-minimizing mark over all measurable functions of the reference is affine, with the responsiveness and intercept of Theorem 6.2. If either Gaussianity or linearity of the adversary’s payoff fails, the optimal mark is in general nonlinear. The restriction $a \in [0, 1]$ never binds: the loss (4.10) is strictly convex in a on all of $\mathbb{R}$ and its unconstrained minimizer lies in $(0, 1)$ (Theorem 6.2), so anchoring ($a < 0$) and amplification ($a > 1$) are dominated.*

Proof. The loss $\mathbb{E}[(M(b) - v)^2]$ over measurable M is minimized pointwise by the conditional expectation $M^*(b) = \mathbb{E}[v \mid b]$, the $L^2(\sigma(b))$ projection of v . We show that against the directional adversary (v, b) is jointly Gaussian, so this projection is affine, and that the affine mark and the directional best response are mutually consistent. Against an affine mark $M(b) = ab + c$ the directional best response is $\delta^*(w) = wa/\kappa$ (Lemma 5.2), a function of the private type w alone, independent of the price path (v, ε) (Remark 5.3). Hence $b = v - \varepsilon + \delta^*(w)$ is the sum of the jointly Gaussian (v, ε) and the independent term $\delta^*(w)$. When (v, ε, w) are jointly Gaussian, (v, b) is jointly Gaussian, so $\mathbb{E}[v \mid b] = \alpha b + \beta$ is affine; the loss-minimizing mark over *all* measurable functions is then affine, and minimizing $\mathbb{E}[(ab + c - v)^2]$ over (a, c) returns the responsiveness and intercept of Theorem 6.2. The argument is a fixed point: the affine mark induces a path-independent Gaussian bias, which makes the projection affine, which is the affine mark. If the type law is non-Gaussian, b is a Gaussian location-mixture and $\mathbb{E}[v \mid b]$ is in general nonlinear; and against the saboteur, whose bias $\delta^*(v)$ conditions on v (Lemma 5.4), b is no longer a Gaussian shift of $v - \varepsilon$, so the optimal mark is again nonlinear: the stated scope. Finally, $a \in [0, 1]$ never binds: by Theorem 6.2 the loss $L/\sigma^2 = 1 - 2Ca + Va^2 + \rho a^4$ is strictly convex on $\mathbb{R}$ (second derivative $2V + 12\rho a^2 > 0$) with unique unconstrained minimizer the cubic root $a^* \in (0, 1)$, so anchoring ($a < 0$) and amplification ($a > 1$) are strictly dominated. $\square$

3.6 Timing, information, and equilibrium

The adversary has a privately known *type* w , the marginal value to the adversary of displacing the mark, drawn from a distribution G with mean $\bar{w}$ and variance $\text{Var}(w) < \infty$. The type captures the adversary’s stake: a position that gains when the boundary is crossed, or a claim on the funding transfer (Section 4 makes w explicit). The game has three stages. At the outset the designer commits to the mark rule (a, c) , the window τ , and a liquidation convention, knowing the type distribution G and the law of v but neither the realized attack nor the price path. Nature then draws the type $w \sim G$ and the fair-value path, and the adversary privately observes its own type and (depending on the objective in Section 5) the current value v . Finally the adversary chooses the bias δ to maximize its payoff net of the cost $\frac{\kappa}{2}\delta^2$, after which the reference (3.4) forms, the mark (3.7) is published, funding is exchanged, and positions crossing their boundaries are liquidated under the chosen convention. The designer minimizes expected social loss (Section 4), the expectation taken over $w \sim G$ and the stationary path of v , anticipating the adversary’s Stage-2 best response. The solution concept is subgame perfection: the adversary best-responds in Stage 2 for each (w, v) and each committed mechanism, and the designer chooses the mechanism that minimizes loss given that best response. Commitment in Stage 0, before w is realized, is what makes the problem one of

robust design: the designer cannot tailor the mark to the realized attack, only to the distribution of attacks, and it is the dispersion of that distribution that will matter.

3.7 Strategies, equilibrium, and well-posedness

The three stages define a Stackelberg game with commitment. We record the strategy spaces, the information each player conditions on, and the solution concept, and then state that the equilibrium exists and the designer's program is well posed; the proof is Appendix A.4.

Information is carried by two σ -algebras. The designer moves at Stage 0 knowing the type law G and the law of v but neither realization; its action is the deterministic triple $(a, c, \tau) \in \mathcal{A} := [0, 1] \times \mathbb{R} \times [0, \infty)$ together with a convention $\zeta \in \{\text{MARK}, \text{OWN}\}$ (Section 11). The adversary moves at Stage 2 knowing its type w and, for the saboteur, the current value v ; its strategy is a measurable map $\delta : \mathcal{W} \times \mathbb{R} \rightarrow \mathbb{R}$, $(w, v) \mapsto \delta(w, v)$, measurable with respect to its information ($\sigma(w)$ for the directional manipulator, $\sigma(w, v)$ for the saboteur). Given (a, c, τ) and a realized (w, v, ε) , the reference (3.4), the mark (3.7), and the error $M(b) - v$ are determined, and payoffs are the adversary's (5.2)/(5.3) and the designer's negative loss (4.10).

Definition 3.6 (Subgame-perfect equilibrium). A subgame-perfect equilibrium is a committed triple (a^*, c^*, τ^*) together with an adversary strategy $\delta^*(\cdot)$ such that (i) for every committed (a, c, τ) and every (w, v) , $\delta^*(w, v)$ maximizes the adversary's Stage-2 payoff; and (ii) (a^*, c^*, τ^*) minimizes $L(a, c, \tau) := \mathbb{E}[(M(b) - v)^2]$ over $\mathcal{A}$, evaluated at the adversary's best response, with the expectation taken over $w \sim G$ and the stationary law of (v, ε) .

Theorem 3.7 (Existence and well-posedness). *Take the directional adversary, or the saboteur on its feasible region $\{(a, \tau) : \mu(\tau)a^2 < 1\}$. Then:*

- (i) *for each committed (a, c, τ) the adversary has a unique best response $\delta^*(w, v)$, jointly measurable in (w, v) and square-integrable;*
- (ii) *the loss $L(a, c, \tau)$ is finite and jointly continuous on $\mathcal{A}$;*
- (iii) *the intercept and responsiveness are pinned down by unique interior solutions $c^*(a, \tau) = -\bar{w}a^2/\kappa(\tau)$ and $a^*(\tau) \in (0, 1)$ (Theorem 6.2), and a finite optimal window $\tau^* \in [0, \infty)$ exists (interior under (7.2), otherwise $\tau^* = 0$; Proposition 7.1);*
- (iv) *consequently a subgame-perfect equilibrium exists, with equilibrium loss $L^* = \min_{\mathcal{A}} L$.*

The content of Theorem 3.7 is that the pointwise best responses of Section 5 assemble into a well-defined measurable map, that integrating the squared mark error against the Gaussian price law and the finite-variance type law yields a finite, continuous objective, and that minimizing it has a solution because the intercept and responsiveness solve interior first-order conditions and the optimized loss is continuous in τ with $L^*(0) < \sigma^2$ and a finite limit σ^2 as $\tau \rightarrow \infty$. The substantive characterizations (how a^* , τ^* , and the convention move with the primitives) are Sections 6–11.

4 The welfare loss

The designer's objective is not assumed. This section derives it from the two channels the mark feeds, using a cross-section of leveraged positions that will reappear, in richer form, in the localization result of Section 11.

4.1 A population of leveraged traders

The welfare loss is built from a cross-section whose boundary density g_0 and funding curvature λ_F were, in the sketch, posited. We derive them from a single population of optimizing traders, so that the two channels the mark feeds (liquidation, indexed by a boundary b_i , and funding, indexed by a size q_i) are two readings of the *same* trader's problem rather than unrelated primitives.

A unit continuum of traders is indexed by a type $\zeta \in Z \subseteq \mathbb{R}^k$ with continuous, strictly positive density f_ζ on the open set Z . The type collects a private valuation $\nu(\zeta)$ (a hedging or speculative motive for holding the perpetual) and a collateral capacity $K(\zeta) > 0$. Facing funding rate f , trader ζ chooses a signed position q to maximize a private payoff net of funding,

$$q^*(\zeta; f) = \arg \max_{q \in \mathbb{R}} \pi(q; \nu(\zeta)) - f q, \quad (4.1)$$

with $\pi(\cdot; \nu)$ smooth and strictly concave, peaked at the efficient size $q^*(\zeta) := q^*(\zeta; f^\circ)$ evaluated at the no-error funding f° . The position is held on margin: with maintenance fraction $m_m \in (0, 1)$, a long of size $q > 0$ entered at price P on collateral K is liquidated when the marking price falls to the level at which equity meets the requirement, $K + q(b - P) = m_m qP$, i.e.

$$b = B(\zeta) := P(1 + m_m) - \frac{K(\zeta)}{q^*(\zeta)}, \quad (4.2)$$

and symmetrically from above for a short. Thus the *same* type ζ delivers both the size $q^*(\zeta)$ that enters the funding channel and the boundary $B(\zeta)$ that enters the liquidation channel; leverage links them, since a larger position on given collateral both enlarges the funding footprint and pulls the boundary toward the current price.

Assumption 2 (Population regularity). The boundary map $B : Z \rightarrow \mathbb{R}$ is continuously differentiable with non-vanishing gradient $\nabla B \neq 0$ on Z , and $|\pi_{qq}(q^*(\zeta); \nu(\zeta))|$ is continuous and bounded away from zero on Z .

Lemma 4.1 (Cross-section regularity). *Under Assumption 2, the cross-sectional distribution of boundaries $b = B(\zeta)$ admits a continuous density g on $B(Z)$, and*

$$g_0 := g(v) > 0 \iff v \in \text{int } B(Z),$$

that is, a positive mass of traders is marginal at the current value. In the scalar case $k = 1$ with B strictly monotone, $g(b) = f_\zeta(B^{-1}(b)) / |B'(B^{-1}(b))|$.

Proof. For $k = 1$, B is a C^1 diffeomorphism onto its image (strict monotonicity from $B' \neq 0$), and the change-of-variables formula gives the stated density, continuous and positive wherever $f_\zeta > 0$, i.e. on $\text{int } B(Z)$. For $k > 1$, B is a submersion, and by the coarea formula the law of $B(\zeta)$ has density $g(b) = \int_{\{B=b\}} f_\zeta |\nabla B|^{-1} d\mathcal{H}^{k-1}$, continuous in b because f_ζ is continuous and the level sets $\{B = b\}$ vary continuously in Hausdorff measure under the submersion property; positivity at $b = v$ holds iff the level set through v carries positive f_ζ -mass, i.e. $v \in \text{int } B(Z)$. $\square$

The localization analysis of Section 11 re-parametrizes this same population by distance to liquidation $s = |B(\zeta) - v| \geq 0$; its density h is the pushforward of f_ζ in the s -coordinate, with $h(0) = g_0$, so the boundary density is one object viewed in two coordinates.

Proposition 4.2 (Leverage, margin, and the liquidation weight). *Higher leverage (a larger position on given collateral) moves each boundary (4.2) toward the entry price, since $\partial B / \partial q^* =$*

$K(\zeta)/q^*(\zeta)^2 > 0$ for a long; a maintenance-margin haircut scales admissible positions down and moves boundaries away. The near-boundary mass $\int_{|b-v| \leq \Delta} g \, db$, and with it the liquidation weight $\lambda_L g_0$ in the welfare coefficient $\frac{1}{2}(\lambda_L g_0 + \lambda_F)$, rises with leverage and falls with the haircut. Margin reaches welfare through the same structural object g_0 as the mark, which is the source of the margin-window substitution developed in the companion paper.

Proof. The sign of $\partial B/\partial q^*$ is read from (4.2); raising leverage contracts the entry-to-boundary gap monotonically, so boundaries relocate toward the price and, by Lemma 4.1, the mass within Δ of v weakly rises; a haircut reverses the relocation. The welfare coefficient is increasing in g_0 by Proposition 4.9, and it substitutes for the window. $\square$

4.2 Endogenous leverage and the boundary distribution

The population of Section 4.1 left the size map q^* , the boundary map B , and the density g as objects of a general payoff π . We now give a fully worked instance: mean-variance traders holding a portfolio of the perpetual and a correlated spot hedge, choosing both position *and* leverage optimally. It delivers closed forms for q^* , the boundary, the density g , and the curvature λ_F , and signs how the welfare coefficient moves with volatility, margin, and the dispersion of beliefs, turning g_0 from a posited constant into a comparative-statics object (Figure 1).

Position choice. Trader i has risk aversion γ_i and a private valuation ν_i (subjective excess drift on the perpetual over a funding interval), and may hold a spot hedge with valuation ν_i^0 , the two returns having per-interval volatilities s, s_0 and basis correlation $\rho_c \in (-1, 1)$. Mean-variance utility over the perpetual position q and hedge q_0 ,

$$\max_{q, q_0} q(\nu_i - f) + q_0 \nu_i^0 - \frac{\gamma_i}{2} (q^2 s^2 + 2\rho_c s s_0 q q_0 + q_0^2 s_0^2),$$

has the closed-form perpetual position

$$q^*(\zeta; f) = \frac{(\nu_i - f) - \rho_c (s/s_0) \nu_i^0}{\gamma_i s_\perp^2}, \quad s_\perp^2 := s^2(1 - \rho_c^2), \quad (4.3)$$

with $s_\perp^2$ the residual (post-hedge) variance. Concentrating out q_0 , the effective curvature is $\pi_{qq} = -\gamma_i s_\perp^2$, so the funding curvature of Section 4.5 aggregates to

$$\lambda_F = \lambda_I^2 \mathbb{E}_\zeta \left[\frac{1}{\gamma_i s^2 (1 - \rho_c^2)} \right], \quad (4.4)$$

larger where traders are less risk-averse or better hedged ($\rho_c^2 \rightarrow 1$), because such traders hold positions more sensitive to funding.

Leverage choice and first-passage liquidation. The boundary (4.2) is set by the buffer $\ell := P - b = K/q - m_m P \geq 0$, the price drop to liquidation for a long. A position is liquidated not when the price *ends* an interval below the boundary but the first time it *touches* it, so the right statistic is the first-passage probability of the marking price to $-\ell$, not the terminal tail. We compute it for the mean-reverting price and optimize the buffer against it.

Lemma 4.3 (First-passage liquidation). *Let the de-meaned marking price over a holding interval of length T be the Ornstein–Uhlenbeck path of (3.1) started at the entry value, and let $s^2 := \varsigma^2 T$ be the*

driving diffusion's interval variance (the exact terminal variance is $\sigma^2(1 - e^{-2\phi T}) = s^2(1 + O(\phi T))$). In the slow-reversion regime $\phi T \ll 1$ the probability that a long with buffer ℓ is liquidated is

$$p_\ell(\ell) := \mathbb{P}\left(\min_{u \leq T} X_u \leq -\ell\right) = 2\Phi(-\ell/s) + O(\phi T),$$

the leading term being the Brownian reflection value, twice the terminal tail, exact in the limit $\phi \rightarrow 0$. The $O(\phi T)$ mean-reversion correction is computed in closed form in Corollary A.9. It is negative for economically relevant collateral buffers (the drift $-\phi X$ opposes excursions away from the anchor and lowers the liquidation probability), although the time-changed Brownian clock can produce a small positive correction for very small buffers below roughly 0.61 s. The non-perturbative law, valid for any ϕT , is Proposition A.7; Figure 11 confirms the leading order across the calibrated regime.

Proof. For a driftless Brownian motion of variance rate ς^2 from 0, the reflection principle gives $\mathbb{P}(\min_{u \leq T} W_u \leq -\ell) = 2\mathbb{P}(W_T \leq -\ell) = 2\Phi(-\ell/s)$, $s = \varsigma\sqrt{T}$. As $\phi \rightarrow 0$ the killed Ornstein–Uhlenbeck transition density converges to the killed-Brownian one, so $p_\ell(\ell) = 2\Phi(-\ell/s) + O(\phi T)$. The mean reversion acts in two opposing directions in the first-passage expansion: the drift $-\phi X$ pulls the path back toward the anchor, opposing the excursions that reach $-\ell$, while the time-changed Brownian representation lengthens the auxiliary Brownian clock by a factor $1 + O(\phi T)$, which slightly enlarges very short-boundary hitting probabilities. The sign of the $O(\phi T)$ correction is therefore not uniform in the buffer; its exact value (negative for $\ell \gtrsim 0.61$ s and positive only for very small buffers below that scale) is computed in Proposition A.7 and Corollary A.9. $\square$

Posting collateral for a larger buffer lowers p_ℓ but ties up capital at opportunity cost r . Per unit position, with liquidation penalty ϖ , the trader solves

$$\min_{\ell \geq \ell_{\text{floor}}} \mathcal{C}(\ell) = r\ell + \varpi p_\ell(\ell), \quad \ell_{\text{floor}} = (m_0 - m_m)P \geq 0,$$

with ℓ_{floor} the regulatory minimum buffer from the initial margin $m_0 \geq m_m$. Using the leading first-passage $p_\ell = 2\Phi(-\ell/s)$, the first-order condition $r = (2\varpi/s)(2\pi)^{-1/2}e^{-\ell^2/2s^2}$ gives the interior optimum

$$\ell^* = s\sqrt{2\log(2\varpi/(rs\sqrt{2\pi}))} \quad \text{when } 2\varpi > rs\sqrt{2\pi}, \quad (4.5)$$

and the corner $\ell^* = \ell_{\text{floor}}$ otherwise; the chosen buffer is $\hat{\ell}_i = \max(\ell_i^*, \ell_{\text{floor}})$. The path-risk correction doubles the effective penalty relative to the terminal-tail heuristic, widening the buffer; the comparative statics are unchanged, $\partial\ell^*/\partial s > 0$, $\partial\ell^*/\partial\varpi > 0$, $\partial\ell^*/\partial r < 0$. The mean-reversion correction to ℓ^* , of order ϕT and contracting the buffer, is given in closed form in Corollary A.9.

The boundary density. A long trader entered at price P_i , distributed across cohorts by the stationary law of the value with density f_P (the $\mathcal{N}(0, \sigma^2)$ of (3.2)), and sits at boundary $B_i = P_i - \hat{\ell}_i$. The cross-sectional density of boundaries at the current value is therefore the mixture

$$g_0 = g(v) = \mathbb{E}_{\hat{\ell}}[f_P(v + \hat{\ell})] > 0, \quad (4.6)$$

shorts contributing $\mathbb{E}_{\hat{\ell}}[f_P(v - \hat{\ell})]$ symmetrically.

Proposition 4.4 (Comparative statics of the welfare coefficient). *With g_0 from (4.6) and λ_F from (4.4), evaluated at v near the entry mode:*

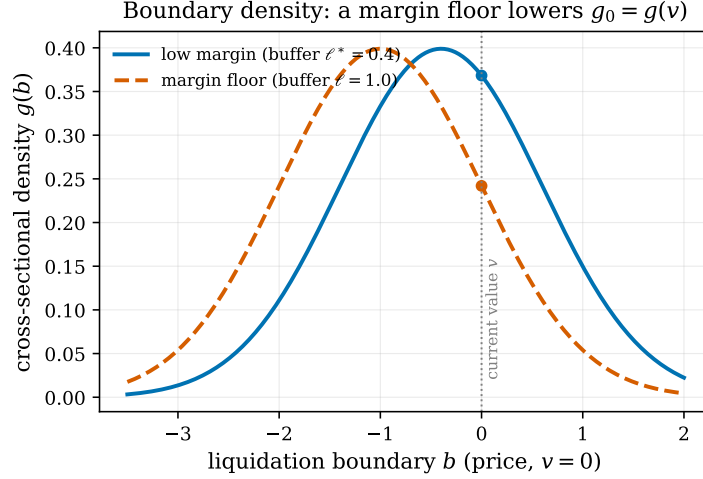


Figure 1: The derived cross-sectional boundary density (4.6) for long positions. A maintenance-margin floor widens the optimal buffer $\hat{\ell}$ of (4.5), sliding boundaries away from the current value v and lowering the marginal-position density $g_0 = g(v)$ (dots). This is the structural channel of Proposition 4.4(a).

- (a) a tighter maintenance margin (a higher floor ℓ_{floor} , hence wider $\hat{\ell}$) lowers g_0 , $\partial g_0 / \partial \ell_{\text{floor}} < 0$, since f_P is decreasing away from its mode; this is Proposition 4.2 made quantitative;
- (b) higher volatility s raises the buffer, lowering g_0 through (4.5), and lowers λ_F through (4.4), so it thins both the liquidation channel and the funding-sensitivity channel;
- (c) cheaper leverage (lower carry r) tightens the buffer, raising g_0 ; greater dispersion of valuations ν_i widens the cross-section of positions and boundaries, raising the near-boundary mass.

Proof. (a) Differentiating (4.6), $\partial g_0 / \partial \ell_{\text{floor}} = \mathbb{E}[f'_P(v + \hat{\ell}) \partial \hat{\ell} / \partial \ell_{\text{floor}}]$; for traders at the corner $\partial \hat{\ell} / \partial \ell_{\text{floor}} = 1$, and $f'_P(v + \hat{\ell}) < 0$ for $v + \hat{\ell}$ above the mode of f_P , giving the sign. (b) $\partial \ell^* / \partial s > 0$ from (4.5) feeds the same channel as (a); $\partial \lambda_F / \partial s < 0$ is immediate from (4.4). (c) $\partial \ell^* / \partial r < 0$ from (4.5) reverses (a); wider ν -dispersion spreads q^* and hence B , raising $\int_{|b-v| \leq \Delta} g$. $\square$

4.3 A continuum of leveraged positions

Index traders by $i \in [0, 1]$ as in Section 4.1, with ζ_i the type and $b_i = B(\zeta_i)$ the maintenance boundary expressed as a threshold on the price that determines fair value: a long is liquidated when the marking price falls below b_i , a short when it rises above. By Lemma 4.1 the cross-sectional distribution of thresholds admits a continuous density $g(\cdot)$ near the current fair value v , and we write

$$g_0 := g(v) > 0 \tag{4.7}$$

for the *mass of marginal positions per unit price* at the current value: the density of positions on the verge of liquidation. This g_0 is the structural object through which the depth of leverage near the boundary enters welfare.

4.4 The liquidation channel

The efficient liquidation rule closes position i when the *fair value* crosses b_i ; the mechanism closes it when the *mark* M crosses b_i . Consider longs (shorts are symmetric). Efficient liquidation occurs on $\{v < b_i\}$; mark liquidation on $\{M < b_i\}$. The two disagree exactly for positions with threshold between v and M . A wrong decision is costly in proportion to how far the position is on the wrong side of the boundary, reflecting bad debt on a missed liquidation and wrongly seized equity or price impact on a false one.

Assumption 3 (Liquidation cost). A position with threshold b_i that is liquidated when it should not be, or kept when it should be liquidated, imposes a deadweight cost $\lambda_L |b_i - v|$, where $\lambda_L > 0$ and $|b_i - v|$ is the distance by which the position is on the wrong side of the boundary at fair value.

Remark 4.5 (The linear cost from bad debt and seized equity). Assumption 3 is the reduced form of an accounting identity. A *missed* liquidation of a long with boundary $b_i \in (v, M]$ keeps a position open while insolvent: equity has fallen below maintenance by $b_i - v$ per unit, absorbed by the insurance fund or counterparties as bad debt, a loss proportional to $b_i - v$. A *false* liquidation of a long with $b_i \in (M, v)$ seizes and force-sells a position still solvent by $v - b_i$ per unit, destroying that equity through penalty and price impact, a loss proportional to $v - b_i$. In both directions the per-position deadweight is $\lambda_L |b_i - v|$, with λ_L the sum of the bad-debt and liquidation-penalty rates, which is Assumption 3. Linearity is the leading-order cost of being on the wrong side of a boundary by a small distance; the quadratic *aggregate* of Lemma 4.6 arises from integrating this linear per-position cost over the $O(e)$ cross-section that is wrongly classified.

Lemma 4.6 (Quadratic liquidation loss). *Under Assumption 3, conditional on a mark error $e := M - v$, the liquidation deadweight is*

$$D_L(e) = \frac{\lambda_L g_0}{2} e^2 + o(e^2) \quad \text{as } e \rightarrow 0,$$

and hence $\mathbb{E}[D_L] = \frac{1}{2} \lambda_L g_0 \mathbb{E}[(M - v)^2] + o(\mathbb{E}[(M - v)^2])$.

Proof. Take $e = M - v > 0$ (the case $e < 0$ is symmetric). The positions on which the mark and the efficient rule disagree are the longs with $b_i \in (v, M]$: efficient closure requires $v < b_i$, which holds, while mark closure requires $M < b_i$, which fails, so these are *missed* liquidations, each insolvent by $b_i - v$. The deadweight is

$$D_L(e) = \int_v^M \lambda_L (b_i - v) g(b_i) db_i = \lambda_L \int_0^e s g(v + s) ds = \lambda_L \left(g_0 \int_0^e s ds + \int_0^e s [g(v + s) - g_0] ds \right).$$

The first term is $\frac{1}{2} \lambda_L g_0 e^2$. For the second, continuity of g gives $\sup_{0 \leq s \leq e} |g(v + s) - g_0| \rightarrow 0$ as $e \rightarrow 0$, so its absolute value is at most $\frac{1}{2} e^2 \sup_{0 \leq s \leq e} |g(v + s) - g_0| = o(e^2)$. Taking expectations over the law of e and noting the symmetric contribution from $e < 0$ (false liquidations of longs with $b_i \in (M, v)$, each solvent by $v - b_i$) yields the claim. $\square$

4.5 The funding channel

Each funding interval, a transfer at rate f proportional to the mark-index gap passes between the sides of the contract; with $I = v$,

$$f = \lambda_I (M - v) + (\text{carry}), \quad \lambda_I > 0, \quad (4.8)$$

the carry term being a constant that the designer's intercept absorbs. A trader choosing position size q trades off private value against the funding it pays or receives. With a smooth, strictly concave private payoff $\pi(q)$ maximized at the efficient size q^* , and funding entering linearly as $-f q$, the privately optimal size solves $\pi'(q) = f$, so a nonzero funding error displaces the position by $\Delta q = -f/\pi''(q^*) + o(f)$ and destroys surplus

$$D_F(f) = \pi(q^*) - \pi(q^* + \Delta q) + f \Delta q = \frac{1}{2} \frac{f^2}{|\pi''(q^*)|} + o(f^2) = \frac{1}{2} \lambda_F (M - v)^2 + o((M - v)^2), \quad (4.9)$$

with $\lambda_F := \lambda_I^2/|\pi''(q^*)| > 0$. The funding deadweight is quadratic in the same mark error. Aggregating over the population of Section 4.1, the funding deadweight is $\mathbb{E}_\zeta[D_F(f; \zeta)] = \frac{1}{2} \lambda_F (M - v)^2 + o((M - v)^2)$ with the population-average curvature

$$\lambda_F = \mathbb{E}_\zeta \left[\frac{\lambda_I^2}{|\pi_{qq}(q^*(\zeta); \nu(\zeta))|} \right],$$

finite and positive by Assumption 2; the single-trader expression is the integrand.

Proposition 4.7 (Collapsing the index is without loss). *Let the honest index be a contemporaneous estimate $I = v + \xi$, with ξ mean-zero and independent of (v, ε, w) : the index reads un-averaged prints, and only the mark is smoothed. Then the funding deadweight is $\frac{1}{2} \lambda_F \mathbb{E}[(M - I)^2] = \frac{1}{2} \lambda_F (\mathbb{E}[(M - v)^2] + \text{Var}(\xi))$, so replacing I by v changes the loss only by the constant $\frac{1}{2} \lambda_F \text{Var}(\xi)$, independent of the instruments (a, c, τ) . The minimizing mechanism is therefore unchanged, and the liquidation channel, which compares the mark to boundaries and never sees I , is untouched. Setting $I = v$ thus isolates the design in the mark without loss.*

Proof. The trader responds to the funding it pays, $f = \lambda_I(M - I) + \text{carry}$, so the distortion surplus is $\frac{1}{2} \lambda_F (M - I)^2$ by (4.9) with $M - I$ for $M - v$. Expanding, $\mathbb{E}[(M - I)^2] = \mathbb{E}[(M - v)^2] - 2\mathbb{E}[(M - v)\xi] + \text{Var}(\xi)$, and $\mathbb{E}[(M - v)\xi] = 0$ since ξ is mean-zero and independent of (v, ε, w) , hence of $M - v$. The cross term vanishes, only a constant is added, and the liquidation deadweight does not involve I . $\square$

When a funding clamp binds, the funding deadweight is truncated below the quadratic, so (4.10) is an upper bound on the funding loss, exact in the slack-clamp regime.

Remark 4.8 (A partially manipulable index). Proposition 4.7 and the analysis take the index as clean. Suppose instead that the trades moving the mark's reference by δ also move the contemporaneous index by $\theta\delta$, with pass-through $\theta \in [0, 1]$: the index is a broad composite the same attack contaminates only partly. The funding gap is then $M - I = (M - v) - \theta\delta + o(\delta)$, so the index moves *with* the attack and the funding error is attenuated, not amplified: on the directional component the effective funding sensitivity falls from λ_I to $(1 - \theta)\lambda_I$, rescaling the funding weight $\lambda_F \mapsto (1 - \theta)^2 \lambda_F$ and the attacker's funding benefit by the same factor, while the quadratic objective (4.10) and every downstream result are preserved with a smaller funding weight. The liquidation channel, which compares the mark to fixed boundaries and never reads I , is untouched. Two readings follow. First, the clean-index normalization is *conservative* for the funding channel: $\theta = 0$ overstates, not understates, the funding loss, so taking $I = v$ cannot manufacture the over-trust region. Second, in the limit $\theta \rightarrow 1$ of an index that co-moves perfectly with the attack, the funding channel closes and the design is carried entirely by liquidation: the case in which contesting the mark is purely about who is liquidated, and the reflexive results survive unchanged because the depth feedback runs through the manipulation cost, not through which channel the error feeds.

4.6 The reduced welfare loss

Proposition 4.9 (Welfare loss). *Under Assumption 3 and the funding structure (4.8)-(4.9), the social deadweight from a mark error is*

$$\mathbb{E}[D_L] + \mathbb{E}[D_F] = \frac{1}{2}(\lambda_L g_0 + \lambda_F) \mathbb{E}[(M - v)^2] + o(\mathbb{E}[(M - v)^2]).$$

Choosing welfare units so that $\frac{1}{2}(\lambda_L g_0 + \lambda_F) = 1$, the designer's loss is

$$L := \mathbb{E}[(M(b) - v)^2], \quad (4.10)$$

the mean squared mark error, to leading order in the size of that error.

Proof. Add Lemma 4.6 and (4.9); both deadweights are $\frac{1}{2}(\cdot)\mathbb{E}[(M - v)^2] + o(\mathbb{E}[(M - v)^2])$, and the normalization sets the coefficient to one. $\square$

Remark 4.10 (Additivity and the absence of double counting). A trader enters both channels (it has a size, distorted by funding, and a boundary, possibly crossed wrongly), but the two deadweights add to leading order rather than overlapping. The mass of wrongfully liquidated traders is itself $O(e)$ (the cross-section between v and M , Lemma 4.6), each carrying an $O(e)$ liquidation loss, so $\mathbb{E}[D_L] = O(e^2)$; the funding distortion acts on the $O(1)$ mass of surviving traders but costs each only $O(e^2)$ in surplus, by the envelope property of (4.9); any interaction (a trader both near its boundary and distorted in size) is of order $O(e) \cdot O(e^2)$ and is absorbed into the remainder. Hence $\mathbb{E}[D_L] + \mathbb{E}[D_F]$ is the sum of Proposition 4.9 to order e^2 , with no term counted twice.

The quadratic objective is thus a consequence of two primitive cost structures, not a modeling shortcut, and its coefficient is interpretable: it rises with the density of marginal leverage g_0 and with the sensitivity of the funding transfer. The leading-order approximation is the statement that the mark error is small relative to the price scale on which the leverage density g varies, which is the empirically relevant case and the one in which the affine mark (3.7) is the right object. We now turn to how the error $M - v$ is generated.

4.7 The stationary flow loss

The objective (4.10) was written at a “representative date.” We make the device precise: the per-date mean squared error is the long-run flow loss an exchange bears under the stationary price law, and the liquidation deadweight is the flow rate of wrongful liquidations averaged along the path.

The fair value (3.1) is taken in its stationary law, under which it is ergodic: for any Ψ with $\mathbb{E}|\Psi(v_0)| < \infty$,

$$\frac{1}{T} \int_0^T \Psi(v_s) ds \xrightarrow{\text{a.s.}} \mathbb{E}[\Psi(v_0)] \quad (T \rightarrow \infty). \quad (4.11)$$

Run the mechanism continuously. At each instant the efficient rule liquidates the positions whose boundary the fair value has crossed, the mechanism those whose boundary the mark has crossed. The two disagree on the cross-section between v_s and M_s , a flow deadweight $D_L(M_s - v_s)$ by Lemma 4.6, and the funding transfer mis-prices by $D_F(M_s - v_s)$ each interval. The error process $e_s = M_s - v_s$ is a measurable functional of the stationary Ornstein–Uhlenbeck path (through the level v_s and its trailing window average), and of the attack type, drawn afresh from G at each marking instant (the discrete-marking convention of Remark 4.11); it is therefore stationary and, the ergodicity passing from the price process to any measurable functional of it, ergodic. The path-functional form of the ergodic theorem (4.11) (Birkhoff’s theorem applied to $\{e_s\}$ rather than

to a function of v_s alone) then makes the long-run average of the flow loss equal to its stationary expectation over the joint law of (v, ε) and $w \sim G$,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T [D_L(M_s - v_s) + D_F(M_s - v_s)] ds = \mathbb{E}[D_L(M - v) + D_F(M - v)], \quad (4.12)$$

the snapshot expectation (4.10) after the normalization of Proposition 4.9. The static analysis is therefore the stationary-flow problem, not a one-shot approximation to it: the representative-date expectation is exactly the time-average loss the exchange experiences. Finiteness of the right-hand side is immediate, $M - v$ is a sum of jointly Gaussian and independent finite-variance terms, with all moments finite (Theorem 3.7(ii)), so the criterion is well defined for every $(a, c, \tau) \in \mathcal{A}$.

Remark 4.11 (Absorbing versus flow errors and the timing convention). The two error directions are not symmetric in time, and (4.12) as written treats them as if they were. A *missed* liquidation ($b_i \in (v, M]$) keeps an insolvent position open and its bad debt accrues as a genuine flow, for which the stationary time-average is exactly right. A *false* liquidation ($b_i \in (M, v)$) destroys solvent equity once and removes the position; it is absorbing, and a literal time-integral of the stock of currently-misclassified positions would count a position that is in fact liquidated at its first crossing and gone. Identity (4.12) is the correct long-run loss under an explicit and standard timing convention: the mark is recomputed at discrete marking instants, liquidation is checked against the just-published mark at those instants, and the stationary population is replenished: a closed position is replaced by an entrant drawn from the same leverage law, maintaining the boundary density g . Under this convention the per-instant loss is the snapshot $\frac{1}{2}\lambda_L g_0 e^2$ in *both* directions, removal and replacement are carried by the stationary g , and the time-average is (4.12). A fully continuous-time treatment of the absorbing direction would replace its snapshot by a first-passage *rate* of mark-crossings times the per-event loss (an object scaling with the marking frequency and the mark's velocity rather than with $\mathbb{E}[e^2]$) which makes the false-liquidation *coefficient* direction-dependent, $\lambda_L^+ \neq \lambda_L^-$, but changes nothing structural. Every result downstream uses only that the loss is a positive-definite quadratic in the mark error: the discount by $\text{Var}(w)$ (Theorem 6.2), the window trade-off (Theorem 7.2), and the reflexive fold (Proposition 9.3) are invariant to how the quadratic's coefficient is split across the two directions, a split that rescales the loss level and shifts the intercept while leaving the responsiveness and the comparative statics intact. We carry the symmetric leading order, exact under the discrete-marking convention and structurally robust to the asymmetric-coefficient refinement.

Definition 4.12 (Small-error regime). Let $\bar{e} := (\mathbb{E}[(M - v)^2])^{1/2}$ be the root-mean-square mark error and $\underline{s}$ the smaller of the price scale on which the boundary density g varies and the curvature scale $|\pi_{qq}/\pi_{qqq}|$ of the private payoff. The leading-order objective (4.10) is the first term of an expansion in $\bar{e}/\underline{s}$: all $o(\cdot)$ remainders in Section 4 are $O(\bar{e}/\underline{s})$ relative to the displayed terms, and the affine mark (3.7) is optimal to that order (Proposition 3.5). The regime is the empirically relevant one in which mark errors are small relative to the dispersion of leverage and of desired positions.

5 Adversaries and the staleness of an averaged mark

The mark error $M(b) - v$ has two sources: the staleness ε baked into the average and the bias δ chosen by the adversary. We solve the adversary's Stage-2 problem for the two objectives and then record the moments of the staleness.

5.1 The adversary's holdings and the type

The type w and its distribution G were introduced as the “marginal value of displacing the mark.” We ground them in the adversary's book, and in doing so show that the directional and saboteur objectives of the next two subsections are the linear and quadratic terms of a single benefit function, not two unrelated models.

An adversary holds a perpetual position of size Q_A , may hold a related claim that pays in a liquidation cascade, and may hold a convex (option-like) claim that pays in dislocation. Displacing the mark by $m := a\delta$ benefits it through three routes, collected in a benefit function $\Pi_A(m)$:

- (F) *Funding.* Holding Q_A , the adversary's funding receipt moves by $-Q_A \Delta f = -Q_A \lambda_I m$ while the clamp slacks, a benefit linear in m with slope $q\lambda_I := -Q_A \lambda_I$ on the profitable side.
- (L) *Liquidation.* A displacement m liquidates the boundary mass $\approx g_0 m$ (Lemma 4.6); at a per-position cascade or auto-deleveraging profit π_A , the benefit is $\pi_A g_0 m$, again linear in m .
- (S) *Dislocation.* A convex (option-like) claim that pays in the *magnitude* of the error (its value flat at $m = 0$, with curvature $2\omega_A > 0$ there) contributes $\omega_A m^2$ to leading order.

Expanding around $m = 0$,

$$\Pi_A(m) = \underbrace{(q\lambda_I + \pi_A g_0)}_{:= w_{\text{dir}}} m + \underbrace{\omega_A}_{:= w_{\text{sab}}} m^2 + o(m^2), \quad (5.1)$$

so the net payoff $\Pi_A(a\delta) - \frac{\kappa}{2}\delta^2$ has a *directional* part with type $w_{\text{dir}} = \Pi'_A(0)$ and a *saboteur* part with type $w_{\text{sab}} = \frac{1}{2}\Pi''_A(0)$. The pure directional payoff (5.2) is the case $w_{\text{sab}} = 0$ (a linear book: a position or a funding claim); the pure saboteur payoff (5.3) is the case $w_{\text{dir}} = 0$ (a symmetric convex claim, paying in dislocation regardless of sign). Section 5.2 solves the first, Section 5.3 the second; a mixed book superposes them, the directional part canceled by the intercept and the saboteur part governing the threshold of Section 8.

The designer does not observe the adversary's book, so w is drawn from the cross-book distribution G , with mean $\bar{w}$ and variance $\text{Var}(w)$ inherited from the dispersion of positions Q_A , cascade exposures π_A , and convex claims ω_A across potential attackers. It is this dispersion (the designer's uncertainty about which book it faces) that the robust mark prices; the mean is a predictable bias the intercept removes for free (Lemma 5.2, Proposition 6.4). The decomposition $w = w_L + q\lambda_I$ into liquidation and funding stakes is exactly the split of w_{dir} in (5.1) into its liquidation and funding components, with $w_L = \pi_A g_0$.

Remark 5.1 (State-dependence of the type). The directional type $w_{\text{dir}} = q\lambda_I + \pi_A g_0$ carries the boundary density $g_0 = g(v)$ and so is in principle a function of the current value, whereas Lemma 5.2 and the orthogonality $a\bar{v}_\tau - v \perp a\delta^*$ in Lemma 6.1 treat w as price-path-independent. The gap is higher-order. The induced cross term in the loss is $2a(a-1)\mathbb{E}[v\delta^*] = 2a^2(a-1)\pi_A \text{Cov}(v, g(v))/\kappa$. For a balanced long-short book g is symmetric in the deviation, $\text{Cov}(v, g(v)) = 0$, and the orthogonality is exact. For a one-sided attack g is asymmetric, but in the small-error regime $g(v) = g_0 + O(v/\underline{s})$, so $\text{Cov}(v, g(v)) = O(\sigma^2/\underline{s})$ and the cross term is suppressed by the same ratio $\bar{e}/\underline{s}$ that controls every remainder in Section 4 (Definition 4.12); at the calibrated scales it is a few percent of the leading staleness term. The clean loss $L/\sigma^2 = 1 - 2Ca + Va^2 + \rho a^4$, and with it “only $\text{Var}(w)$ survives, the mean absorbed for free”, is therefore the leading order: exact for a balanced book, and correct to the order of the expansion for a one-sided one.

5.2 The directional manipulator

The directional manipulator profits from the *displacement* of the mark, for instance because it holds a position that gains when longs are liquidated, or a claim on the funding transfer. Its type w is the marginal value of a unit of mark displacement; given responsiveness a , a bias δ moves the mark by $a\delta$, so its payoff is

$$U_{\text{dir}}(\delta; w) = w(a\delta) - \frac{\kappa}{2}\delta^2. \quad (5.2)$$

Lemma 5.2 (Directional best response). *The payoff (5.2) is strictly concave with unique maximizer*

$$\delta^*(w) = \frac{w a}{\kappa},$$

which is linear in w and independent of v . The induced bias has mean $\bar{w}a/\kappa$ and variance $a^2 \text{Var}(w)/\kappa^2$, and contributes a mark displacement $a\delta^ = wa^2/\kappa$.*

Proof. $\partial_\delta U_{\text{dir}} = wa - \kappa\delta$ and $\partial_\delta^2 U_{\text{dir}} = -\kappa < 0$; setting the derivative to zero gives $\delta^* = wa/\kappa$. Mean, variance, and displacement follow by linearity, using that w is independent of the price path. $\square$

The crucial feature is that δ^* varies with the private type w but not with the fair value v . Its mean is a predictable bias that the designer's intercept will cancel for free; only its variance, driven by the dispersion $\text{Var}(w)$ the designer faces at Stage 0, will survive in the loss.

Remark 5.3 (The directional attack is detail-free in the price path). Because $\delta^* = wa/\kappa$ depends on neither v nor the staleness ε (Lemma 5.2), the directional results hold whatever the adversary observes about the current price: its value, its trailing average, or nothing. Only the saboteur, whose payoff is the error itself, conditions its attack on v . The information structure of Section 3.7 therefore bites only through the saboteur channel; the directional discount of Theorem 6.2 is robust to the adversary's price-path information.

5.3 The saboteur

The saboteur profits from the mark's *error* itself, for instance to maximize false liquidations or to ignite a cascade. Observing v , it chooses δ to maximize a payoff increasing in the squared mark error,

$$U_{\text{sab}}(\delta; w, v) = w(M(v + \delta) - v)^2 - \frac{\kappa}{2}\delta^2. \quad (5.3)$$

Lemma 5.4 (Saboteur best response). *Let $\mu := 2w/\kappa$ and fix $M(b) = ab + c$. If $\mu a^2 < 1$, the payoff (5.3) is strictly concave in δ with unique maximizer*

$$\delta^*(v) = \frac{\mu a}{1 - \mu a^2}[(a - 1)v + c], \quad \text{inducing the error} \quad M(v + \delta^*) - v = \frac{(a - 1)v + c}{1 - \mu a^2}.$$

If $\mu a^2 \geq 1$ the payoff is convex and unbounded above in δ : the saboteur's problem has no interior solution and a mark with responsiveness $a \geq \mu^{-1/2}$ is exploitable without bound: an unbudgeted idealization, regularized to a finite budget corner with finite profit and loss in Proposition 8.3.

Proof. Write $e(\delta) = M(v + \delta) - v = (a - 1)v + c + a\delta$, so $U_{\text{sab}} = w e^2 - \frac{\kappa}{2}\delta^2$. Then $\partial_\delta U_{\text{sab}} = 2wea - \kappa\delta$ and $\partial_\delta^2 U_{\text{sab}} = 2wa^2 - \kappa = \kappa(\mu a^2 - 1)$. If $\mu a^2 < 1$ the payoff is strictly concave; the first-order condition $2wea = \kappa\delta$, i.e. $\delta = \mu a e$, combined with $e = (a - 1)v + c + a\delta = (a - 1)v + c + \mu a^2 e$, gives $e(1 - \mu a^2) = (a - 1)v + c$ and the stated e and $\delta^* = \mu a e$. If $\mu a^2 \geq 1$ the payoff is convex, and since e grows linearly in δ while the cost grows quadratically with a coefficient no larger, $U_{\text{sab}} \rightarrow \infty$ as $|\delta| \rightarrow \infty$. $\square$

The contrast with the directional case is sharp: the saboteur's bias depends on v , and its effect is *self-reinforcing* in responsiveness, since a more responsive mark passes more of the induced error through, which raises the marginal return to injecting more bias. This feedback is what produces a threshold rather than a smooth discount in Section 8.

5.4 Staleness moments

The remaining input is the staleness $\varepsilon = v - \bar{v}_\tau$ of the averaged reference. Because the average is taken of the same mean-reverting path that determines fair value, staleness is not independent noise; the following lemma makes its moments, and their dependence on the window, explicit.

Lemma 5.5 (Staleness moments). *Let $x := \phi\tau$. The window average (3.3) satisfies*

$$C(x) := \frac{\text{Cov}(v, \bar{v}_\tau)}{\sigma^2} = \frac{1 - e^{-x}}{x}, \quad V(x) := \frac{\text{Var}(\bar{v}_\tau)}{\sigma^2} = \frac{2}{x^2}(x - 1 + e^{-x}),$$

with $0 < C(x) \leq 1$, $0 < V(x) \leq 1$, and $V(x) > C(x)$ for all $x > 0$. The staleness gap obeys

$$\frac{\text{Var}(\varepsilon)}{\sigma^2} = 1 - 2C(x) + V(x), \quad \frac{\text{Cov}(v, \varepsilon)}{\sigma^2} = 1 - C(x) > 0.$$

As $x \rightarrow 0$, $C(x) = 1 - \frac{x}{2} + O(x^2)$ and $V(x) = 1 - \frac{x}{3} + O(x^2)$, so $\text{Var}(\varepsilon)/\sigma^2 = \frac{2}{3}x + O(x^2) = \varsigma^2\tau/(3\sigma^2) + O(\tau^2)$. As $x \rightarrow \infty$, $C, V \rightarrow 0$ and the average is uninformative.

Proof. Using (3.2), $\text{Cov}(v, \bar{v}_\tau) = \frac{1}{\tau} \int_0^\tau \sigma^2 e^{-\phi u} du = \sigma^2(1 - e^{-\phi\tau})/(\phi\tau)$, which is C . The same computation gives

$$\text{Var}(\bar{v}_\tau) = \frac{1}{\tau^2} \int_0^\tau \int_0^\tau \sigma^2 e^{-\phi|u-s|} du ds = \sigma^2 \frac{2}{(\phi\tau)^2} (\phi\tau - 1 + e^{-\phi\tau}),$$

which is V . Then $\text{Var}(\varepsilon) = \text{Var}(v) + \text{Var}(\bar{v}_\tau) - 2\text{Cov}(v, \bar{v}_\tau)$ and $\text{Cov}(v, \varepsilon) = \text{Var}(v) - \text{Cov}(v, \bar{v}_\tau)$ give the stated expressions. For $V > C$, compute directly

$$V(x) - C(x) = \frac{2}{x^2}(x - 1 + e^{-x}) - \frac{1 - e^{-x}}{x} = \frac{1}{x^2}(x - 2 + (2 + x)e^{-x}),$$

and the bracket $\psi(x) = x - 2 + (2 + x)e^{-x}$ has $\psi(0) = 0$ and $\psi'(x) = 1 - (1 + x)e^{-x} > 0$ for $x > 0$, since $(1 + x)e^{-x} < 1$ there; hence $\psi > 0$ and $V > C$. The expansions follow from $e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \dots$. $\square$

The positive covariance $\text{Cov}(v, \varepsilon) = \sigma^2(1 - C) > 0$ is the content of treating staleness as a derived object: a price that has risen recently is both high and above its own trailing average, so the staleness gap carries directional information and co-moves with what the mark must track. Modeling staleness as independent noise would set this covariance to zero and overstate the harm of averaging.

6 The robust mark

We now combine the welfare loss (4.10), the directional best response of Lemma 5.2, and the staleness moments of Lemma 5.5 to characterize the optimal responsiveness for a fixed construction, that is, for a fixed window τ and hence a fixed triple

$$C := C(\phi\tau), \quad V := V(\phi\tau), \quad \rho := \rho(\tau) = \frac{\text{Var}(w)}{\kappa(\tau)^2 \sigma^2}.$$

The scalar ρ is the *manipulability index*: the variance of the manipulation incentive, scaled by the squared robustness and the value variance.

6.1 The loss as a function of responsiveness

Lemma 6.1 (Reduced loss). *Against the directional manipulator, with the intercept set to $c^* = -\bar{w}a^2/\kappa$, the loss (4.10) is*

$$\frac{L(a, \tau)}{\sigma^2} = \underbrace{\frac{1 - 2C a + V a^2}{\sigma^2}}_{\text{tracking, incl. staleness}} + \underbrace{\frac{\rho a^4}{\sigma^2}}_{\text{manipulation}}. \quad (6.1)$$

The tracking block equals $\text{Var}(a\bar{v}_\tau - v)/\sigma^2$; it is $(1 - a)^2$ when $\tau = 0$ and rises toward 1 as $\tau \rightarrow \infty$.

Proof. By (3.4) and (3.7), $M(b) - v = a(\bar{v}_\tau + \delta) + c - v = (a\bar{v}_\tau - v) + a\delta + c$. Under Lemma 5.2 the manipulator plays $\delta^* = wa/\kappa$, independent of the price path, so $a\bar{v}_\tau - v$ and $a\delta^*$ are uncorrelated. Hence

$$\mathbb{E}[(M - v)^2] = (\mathbb{E}[a\bar{v}_\tau - v] + a\mathbb{E}[\delta^*] + c)^2 + \text{Var}(a\bar{v}_\tau - v) + a^2 \text{Var}(\delta^*).$$

Now $\mathbb{E}[a\bar{v}_\tau - v] = 0$ and $a\mathbb{E}[\delta^*] = \bar{w}a^2/\kappa$, so $c^* = -\bar{w}a^2/\kappa$ annihilates the mean term. The variance term is $a^2 \text{Var}(\delta^*) = a^2 \cdot a^2 \text{Var}(w)/\kappa^2 = \rho a^4 \sigma^2$. Finally, since $\bar{v}_\tau = v - \varepsilon$, $\text{Var}(a\bar{v}_\tau - v) = \text{Var}((a - 1)v - a\varepsilon)$, and using $\text{Var}(v) = \sigma^2$, $\text{Var}(\bar{v}_\tau) = V\sigma^2$, $\text{Cov}(v, \bar{v}_\tau) = C\sigma^2$ gives $\text{Var}(a\bar{v}_\tau - v) = \sigma^2(a^2V - 2aC + 1)$. Dividing by σ^2 yields (6.1). $\square$

Equation (6.1) isolates the economics. Staleness enters through the pair (C, V) : the covariance C rewards responsiveness, the dispersion V penalizes it. Manipulation enters *quartically* because the induced bias has variance $a^2 \text{Var}(w)/\kappa^2$ and the mark scales that bias by a further factor a^2 . Only $\text{Var}(w)$, the spread of the manipulation the designer anticipates, appears; its mean $\bar{w}$ has been removed by the intercept, at no cost, exactly as Lemma 5.2 foreshadowed.

6.2 Optimal responsiveness

Theorem 6.2 (Optimal mark responsiveness). *For fixed τ , the loss (6.1) is strictly convex in a , and the welfare-optimal responsiveness a^* is the unique root in $(0, 1)$ of the cubic*

$$2\rho a^3 + V a - C = 0, \quad (6.2)$$

given in closed form by

$$a^* = U^{1/3} - \frac{V}{6\rho} U^{-1/3}, \quad U = \frac{C}{4\rho} + \sqrt{\left(\frac{C}{4\rho}\right)^2 + \left(\frac{V}{6\rho}\right)^3}. \quad (6.3)$$

It is strictly decreasing in the manipulability index ρ and in the dispersion V , and strictly increasing in the covariance C . When $\tau = 0$ (so $C = V = 1$), it solves $1 - a = 2\rho a^3$. The mark strictly underweights the reference, $a^* < 1$, whenever it is manipulable ($\rho > 0$) or stale ($V < 1$).

Proof. From (6.1), $\partial_a(L/\sigma^2) = -2C + 2Va + 4\rho a^3$ and $\partial_a^2(L/\sigma^2) = 2V + 12\rho a^2 > 0$, so L is strictly convex in a and the first-order condition is (6.2). Let $f(a) = 2\rho a^3 + Va - C$. Then f is strictly increasing, $f(0) = -C < 0$, and $f(1) = 2\rho + V - C > 0$ because $V > C$ by Lemma 5.5; hence a unique root $a^* \in (0, 1)$. The cubic (6.2) is depressed (no quadratic term), with discriminant making the radicand in (6.3) positive, so Cardano's formula gives the single real root (6.3); the absence of a negative cube root makes it numerically stable. For the comparative statics, differentiate (6.2) implicitly: $(6\rho a^2 + V) da + a^3 d\rho + a dV - dC = 0$, and $6\rho a^2 + V > 0$, so $\partial a^*/\partial \rho = -a^3/(6\rho a^2 + V) < 0$, $\partial a^*/\partial V = -a/(6\rho a^2 + V) < 0$, and $\partial a^*/\partial C = 1/(6\rho a^2 + V) > 0$. Finally $a^* = 1$ would require $f(1) = 2\rho + V - C = 0$, impossible when $\rho > 0$ or $V < 1$ (the latter forcing $V - C < 1 - C \leq 0$ only if $C \geq 1$; but $C \leq 1$ with $C = 1$ only at $\tau = 0$, where $V = 1$ too, so $V < 1 \Rightarrow$ either $\rho > 0$ or the strict inequality $f(1) > 0$ holds), giving $a^* < 1$. $\square$

Theorem 6.2 is the paper’s central result, and the cubic (6.2) states it in economic terms: its three terms are the three forces acting on responsiveness. The covariance C is the marginal value of listening, the rate at which a more responsive mark tracks fair value. The dispersion V is the marginal cost of staleness, the error a lagging average imports. And ρa^3 is the marginal cost of manipulation, rising steeply in responsiveness because a mark that listens harder both invites a larger bias and passes more of it through. The optimum sets the value of listening equal to the sum of the two costs. The closed form (6.3) makes this operational rather than merely existential: an exchange that can estimate three numbers from its own data, how well its averaged reference covaries with value (C), how dispersed that average is (V), and how manipulable it is (ρ), reads its optimal responsiveness off directly, with no simulation (Figure 2). The comparative statics are then the ones intuition would demand once the problem is posed this way: trust the reference less when it is easier to move or staler, more when it tracks value better. And the conclusion that names the paper follows at once, $a^* < 1$ whenever the reference is manipulable or stale, so a fully trusting mark is optimal only for a reference that is both unmanipulable and contemporaneous: a corner no live venue occupies. A scope note the billing requires: Theorem 6.2 characterizes the best *affine* mark. By Proposition 3.5 this is the global optimum over all measurable rules when the type law is Gaussian, and by Theorem 14.2 it is the minimax-robust optimum when the type law G is known only through its variance. The latter is the operative case: a designer uncertain about G cannot implement the unrestricted optimum against a particular non-Gaussian G without knowing it, so the responsiveness characterized here, robust to not knowing G , is the right central object, while the type-specific nonlinear optimum is the companion’s. “Central” refers to the affine rule under type uncertainty, not to a full characterization of the unrestricted optimum for an arbitrary known G .

Corollary 6.3 (Price of manipulability). *The optimized loss $L^* = L(a^*, \tau)$ is strictly increasing in ρ and bounded above by σ^2 . As $\rho \rightarrow \infty$ (a reference of vanishing depth) or as $\tau \rightarrow \infty$ (so $C, V \rightarrow 0$), $a^* \rightarrow 0$ and $L^* \uparrow \sigma^2$, the loss from ignoring the reference and marking to the anchor. The naive mark $a = 1$ has loss $\sigma^2(1 - 2C + V + \rho) = \sigma^2(\text{Var}(\varepsilon)/\sigma^2 + \rho)$, unbounded in ρ ; the robust mark caps the loss at the value of the reference’s information.*

Proof. By the envelope theorem $\partial L^*/\partial \rho = \sigma^2 a^{*4} > 0$. As $\rho \rightarrow \infty$, (6.2) forces $a^* \rightarrow 0$, and $L(0, \tau) = \sigma^2$; monotonicity gives $L^* \uparrow \sigma^2$. The same limit holds as $C, V \rightarrow 0$. The naive-mark loss is (6.1) at $a = 1$, namely $\sigma^2(1 - 2C + V + \rho)$, and $1 - 2C + V = \text{Var}(\varepsilon)/\sigma^2$ by Lemma 5.5. $\square$

6.3 Worst-case manipulation

The discount in Theorem 6.2 is driven by $\text{Var}(w)$, a property of the distribution G . The same conclusion holds if the designer, rather than knowing G , knows only that w lies in a bounded band and minimizes worst-case loss.

Proposition 6.4 (Robust discount). *Suppose the designer knows only that $w \in [\bar{w} - h, \bar{w} + h]$ and minimizes $\max_w \mathbb{E}[(M - v)^2]$ over the band, the remaining expectation taken over the price path. Then the optimal intercept cancels the midpoint $\bar{w}$, and the optimal responsiveness solves (6.2) with h^2 in place of $\text{Var}(w)$ in ρ .*

Proof. Fix (a, c) . For a fixed type w , the manipulator plays $\delta^* = wa/\kappa$ (Lemma 5.2), so the mark error is $(a\bar{v}_\tau - v) + wa^2/\kappa + c$, and its mean square over the price path is $\text{Var}(a\bar{v}_\tau - v) + (wa^2/\kappa + c)^2$, the cross term vanishing as in Lemma 6.1. The worst case over $w \in [\bar{w} - h, \bar{w} + h]$ maximizes the

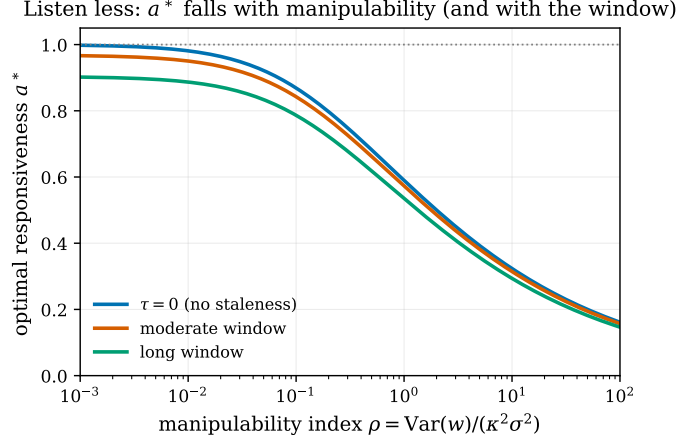


Figure 2: The optimal responsiveness a^* of Theorem 6.2, the unique root in $(0, 1)$ of the cubic (6.2), as a function of the manipulability index ρ . The mark “listens less” as the reference becomes easier to move ($a^* \downarrow$ in ρ) and as the averaging window lengthens (lower covariance C and dispersion V shift the curves down). A fully responsive mark, $a^* = 1$, is optimal only in the unmanipulable, un-stale limit $\rho \rightarrow 0$, $\tau \rightarrow 0$.

second term, attained at an endpoint:

$$\max_w \mathbb{E}[(M - v)^2] = \text{Var}(a\bar{v}_\tau - v) + \max \left\{ \left(\frac{(\bar{w}-h)a^2}{\kappa} + c \right)^2, \left(\frac{(\bar{w}+h)a^2}{\kappa} + c \right)^2 \right\}.$$

The inner maximum is minimized over c when the two endpoint magnitudes are equal, i.e. $c^* = -\bar{w}a^2/\kappa$, leaving $(ha^2/\kappa)^2 = h^2a^4/\kappa^2$. The worst-case loss is therefore $\sigma^2(1 - 2Ca + Va^2) + h^2a^4/\kappa^2$, which is (6.1) with h^2 in place of $\text{Var}(w)$; optimizing in a gives (6.2) with the substituted index. $\square$

It is therefore the *spread* of the designer’s uncertainty about the manipulation, whether a variance or a band half-width, and not its location, that forces the mark to listen less. A manipulation the designer can predict is neutralized by the intercept; only the part it cannot predict is costly.

7 The optimal averaging window

The construction is itself a choice. The designer picks $\tau \geq 0$, trading manipulation against staleness, and we write

$$L^*(\tau) = \min_{a \in [0,1]} L(a, \tau), \quad a^*(\tau) = \arg \min_a L(a, \tau)$$

for the optimized loss and its minimizer from Theorem 6.2.

7.1 The empirically relevant regime

Mark windows run from seconds to minutes while fair values revert over hours, so $x = \phi\tau \ll 1$. By Lemma 5.5, $C(x) = 1 - \frac{x}{2} + O(x^2)$ and $V(x) = 1 - \frac{x}{3} + O(x^2)$, and (6.1) reduces to the *locally-Brownian loss*

$$\frac{L(a, \tau)}{\sigma^2} = (1 - a)^2 + \phi\tau a \left(1 - \frac{a}{3} \right) + \rho(\tau) a^4 + O((\phi\tau)^2), \quad (7.1)$$

in which staleness is a linear-in- τ penalty with positive, a -dependent slope $\phi a(1 - a/3)$: averaging a fraction of the reversion time costs tracking error at rate $\varsigma^2/3$ per unit window, by Lemma 5.5. We work with the leading terms of (7.1) throughout this section; the analysis specializes the exact loss (6.1) and all statements are understood to hold up to the $O((\phi\tau)^2)$ remainder.

Assumption 4 (Iso-elastic robustness). The cost of moving the reference is iso-elastic in the window,

$$\kappa(\tau) = \kappa_0 (1 + \tau/\tau_0)^\beta, \quad \beta > 0, \tau_0 > 0,$$

so the manipulability index $\rho(\tau) = \rho_0 (1 + \tau/\tau_0)^{-2\beta}$ is strictly decreasing and strictly convex, with $\rho_0 = \text{Var}(w)/(\kappa_0^2 \sigma^2)$.

The iso-elastic form is the member of the derived class of Section 3.4 (Proposition 3.2) in which the carry rate of holding the print displaced grows as the $(\beta - 1)$ power of elapsed attack duration; $\beta = 1$ recovers the constant-carry robustness $\kappa(\tau) = \kappa_0 + \gamma\tau$ of Lemma 3.1. It places a single primitive on the manipulation technology: holding the reference displaced over a longer window raises the required expenditure polynomially, with elasticity β at scale τ_0 .

7.2 Existence

Proposition 7.1 (Existence of an interior optimal window). *Under Assumption 4, in the regime (7.1), an interior optimal window $\tau^* \in (0, \infty)$ exists if and only if the marginal robustness gain at the instantaneous mark exceeds the marginal staleness cost,*

$$|\rho'(0)| a^*(0)^3 > \phi \left(1 - \frac{a^*(0)}{3}\right), \quad |\rho'(0)| = \frac{2\beta\rho_0}{\tau_0}, \quad (7.2)$$

with $a^*(0)$ the root of $1 - a = 2\rho_0 a^3$. Otherwise the instantaneous mark $\tau^* = 0$ is optimal.

Proof. L^* is continuous on $[0, \infty)$ and differentiable on $(0, \infty)$; by the envelope theorem its derivative is the partial of (7.1) in τ evaluated at $a^*(\tau)$,

$$\frac{1}{\sigma^2} \frac{dL^*}{d\tau} = a^*(\tau) \Xi(\tau), \quad \Xi(\tau) := \phi \left(1 - \frac{a^*(\tau)}{3}\right) + \rho'(\tau) a^*(\tau)^3. \quad (7.3)$$

Since $a^* > 0$, the sign of $dL^*/d\tau$ is that of Ξ . At $\tau = 0$, $\Xi(0) = \phi(1 - a^*(0)/3) - |\rho'(0)| a^*(0)^3$, negative exactly under (7.2), so L^* strictly decreases at the instantaneous mark. As $\tau \rightarrow \infty$, $\rho'(\tau) \rightarrow 0$ and $a^*(\tau) \rightarrow 0$, so $\Xi(\tau) \rightarrow \phi > 0$ and L^* eventually increases. By continuity Ξ vanishes at some interior τ^* , a stationary point of L^* that is a local minimum since L^* falls before it and rises after. If (7.2) fails, $\Xi(0) \geq 0$; we show $\Xi \geq 0$ throughout, so $\tau = 0$ is optimal. The condition $\Xi(\tau) \geq 0$ fails only if Ξ dips below zero, which by $\Xi(\infty) = \phi > 0$ would require two sign changes and hence an interior local maximum of L^* ; but Proposition 7.3 below shows every interior stationary point of L^* is a strict local minimum in this regime, excluding a maximum. Hence $\Xi \geq 0$ and $\tau^* = 0$. $\square$

Condition (7.2) is the economic test for whether to smooth the mark at all, and it compares two marginal quantities at the instantaneous reference. The left side is the robustness the first instant of averaging buys: the rate $|\rho'(0)|$ at which manipulability falls as the window opens, felt at the responsiveness $a^*(0)^3$ with which manipulation enters the loss. The right side is the staleness that same instant adds: the reversion rate ϕ at which the value drifts away from a lagging average. When a reference is already deep or attached to a fast-moving value (ρ_0 small, ϕ large), the staleness blade dominates from the origin, the test fails, and the exchange optimally publishes the instantaneous

mark and smooths not at all. When the reference is thin and the value slow, smoothing repays its staleness immediately, and the optimal window is interior. The same marginal comparison governs that interior optimum below, now balanced away from the origin rather than at it.

7.3 Characterization and comparative statics

Theorem 7.2 (The optimal window). *Under Assumption 4 and (7.2), an interior optimum τ^* satisfies, jointly with (6.2),*

$$\phi\left(1 - \frac{a^*}{3}\right) = -\rho'(\tau^*) a^{*3} = \frac{2\beta \rho(\tau^*)}{\tau_0 + \tau^*} a^{*3}. \quad (7.4)$$

At any such optimum, the window is strictly increasing in the manipulation incentive $\text{Var}(w)$ and, at a fixed value scale σ^2 , strictly decreasing in the instantaneous variance $\varsigma^2 = 2\phi\sigma^2$ (equivalently, in ϕ):

$$\frac{d\tau^*}{d\text{Var}(w)} > 0 \quad \text{whenever} \quad \phi(\tau_0 + (1 + \beta)\tau^*) < 3\beta, \quad \frac{d\tau^*}{d\varsigma^2} < 0.$$

The first condition is implied by the slow-mean-reversion regime and holds throughout the calibrated region.

Proof. Equation (7.4) is $\Xi(\tau^*) = 0$ from (7.3) with $\rho' = -2\beta\rho/(\tau_0 + \tau)$. Treat τ^* as defined by $\Xi(\tau; \theta) = 0$ for a parameter θ , where $a^*(\tau; \theta)$ solves the inner first-order condition $F := 2(1 - a) - \phi\tau(1 - 2a/3) - 4\rho(\tau)a^3 = 0$. By the implicit function theorem

$$\frac{d\tau^*}{d\theta} = -\frac{\partial\Xi/\partial\theta}{\partial\Xi/\partial\tau}\Big|_{\tau^*},$$

and $\partial\Xi/\partial\tau|_{\tau^*} > 0$ by Proposition 7.3 (the optimum is a strict local minimum), so $\text{sign}(d\tau^*/d\theta) = -\text{sign}(\partial\Xi/\partial\theta)$. Write $G_a := 2 - \frac{2\phi\tau}{3} + 12\rho a^{*2} > 0$; from $F = 0$, $\partial a^*/\partial\rho_0 = -4(1 + \tau/\tau_0)^{-2\beta} a^{*3}/G_a < 0$ and $\partial a^*/\partial\phi = -\tau(1 - 2a^*/3)/G_a < 0$.

Volatility. Hold σ^2 fixed and vary $\varsigma^2 = 2\phi\sigma^2$, i.e. vary ϕ . Then

$$\frac{\partial\Xi}{\partial\phi} = \left(1 - \frac{a^*}{3}\right) + \frac{\partial a^*}{\partial\phi} \left(-\frac{\phi}{3} + 3\rho' a^{*2}\right).$$

The bracket $(-\phi/3 + 3\rho' a^{*2}) < 0$ since $\rho' < 0$, and $\partial a^*/\partial\phi < 0$, so their product is positive; with $(1 - a^*/3) > 0$, $\partial\Xi/\partial\phi > 0$ and $d\tau^*/d\varsigma^2 < 0$.

Manipulation. Parametrize by $\rho_0 \propto \text{Var}(w)$. Writing $p := (1 + \tau/\tau_0)^{-2\beta}$ and $u := 1 + \tau/\tau_0$, so $\rho = \rho_0 p$, $\rho' = -(2\beta/\tau_0)\rho_0 p/u$, and $\partial\rho'/\partial\rho_0 = -(2\beta/\tau_0)p/u$,

$$\frac{\partial\Xi}{\partial\rho_0} = \frac{\partial a^*}{\partial\rho_0} \left(-\frac{\phi}{3} + 3\rho' a^{*2}\right) + \frac{\partial\rho'}{\partial\rho_0} a^{*3} = p a^{*3} \left\{ \frac{4\phi}{3G_a} + \frac{24\beta\rho_0 p a^{*2}}{\tau_0 u G_a} - \frac{2\beta}{\tau_0 u} \right\},$$

using $\partial a^*/\partial\rho_0 = -4p a^{*3}/G_a$. Multiplying the brace by $\tau_0 u G_a > 0$ and substituting $G_a = 2 - \frac{2\phi\tau}{3} + 12\rho_0 p a^{*2}$, the term $24\beta\rho_0 p a^{*2}$ cancels against the same term in $2\beta G_a$, leaving the brace negative if and only if $\frac{4\phi\tau_0 u}{3} < 4\beta - \frac{4\beta\phi\tau}{3}$, that is $\phi(\tau_0 u + \beta\tau) < 3\beta$. Since $\tau_0 u = \tau_0 + \tau$, this is $\phi(\tau_0 + (1 + \beta)\tau) < 3\beta$. Under that condition $\partial\Xi/\partial\rho_0 < 0$ and $d\tau^*/d\text{Var}(w) > 0$. In the slow-mean-reversion regime $\phi\tau, \phi\tau_0 \ll 1$, the left side is small and the condition holds. $\square$

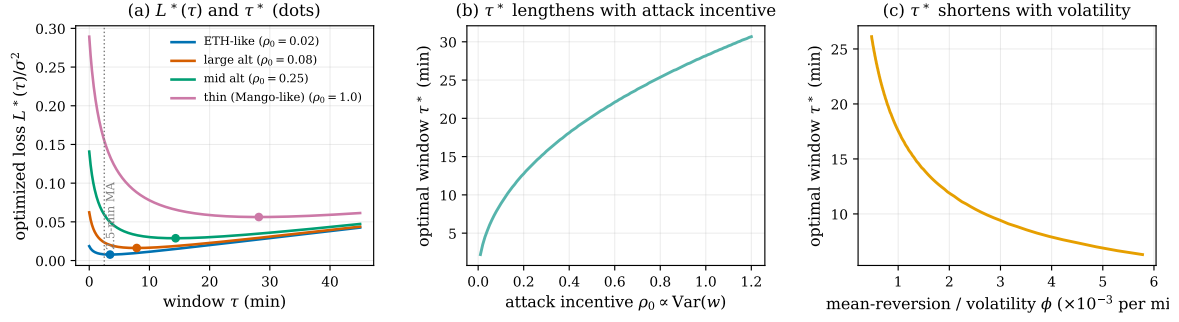


Figure 3: The optimal averaging window of Theorem 7.2, computed from the loss (6.1) with the iso-elastic robustness ($\beta = 0.5$, $\tau_0 = 1$ min) and an eight-hour anchor half-life ($\phi = \ln 2/480$ per min), a representative calibration. (a) $L^*(\tau)$ for a depth-ranked panel; the interior minimum τ^* (dots) lengthens from ≈ 3.5 min on an ETH-like reference, comparable to the short moving averages venues run (dotted), to ≈ 28 min on a thin, Mango-like one. (b) τ^* rises with the attack incentive $\rho_0 \propto \text{Var}(w)$. (c) τ^* falls with volatility ϕ . The two panels are the signed comparative statics of Theorem 7.2.

The interior window equates the two margins that Condition (7.2) compared at the origin: the left side of (7.4) is the marginal tracking error from averaging an instant longer, the right side the marginal fall in the manipulation loss as the reference grows harder to move. Its comparative statics are the ones an exchange would want to be true. A reference more valuable to attack, a larger $\text{Var}(w)$, a fatter tail of manipulation incentives, is optimally averaged *longer*, because the manipulation blade of the trade-off has lengthened relative to the staleness blade. A more volatile underlying is averaged *shorter*, because staleness now accrues faster against the lagging average. These are signed predictions, not parameters to be fit after the fact: across a cross-section of perpetuals, the thin, heavily-targeted, slowly-drifting references should carry the longest marks and the deep, quiet, fast-moving ones the shortest. A representative calibration attaches magnitudes, with the optimal window running from a few minutes on a deep, ETH-like reference (in line with the short moving averages venues actually publish) to roughly half an hour on a thin, Mango-like one (Figure 3).

7.4 Uniqueness

Proposition 7.3 (Uniqueness of the optimal window). *In the regime (7.1), with $u = 1 + \tau/\tau_0$ and $G_a = 2 - \frac{2\phi\tau}{3} + 12\rho a^{*2} > 0$, at any interior stationary point of L^* ,*

$$\frac{1}{\sigma^2} \frac{d^2 L^*}{d\tau^2} = a^* \phi \left[\frac{(2\beta + 1)(1 - \frac{a^*}{3})}{\tau_0 u} - \phi \frac{(3 - \frac{2a^*}{3})(9 - 2a^*)}{3 a^* G_a} \right]. \quad (7.5)$$

Hence every interior stationary point is a strict local minimum, and the optimal window is unique, whenever

$$\phi < \phi^* := \inf_{\tau} \frac{3(2\beta + 1)(1 - \frac{a^*}{3}) a^* G_a}{\tau_0 u (3 - \frac{2a^*}{3})(9 - 2a^*)} > 0. \quad (7.6)$$

This is the slow-mean-reversion condition; the calibration confirms $\phi \ll \phi^*$.

Proof. By (7.3), $\frac{1}{\sigma^2}L^{*'} = a^*\Xi$, so at a stationary point ($\Xi = 0$, $a^* > 0$) one has $\frac{1}{\sigma^2}L^{*''} = a^* d\Xi/d\tau$. Differentiating $\Xi = \phi(1 - a^*/3) + \rho'a^{*3}$ totally,

$$\frac{d\Xi}{d\tau} = \rho''a^{*3} + \frac{da^*}{d\tau}(3\rho'a^{*2} - \frac{\phi}{3}), \quad \frac{da^*}{d\tau} = -\frac{\Upsilon}{G_a}, \quad \Upsilon := \phi(1 - 2a^*/3) + 4\rho'a^{*3},$$

the second from differentiating the inner first-order condition $F = 0$. At the stationary point $\rho'a^{*3} = -\phi(1 - a^*/3)$; substituting this into Υ gives $\Upsilon = -\phi(3 - 2a^*/3)$, hence $da^*/d\tau = \phi(3 - 2a^*/3)/G_a > 0$, and into $3\rho'a^{*2} = -3\phi(1 - a^*/3)/a^*$ gives $3\rho'a^{*2} - \phi/3 = -\phi(9 - 2a^*)/(3a^*)$. By Assumption 4, $\rho''a^{*3} = (2\beta + 1)\phi(1 - a^*/3)/(\tau_0 u)$ at the stationary point. Collecting,

$$\frac{d\Xi}{d\tau} = \frac{(2\beta + 1)\phi(1 - a^*/3)}{\tau_0 u} - \frac{\phi(3 - 2a^*/3)}{G_a} \cdot \frac{\phi(9 - 2a^*)}{3a^*},$$

which is (7.5) after multiplying by a^* . The first term is $O(\phi)$ and the second $O(\phi^2)$, so the bracket is positive for ϕ below (7.6). With every interior stationary point a strict local minimum, two interior minima would require an interposed maximum, a non-minimal stationary point, which is excluded; uniqueness follows, and existence is Proposition 7.1. $\square$

The window is thus a genuine instrument with a unique interior setting and signed comparative statics: a more manipulable or more valuable-to-attack reference is averaged longer, a more volatile one shorter. None of this has a counterpart in a model with fixed reference robustness.

8 Sabotage and admissibility

We now turn to the saboteur of Section 5.3, whose payoff rises in the mark error itself. The relevant statistic is the *sabotage index*

$$\mu(\tau) := \frac{2w}{\kappa(\tau)}, \quad \text{so that, under Assumption 4, } \mu(\tau) = \mu_0 (1 + \tau/\tau_0)^{-\beta}, \quad \mu_0 = \frac{2w}{\kappa_0}.$$

Fix a window, hence $\mu := \mu(\tau)$, and recall from Lemma 5.4 that the saboteur's problem is well posed only on $\mu a^2 < 1$.

8.1 The saboteur loss and the threshold

Proposition 8.1 (Sabotage threshold). *Against the saboteur, the optimal intercept is $c^* = 0$, and the loss as a function of responsiveness is*

$$\frac{L(a)}{\sigma^2} = \frac{(1 - a)^2}{(1 - \mu a^2)^2} \quad \text{on } \{a : \mu a^2 < 1\}. \quad (8.1)$$

Its minimizer and value are:

- (a) *If $\mu \leq 1$, the fully responsive mark $a^* = 1$ is feasible and optimal, with zero loss: a weak saboteur cannot profitably move a responsive, well-built mark.*
- (b) *If $\mu > 1$, full responsiveness is infeasible and*

$$a^* = 1 - \sqrt{1 - 1/\mu} \in (0, 1), \quad \frac{L^*}{\sigma^2} = \frac{(1 + \sqrt{1 - 1/\mu})^2}{4},$$

with a^ strictly decreasing in μ and L^* rising from $\sigma^2/4$ at $\mu = 1$ to σ^2 as $\mu \rightarrow \infty$.*

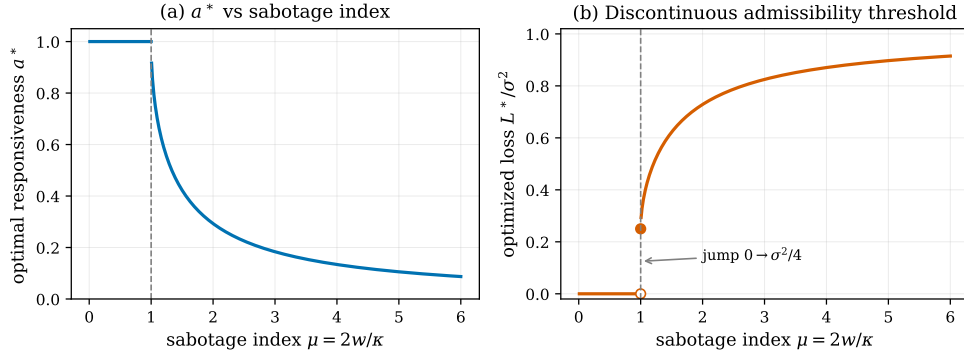


Figure 4: The sabotage threshold of Proposition 8.1. (a) The optimal responsiveness is $a^* = 1$ while the reference is sabotage-safe ($\mu \leq 1$) and falls below one the instant it is not. (b) The optimized loss jumps discontinuously from 0 to $\sigma^2/4$ as the sabotage index $\mu = 2w/\kappa$ crosses one: under-weighting arrives as a knife-edge, not a gradual discount. This is the derived “deep spot market” threshold (Corollary 8.2); a longer window lowers $\mu(\tau)$ and can push a thin reference back below one.

(c) The optimal loss jumps from 0 to $\sigma^2/4$ as μ crosses 1: under sabotage, under-weighting arrives as a discontinuous transition, not a gradual discount.

Proof. By Lemma 5.4, with intercept c the induced error is $e^* = [(a-1)v + c]/(1 - \mu a^2)$ on $\mu a^2 < 1$, so $L = \mathbb{E}[(e^*)^2] = [(a-1)^2\sigma^2 + c^2]/(1 - \mu a^2)^2$, minimized over c at $c^* = 0$ (the saboteur’s bias has zero unconditional mean), giving (8.1). On $a \in [0, 1)$ with $\mu a^2 < 1$ both numerator and denominator are positive, so minimizing L/σ^2 is equivalent to minimizing $\varphi(a) := (1-a)/(1 - \mu a^2)$. Differentiating,

$$\varphi'(a) = \frac{-(1 - \mu a^2) + (1-a)2\mu a}{(1 - \mu a^2)^2} = \frac{-(1 - 2\mu a + \mu a^2)}{(1 - \mu a^2)^2} = \frac{-(\mu a^2 - 2\mu a + 1)}{(1 - \mu a^2)^2}.$$

The numerator root solves $\mu a^2 - 2\mu a + 1 = 0$, i.e. $a = 1 \pm \sqrt{1 - 1/\mu}$, real only for $\mu \geq 1$; the root in $(0, 1)$ is $a^* = 1 - \sqrt{1 - 1/\mu}$, and one checks $\mu a^{*2} < 1$ there, so it is feasible. For $\mu \leq 1$ the numerator $\mu a^2 - 2\mu a + 1 = \mu(a-1)^2 + (1-\mu) \geq 0$ has no interior root, $\varphi' \leq 0$, and φ is minimized at the feasible corner $a = 1$, where $L = 0$. For $\mu > 1$, φ' changes sign from negative to positive at a^* , a minimum. Evaluating with $s := \sqrt{1 - 1/\mu}$: $1 - a^* = s$, and since $a^{*2} = (1-s)^2 = 1 - 2s + s^2$ with $\mu s^2 = \mu - 1$, one gets $1 - \mu a^{*2} = 2(1 - \mu + \mu s)$. The identity $\varphi(a^*) = s/[2(1 - \mu + \mu s)] = (1+s)/2$ holds because $(1+s)(1 - \mu + \mu s) = 1 - \mu + \mu s + s - \mu s + \mu s^2 = s$ on using $\mu s^2 = \mu - 1$. Hence $L^*/\sigma^2 = \varphi(a^*)^2 = (1+s)^2/4$. As $\mu \downarrow 1$, $s \rightarrow 0$ and $L^*/\sigma^2 \rightarrow 1/4$; as $\mu \rightarrow \infty$, $s \rightarrow 1$ and $L^*/\sigma^2 \rightarrow 1$. Monotonicity of a^* and L^* in μ is immediate. The jump in (c) compares the corner value 0 for $\mu \leq 1$ with the limit $1/4$ for $\mu \downarrow 1$. $\square$

The discontinuity is the economic content. A directional manipulator produces a smooth quartic penalty and a smooth discount; a saboteur, whose injected error is amplified by responsiveness, produces a knife-edge (Figure 4). Below the threshold the mark can be fully responsive and lose nothing; the instant the reference becomes exploitable, the best attainable mark must discard a quarter of the value’s variance.

8.2 Admissibility

Corollary 8.2 (Admissibility and the deep-spot-market rule). *A fully responsive mark is sabotage-safe on a given underlying if and only if $\kappa(\tau) \geq 2w$, equivalently $\mu(\tau) \leq 1$. Under Assumption 4, when $\mu_0 > 1$ the admissibility window is*

$$\tau_{\text{adm}} = \tau_0(\mu_0^{1/\beta} - 1) > 0,$$

and $\tau_{\text{adm}} = 0$ when $\mu_0 \leq 1$. A deep spot ($\mu_0 \leq 1$) admits an instantaneous mark; a thin one requires $\tau \geq \tau_{\text{adm}}$, so window length substitutes for depth. If the reference cannot be made safe at any feasible window, the perpetual is admissible only with strict under-weighting and an irreducible loss of at least $\sigma^2/4$.

Proof. Safety at $a = 1$ requires $\mu(\tau) \leq 1$ by Proposition 8.1(a), i.e. $\kappa(\tau) \geq 2w$. Under Assumption 4, $\mu(\tau) = \mu_0(1 + \tau/\tau_0)^{-\beta} \leq 1 \iff (1 + \tau/\tau_0)^\beta \geq \mu_0 \iff \tau \geq \tau_0(\mu_0^{1/\beta} - 1)$ when $\mu_0 > 1$, and holds at $\tau = 0$ when $\mu_0 \leq 1$. The irreducible loss is the $\mu \downarrow 1$ value $\sigma^2/4$ from Proposition 8.1(c). $\square$

Corollary 8.2 gives a derived basis for the requirement, used by regulators approving onshore perpetuals, that the underlying have a deep, active, and continuous spot market, and it supplies a lever: a regulator may admit a perpetual on a thinner underlying by mandating a longer mark window, accepting the staleness that Theorem 7.2 prices. The threshold is in the sabotage index $\mu_0 = 2w/\kappa_0$, the ratio of the strongest sabotage incentive to the instantaneous robustness, so depth (κ_0) and the prize (w) enter as a single sufficient statistic.

8.3 Finite attacker capital removes the singularity

The loss (8.1) diverges as $\mu a^2 \rightarrow 1$, and Lemma 5.4 calls a fully responsive mark on a thin reference “exploitable without bound.” Both are artifacts of an *unbudgeted* attacker: an unbounded dislocation requires unbounded capital, and the manipulation cost $\frac{\kappa}{2}\delta^2$ is what that capital pays. With any finite capital, which every attacker has, the singularity is an idealization, and the model is economically well posed throughout.

Proposition 8.3 (Capital-bounded sabotage). *Let the saboteur face a capital stock B , so the feasible dislocation is capped at $|\delta| \leq \bar{\delta} := \sqrt{2B/\kappa}$. Write $e_0 = (a - 1)v$ at the optimal intercept $c^* = 0$. Then for every $a \in [0, 1]$ and every sabotage index μ :*

- (i) *the saboteur’s problem is well posed with finite profit; the best response is the interior $\delta^* = \mu a e_0 / (1 - \mu a^2)$ when $\mu a^2 < 1$ and $|\delta^*| \leq \bar{\delta}$, and the budget corner $\delta = \bar{\delta} \text{sgn}(e_0)$ otherwise;*
- (ii) *the loss is finite everywhere (the budget cap removes the $\mu a^2 \rightarrow 1$ pole of (8.1)), and continuous in (a, μ) on $[0, 1] \times [0, \infty)$ except at the doubly-degenerate corner $(a, \mu) = (1, 1)$, where a fully responsive mark meets the exact sabotage threshold: there the saboteur’s payoff is flat in δ and the loss depends on the direction of approach, the worst-case value $\bar{\delta}^2$ exceeding the interior limit. Where the budget binds it is*

$$\frac{L_c(a)}{\sigma^2} = (1 - a)^2 + 2a \frac{\bar{\delta}}{\sigma} (1 - a) \sqrt{\frac{2}{\pi}} + a^2 \frac{\bar{\delta}^2}{\sigma^2}, \quad (8.2)$$

independent of μ , so the $\mu a^2 \rightarrow 1$ pole of (8.1) is removed; in particular a fully responsive mark on a thin reference ($\mu > 1$) is admissible at the finite loss $L_c(1) = \bar{\delta}^2$, not infeasible;

(iii) the singular loss (8.1) and the knife-edge admissibility of Corollary 8.2 are the infinite-capital limit $\bar{\delta} \rightarrow \infty$: the threshold $\mu = 1$ is the worst-case rule for an unbudgeted attacker, and finite capital both regularizes it and renders a thin reference admissible at a cost the capital sets.

Proof. (i) The payoff is $\pi(\delta) = w(e_0 + a\delta)^2 - \frac{\kappa}{2}\delta^2 = \frac{\kappa}{2}[\mu(e_0 + a\delta)^2 - \delta^2]$, quadratic with second derivative $\kappa(\mu a^2 - 1)$. On $\mu a^2 < 1$ it is concave with interior maximizer $\delta^* = \mu a e_0 / (1 - \mu a^2)$ (Lemma 5.4); the budget binds iff $|\delta^*| > \bar{\delta}$, and since $\delta^* \propto e_0$ the constrained maximizer is the corner $\delta = \bar{\delta} \text{sgn}(e_0)$. On $\mu a^2 \geq 1$ the payoff is convex, so its maximum over $[-\bar{\delta}, \bar{\delta}]$ is at an endpoint; the common $-\frac{\kappa}{2}\delta^2$ term makes the saboteur choose $\text{sgn}(\delta) = \text{sgn}(e_0)$ to enlarge $(e_0 + a\delta)^2$, the same corner. In all cases $|\delta| \leq \bar{\delta}$ and the profit is finite. (ii) At the corner the induced error is $e = e_0 + a\bar{\delta} \text{sgn}(e_0) = \text{sgn}(e_0)(|e_0| + a\bar{\delta})$, with $|e_0| = (1 - a)|v|$; hence $\mathbb{E}[e^2] = \mathbb{E}[(1 - a)|v| + a\bar{\delta}]^2 = (1 - a)^2\sigma^2 + 2a\bar{\delta}(1 - a)\mathbb{E}|v| + a^2\bar{\delta}^2$, and $\mathbb{E}|v| = \sigma\sqrt{2/\pi}$ gives (8.2), free of μ . For $\mu a^2 < 1$ the loss is the interior integrand $(1 - a)^2 v^2 / (1 - \mu a^2)^2$ over $\{|v| < v_c\}$, where the budget slacks ($v_c = \bar{\delta}(1 - \mu a^2) / (\mu a(1 - a))$), plus the corner integrand over $\{|v| \geq v_c\}$; as $\mu a^2 \rightarrow 1^-$ with a bounded away from 1, $v_c \rightarrow 0$ and the interior contribution vanishes, since its integrand is $O(1)$ there over a window of width $O(1 - \mu a^2)$, so $L \rightarrow L_c$ continuously and the pole is removed. This continuity fails only at the corner $(a, \mu) = (1, 1)$, where the two degeneracies coincide: along $\mu = 1$ the window stays open ($v_c = \bar{\delta}(1 + a)/a \rightarrow 2\bar{\delta}$), the interior error $-v/(1 + a)$ survives, and $L \rightarrow \mathbb{E}[\min(v^2/4, \bar{\delta}^2)] < \bar{\delta}^2$, whereas along $\mu a^2 = 1$ the all-corner value $\bar{\delta}^2$ obtains; at the corner the saboteur's payoff is flat in δ and the loss is direction-dependent. At $a = 1$, $e_0 = 0$ and the corner gives $e = \bar{\delta}$ for $\mu > 1$ (the saboteur exhausts its budget against a fully responsive mark), so $L_c(1) = \bar{\delta}^2$. (iii) As $\bar{\delta} \rightarrow \infty$ the budget never binds on $\mu a^2 < 1$ and the interior loss is $\sigma^2(1 - a)^2 / (1 - \mu a^2)^2$, recovering (8.1); on $\mu a^2 \geq 1$, $L_c \rightarrow \infty$, recovering the infeasibility. The admissibility rule $\mu \leq 1$ is thus the $\bar{\delta} \rightarrow \infty$ worst case; for finite $\bar{\delta}$ the loss is everywhere finite, and the designer's choice between full responsiveness (loss $\bar{\delta}^2$ when $\mu > 1$) and under-weighting is a smooth comparison rather than a knife-edge. $\square$

The capital bound restores economic sense in the region the bare model leaves singular: an attack costs capital, so its size is bounded and its damage finite. The admissibility threshold $\mu \leq 1$ survives as the conservative, capital-free worst case (the right object for a regulator who will not assume a budget on the attacker), while Proposition 8.3 prices what a known capital stock buys: a thin reference, inadmissible under the worst case, is admissible at a loss that scales with the square of the attacker's reach $\bar{\delta}$.

Remark 8.4 (Scope of the sabotage results). Two scope points the sabotage analysis requires. *The global functional form is not load-bearing.* The payoff (5.3) is the leading quadratic term of the saboteur's return in the mark error (Section 5.1), valid for displacements small on the curvature scale of the underlying payoff. The threshold $\mu a^2 = 1$ and the jump of Proposition 8.1 are properties of this local model, and reading the convexity beyond it as an exact global payoff would extrapolate the quadratic outside its validity, except that the capital bound of Proposition 8.3 caps the dislocation at $\bar{\delta} = \sqrt{2B/\kappa}$. When $\bar{\delta}$ lies within the Taylor radius the analysis never leaves the locally valid region: the saboteur's reach is finite, the loss (8.2) is the local quadratic evaluated at a bounded corner, and the $\mu a^2 \rightarrow 1$ singularity is the infinite-capital idealization rather than a prediction. *The admissibility cutoff is the affine-mark frontier.* Against the saboteur the unrestricted optimal mark is nonlinear (Proposition 3.5), and a nonlinear mark (saturating or clamping the pass-through where the induced error is largest) can be sabotage-safe at depths the affine cutoff $\mu \leq 1$ rejects. The deep-spot rule is therefore a *sufficient* condition for affine-mark safety and a conservative one for policy: a regulator applying it never admits an unsafe affine reference, but may reject one a better-designed nonlinear mark would render safe. The cutoff is a clean implementable bound, deliberately on the conservative side of the nonlinear frontier the companion characterizes.

9 Endogenous reference depth and the manipulation trap

The robust mark still rests on a cost of moving the reference that, however microfounded from an averaging operator, is exogenous to the design. This section closes that gap from both ends. It first derives the manipulation cost from a market-microstructure primitive, a resilient reference book, so the convex, window-increasing cost is a consequence rather than a posit. It then lets the reference's depth itself respond to the design, and shows that reliance and manipulability are jointly determined through a liquidity feedback that the reduced form cannot see: the reference market can have multiple equilibria, and relying on it too heavily can tip it from a deep, honest regime into a thin, manipulated one. That feedback is a genuinely new force, distinct from the intuitive injunction to discount a manipulable price.

9.1 The cost of moving the reference, derived

Proposition 9.1 (Manipulation cost from book resilience). *Let the reference's deviation from fundamental obey $d_t = \int_0^t G(t-u) \dot{x}_u du$ for a resilient impact kernel G with instantaneous depth $G(0)^{-1} = \kappa_0$, where $\dot{x}$ is the manipulator's trading intensity. The minimal cost of biasing the τ -window mark $\frac{1}{\tau} \int_0^\tau d_u du$ by δ is $\frac{1}{2} \kappa(\tau) \delta^2$ with $\kappa(\tau)$ increasing in τ . For exponential resilience $G(t) = \kappa_0^{-1} e^{-t/\tau_0}$, holding a constant dislocation gives $\kappa(\tau) = \kappa_0(1 + 2\tau/\tau_0)$; for power-law resilience with recovery exponent $1 - \beta$, $\kappa(\tau) = \kappa_0(1 + \tau/\tau_0)^\beta$, the iso-elastic form of Section 7, in which κ_0 is the instantaneous depth, τ_0 the resilience timescale, and β the resilience exponent. The convexity in δ and the monotonicity in τ are properties of any resilient book.*

Proof. To sustain a constant dislocation $d_u = \delta$ over $[0, \tau]$ against exponential resilience, the manipulator establishes the impact with an initial impulse $\kappa_0 \delta$ bought at average premium $\delta/2$, cost $\frac{1}{2} \kappa_0 \delta^2$, then replenishes at rate $\kappa_0 \delta / \tau_0$ to offset decay, paying δ per unit, a flow cost $\kappa_0 \delta^2 / \tau_0$ per unit time and $\kappa_0 \delta^2 \tau / \tau_0$ over the window; the total $\frac{1}{2} \kappa_0 \delta^2 (1 + 2\tau/\tau_0)$ gives $\kappa(\tau) = \kappa_0(1 + 2\tau/\tau_0)$. The constant path is cost-minimizing among those with average δ because the instantaneous cost is convex. For a power-law impact kernel $G(t) = \kappa_0^{-1} (1 + t/\tau_0)^{-\beta}$, the constant-impact strategy holding $d_u = \delta$ on $[0, \tau]$ requires, in the regime $\tau \gg \tau_0$ where $G(t) \sim \kappa_0^{-1} (t/\tau_0)^{-\beta}$, a trading rate solving the Volterra equation $\delta = \int_0^t G(t-r) \dot{x}_r dr$. In Laplace transforms $\hat{x}(p) = \delta / (p \hat{G}(p))$ with $\hat{G}(p) = \kappa_0^{-1} \tau_0^\beta \Gamma(1 - \beta) p^{\beta-1}$, so $\hat{x}(p) \propto \delta \kappa_0 \tau_0^{-\beta} p^{-\beta}$ and $\dot{x}_t \propto \delta \kappa_0 t^{\beta-1}$; the acquired position $X(\tau) = \int_0^\tau \dot{x} \propto \delta \kappa_0 \tau^\beta$ and the cost $\delta X(\tau) \propto \kappa_0 \delta^2 \tau^\beta$, giving $\kappa(\tau) \propto \kappa_0 (\tau/\tau_0)^\beta$: the iso-elastic form, regularized to $\kappa_0(1 + \tau/\tau_0)^\beta$ with $\kappa(0) = \kappa_0$. The exponent is intrinsic to the kernel, not an artifact of the constant-impact path. Let $C^*(\delta, \tau) = \min \left\{ \frac{1}{2} \iint_{[0, \tau]^2} G(|t-s|) \dot{x}_t \dot{x}_s dt ds : \frac{1}{\tau} \int_0^\tau d_u du = \delta \right\}$ be the cost-optimal program, with $d_u = \int_0^u G(u-r) \dot{x}_r dr$. In the pure-power regime $G(t) = \kappa_0^{-1} \tau_0^\beta t^{-\beta}$ is homogeneous of degree $-\beta$. Substitute $t = \tau t'$ and $\dot{x}_t = \delta \kappa_0 \tau_0^{-\beta} \tau^{\beta-1} \dot{y}_{t'}$: then $d_{\tau u'} = \delta \int_0^{u'} |u' - r'|^{-\beta} \dot{y}_{r'} dr'$, so the constraint $\frac{1}{\tau} \int_0^\tau d = \delta$ becomes the τ - and δ -free constraint $\int_0^1 \int_0^{u'} |u' - r'|^{-\beta} \dot{y}_{r'} dr' du' = 1$ on the rescaled path, and the cost becomes $\kappa_0 \delta^2 (\tau/\tau_0)^\beta \cdot \frac{1}{2} \iint_{[0, 1]^2} |t' - s'|^{-\beta} \dot{y}_{t'} \dot{y}_{s'} dt' ds'$. The minimization over $\dot{y}$ subject to the τ -free constraint is therefore a pure number $\frac{1}{2} c_\beta$, independent of δ, τ , giving $C^*(\delta, \tau) = \frac{1}{2} c_\beta \kappa_0 \delta^2 (\tau/\tau_0)^\beta$ exactly: the exponent β is fixed by the homogeneity degree of G and only the prefactor c_β by the optimization, so the constant-impact path overstates the constant but recovers the exponent. Regularizing G to $\kappa_0^{-1} (1 + t/\tau_0)^{-\beta}$ gives $\kappa(0) = \kappa_0$ and the iso-elastic $\kappa(\tau) = \kappa_0(1 + \tau/\tau_0)^\beta$. Convexity in δ holds because the cost is a positive-definite quadratic form in the trading path, G being completely monotone hence of positive type. $\square$

The robustness technology of the paper is therefore not assumed: κ_0 is the inverse instantaneous price impact of the reference venue, τ_0 its resilience time, and β its resilience exponent, all

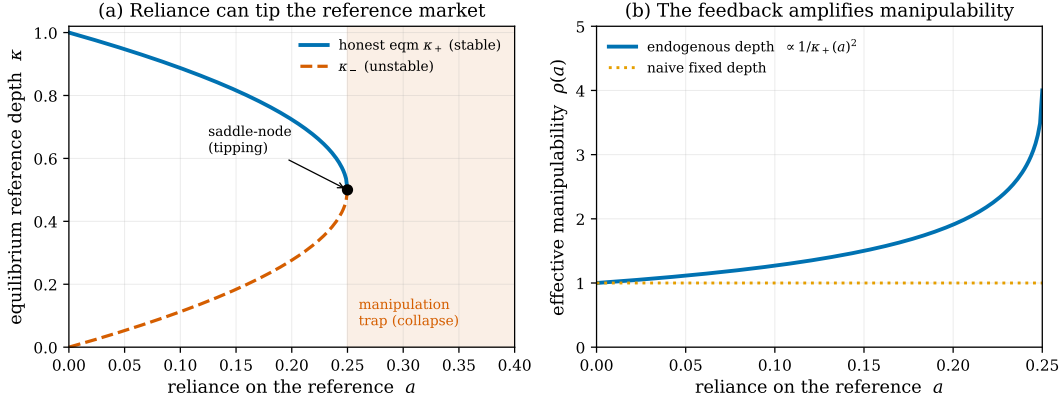


Figure 5: Endogenous reference depth (Proposition 9.2). (a) Equilibrium depth against reliance: a stable honest branch κ_0^+ and an unstable thin branch κ_0^- meet in a saddle-node at a_{tip} , beyond which no active-liquidity equilibrium exists and the reference collapses. The honest branch falls as reliance rises. (b) The effective manipulability $\rho(a) \propto 1/\kappa_0^+(a)^2$ rises with reliance through the depth feedback and diverges at the tipping point, far above the naive fixed-depth value the reduced form assumes.

measurable microstructure objects. But the derivation took depth as given. The design relies on the reference; the reference's depth is supplied by traders who are themselves exposed to the manipulator. Letting that depth respond is where the new economics lies.

9.2 Endogenous depth and the manipulation trap

Proposition 9.2 (The manipulation trap). *Suppose the instantaneous depth κ_0 is supplied competitively by liquidity providers who earn a gross return s per unit depth but lose to the manipulator through adverse selection an amount $\ell_0 \delta^*$ increasing in the manipulator's bias, with linear supply $\kappa_0 = R/c_s$ in the net return $R = s - \ell_0 \delta^*$. With the best response $\delta^* = a\bar{w}/\kappa_0$ (reliance a , representative type $\bar{w}$), equilibrium depth solves*

$$c_s \kappa_0^2 - s \kappa_0 + \ell_0 \bar{w} a = 0, \quad \kappa_0^\pm = \frac{s \pm \sqrt{s^2 - 4c_s \ell_0 \bar{w} a}}{2c_s}.$$

For reliance below $a_{\text{tip}} = s^2/(4c_s \ell_0 \bar{w})$ there are two equilibria: a deep, honest reference κ_0^+ (stable) and a thin, manipulated one κ_0^- (unstable); at a_{tip} they merge in a saddle-node, and for $a > a_{\text{tip}}$ no active-liquidity equilibrium exists and the reference collapses to a manipulated floor. The honest depth κ_0^+ is strictly decreasing in reliance: relying more on the reference thins the very market that keeps it honest.

Proof. Substituting $\delta^* = a\bar{w}/\kappa_0$ into $c_s \kappa_0 = s - \ell_0 a\bar{w}/\kappa_0$ and clearing the denominator gives the quadratic, with roots $\kappa_0^\pm$ real iff the discriminant $s^2 - 4c_s \ell_0 \bar{w} a \geq 0$, i.e. $a \leq a_{\text{tip}}$. With $\mathcal{Q}(\kappa_0) = c_s \kappa_0^2 - s \kappa_0 + \ell_0 \bar{w} a$, $\mathcal{Q}'(\kappa_0^+) = 2c_s \kappa_0^+ - s > 0$ and $\mathcal{Q}'(\kappa_0^-) < 0$, so under the natural tâtonnement (depth rises when the net return is positive) κ_0^+ is stable and κ_0^- unstable. Implicit differentiation gives $d\kappa_0^+/da = -\ell_0 \bar{w}/(2c_s \kappa_0^+ - s) < 0$. $\square$

The intuitive reading of the paper (a manipulable price should be discounted) treats depth as a primitive. Proposition 9.2 says it is not: depth is supplied by traders who lose to the manipulator,

so it falls as reliance rises, and past a point it collapses. The reference's manipulability is thus partly chosen by the designer, not inherited, and the choice has a catastrophic region. This is a feedback the reduced-form model is structurally unable to express, because there κ_0 is a parameter.

The linear supply and adverse selection of Proposition 9.2 are a minimal device, not the source of the fold. The tipping survives any increasing specification.

Proposition 9.3 (Robustness of the trap). *Let depth be supplied by any increasing differentiable S ($S' > 0$) and adverse selection by any increasing differentiable Λ ($\Lambda' > 0$), so equilibrium depth solves the fixed point $\kappa_0 = \Psi(\kappa_0; a) \equiv S(s - \Lambda(a\bar{w}/\kappa_0))$, with Ψ increasing in κ_0 and decreasing in a . Let $\Gamma(\kappa_0; a) = S' \Lambda' a\bar{w}/\kappa_0^2 = \partial_{\kappa_0} \Psi$ be the feedback loop gain. Then (i) where $\Gamma < 1$ throughout, the equilibrium is unique; (ii) where $\Gamma > 1$ on an interval of depths, there are multiple equilibria, the largest (honest) one stable; (iii) since Γ increases in a , there is a tipping reliance a_{tip} , characterized by the tangency $\Psi(\kappa_0; a_{\text{tip}}) = \kappa_0$ with $\partial_{\kappa_0} \Psi = \Gamma = 1$, at which the honest and unstable equilibria collide in a saddle-node and above which the honest equilibrium ceases to exist. The linear case of Proposition 9.2 is the instance $S(R) = R/c_s$, $\Lambda(\delta) = \ell_0 \delta$, with $a_{\text{tip}} = s^2/(4c_s \ell_0 \bar{w})$.*

Remark 9.4 (A positive depth floor). Economic admissibility requires the supply S to be bounded below by a positive residual depth $\underline{\kappa} > 0$: some liquidity (fee-driven, passive, or non-strategic) is always present, so price impact is finite however heavily the reference is relied upon. The collapse of the honest branch past the fold is then to $\underline{\kappa}$, a thin but positive depth, not to the singular $\kappa = 0$ of the bare linear extrapolation $S(R) = R/c_s$ (which, read literally, would make $R < 0$ and depth negative); the manipulation cost $\frac{\kappa}{2} \delta^2$ stays bounded below and no attack is ever infinitely cheap. The linear instance of Proposition 9.2 is to be read with this floor in force (κ_0^+ the honest root and $\underline{\kappa}$ the collapsed state), so that the two branches the fold separates are both economically admissible depths.

Proof. $\partial_{\kappa_0} \Psi = S' \cdot (-\Lambda') \cdot (-a\bar{w}/\kappa_0^2) = \Gamma > 0$ and $\partial_a \Psi = S' \cdot (-\Lambda') \cdot (\bar{w}/\kappa_0) < 0$, so Ψ rises in κ_0 and falls in a . The map $\Psi(\cdot; a)$ is increasing and carries the compact interval $[\underline{\kappa}, \bar{\kappa}]$ into itself, with $\bar{\kappa} = S(s)$ the saturation depth and $\underline{\kappa} \geq 0$ the floor reached as $\kappa_0 \downarrow 0$ sends the bias $\delta^* = a\bar{w}/\kappa_0$, hence the adverse selection, to its maximum; by Tarski's theorem its fixed-point set is a nonempty complete lattice with a greatest element $\kappa_0^+(a)$, the honest branch, and a least element, the manipulated floor. Where $\Gamma < 1$ throughout, Ψ is a contraction and the fixed point is unique; where $\Gamma > 1$ on an interval the greatest and least fixed points are distinct, separated by an unstable middle fixed point at which Ψ crosses the diagonal from below. Because Ψ is decreasing in a , its greatest fixed point is non-increasing in a ; as a rises the honest and middle fixed points approach and at a_{tip} become tangent, the tangency $\Psi = \kappa_0$, $\Psi' = 1$ being the saddle-node, beyond which the honest branch is absent and the greatest fixed point drops discontinuously to the floor. For the linear forms $\Psi(\kappa_0) = (s - \ell_0 a\bar{w}/\kappa_0)/c_s$, and tangency with the diagonal is the double root of $c_s \kappa_0^2 - s \kappa_0 + \ell_0 \bar{w} a = 0$, i.e. vanishing discriminant at $a_{\text{tip}} = s^2/(4c_s \ell_0 \bar{w})$. $\square$

The trap is thus a feature of the feedback's strength, the loop gain exceeding one, rather than of any functional form (Figure 5). Weak adverse selection or inelastic liquidity supply ($\Gamma < 1$) yields a unique, safe equilibrium with no fold; strong feedback ($\Gamma > 1$ over a range) yields the multiplicity and the tipping point, and a richer liquidity-provision model changes the shape of the branches and the location of a_{tip} but not the existence of the saddle-node. The linear specification is the least structure that exhibits it, verified to persist under saturating supply and convex adverse selection.

9.3 Design under endogenous depth

Proposition 9.5 (Reflexive manipulability and the design response). *At the honest equilibrium the effective manipulability is $\rho(a) = \text{Var}(w)/(\kappa_0^+(a)^2\sigma^2)$, with $d\rho/da = -2\rho(d\kappa_0^+/da)/\kappa_0^+ > 0$, rising with reliance through the depth feedback and diverging as $a \uparrow a_{\text{tip}}$ (where $d\kappa_0^+/da \rightarrow -\infty$). The robust-mark first-order condition becomes $2\rho(a)a^3 + Va - C = 0$ with ρ now a function of a ; the extra marginal cost $\rho'(a)a^4 > 0$ lowers the optimal reliance below the fixed-depth optimum of Theorem 6.2 and bounds it away from a_{tip} . The honest equilibrium exhibits hysteresis: a marginal deterioration in the reference market (a rise in manipulator wealth $\bar{w}$ or a fall in the liquidity return s) that crosses the fold collapses depth discontinuously, and restoring the primitive does not restore depth until the lower fold is recrossed.*

Proof. Differentiating $\rho(a) = \text{Var}(w)/(\kappa_0^+(a)^2\sigma^2)$ gives $d\rho/da = -2\rho(d\kappa_0^+/da)/\kappa_0^+$, positive since $d\kappa_0^+/da < 0$ (Proposition 9.2); as $a \uparrow a_{\text{tip}}$, $2c_s\kappa_0^+ - s \rightarrow 0^+$ so $d\kappa_0^+/da \rightarrow -\infty$ and $d\rho/da \rightarrow +\infty$. The objective $L/\sigma^2 = 1 - 2Ca + Va^2 + \rho(a)a^4$ has dL/da augmented by $\rho'(a)a^4 > 0$ relative to the fixed- ρ condition, so its root falls; the divergence of ρ' at a_{tip} makes the marginal cost unbounded there, bounding a^* below it. Hysteresis follows from the branch structure of Proposition 9.2. For a primitive below the fold, the honest equilibrium κ_0^+ and the unstable κ_0^- coexist with κ_0^+ stable under the tâtonnement; a deterioration (a rise in $\bar{w}$ or a fall in s) that drives the discriminant $s^2 - 4c_s\ell_0\bar{w}a$ negative annihilates both in a saddle-node and collapses depth to the floor. Reversing the deterioration restores the discriminant and re-creates $\kappa_0^\pm$, but the state, now resting at the floor, lies below the unstable κ_0^- , which is the basin boundary (Proposition 10.13; the drift $\gamma(\Psi - \kappa)$ is negative between the floor and κ_0^-); the deterministic dynamics therefore do not return to κ_0^+ until a further, opposite move pushes the state back above κ_0^- , the lower fold of the reverse path. Existence of the interior minimizer $a_R < a_{\text{tip}}$ is made rigorous in Proposition 10.1. $\square$

The contribution is therefore not the observation that manipulable references should be discounted, which is intuitive and which the reduced-form shrinkage of the robust mark already encodes. It is that a designed reference is *reflexive*: the act of relying on it changes its manipulability, through the liquidity that responds to the manipulation reliance invites, and this feedback generates multiplicity, a tipping point, and hysteresis, phenomena absent from the price-taking intuition and from any model that holds depth fixed. The design problem is then not only how much to trust the reference but how to keep it in the basin of the honest equilibrium. The window reappears here with a second role: by raising the cost of manipulation it lowers δ^* , shrinks the adverse selection $\ell_0\delta^*$, and so widens the region of reliance over which the honest equilibrium survives.

9.4 Endogenous attacker entry

The fold of Proposition 9.3 treats the attacker population as fixed: a given dispersion $\text{Var}(w)$ of a drawn type. But the prize is open interest near a boundary, and a thinner book makes manipulation cheaper, so a reference that thins draws *more* attackers: a second feedback running alongside reliance→depth. Endogenizing it through free entry shows that it compounds the fold rather than relaxing it.

Let a mass n of attackers, each drawn from the same type law G that disperses the directional adversary of Section 5.2, contest the reference. Their aggregate adverse selection on providers scales with the *mean* attack $n\bar{w}$ (the sum of n independent draws), so the depth map becomes $\kappa = \Psi(\kappa, n; a)$ with $\partial_n\Psi < 0$, while the *dispersion* $\text{Var}(w)$ of each draw is what the mark discounts (Theorem 6.2). These are the two moments of one law G , now carried by an endogenous mass: the fold sees the mean, the discount sees the variance, and a single heterogeneous-entrant model

holds both. The “identical attackers” reading is the degenerate case $G = \delta_{\bar{w}}$ and is without loss for the fold, which depends on the aggregate mean alone; retaining the dispersion is what keeps the discount of Section 6 and the entry-fold here in the same model rather than two. Entry is free: an attacker pays a cost F and earns a manipulation profit that rises as the book thins, cheaper to move, and falls in the number of rivals contesting the same liquidation pool, a congestion of elasticity $q > 0$. The zero-profit condition on *expected* profit pins the entrant mass $n(\kappa; a)$, decreasing in depth ($n'(\kappa) < 0$: a thinner book draws entry).

Proposition 9.6 (Entry compounds the fold). *Along the free-entry equilibrium the effective loop gain decomposes additively,*

$$\Gamma_{\text{tot}} = \Gamma_{\text{depth}} + \Gamma_{\text{entry}}, \quad \Gamma_{\text{entry}} = (\partial_n \Psi) n'(\kappa) > 0,$$

where $\Gamma_{\text{depth}} = \partial_\kappa \Psi|_n$ is the loop gain a fixed-attacker analysis measures. Under prize-splitting congestion of elasticity q , $\Gamma_{\text{entry}} = \Gamma_{\text{depth}}/q$, so $\Gamma_{\text{tot}} = (1 + 1/q) \Gamma_{\text{depth}}$ and the reference reaches its fold $\Gamma_{\text{tot}} = 1$ when

$$\Gamma_{\text{depth}} = \frac{q}{q+1} < 1,$$

strictly below the fixed-attacker threshold of one. Let $\alpha > 0$ denote the scale of the per-attacker prize in the free-entry condition. With linear supply the honest depth then folds at $\kappa_{\text{fold}} = s(q+1)/(c_s(2q+1))$ and a reliance $a_{\text{tip}}^{\text{entry}} < a_{\text{tip}}$, the fixed-point equation clearing for $q = 1$ to the cubic $c_s \kappa^3 - s \kappa^2 + (\ell_0 \alpha \bar{w}^3/F) a^3 = 0$. As $q \rightarrow \infty$ (a rigid attacker population), entry is inelastic, $\Gamma_{\text{entry}} \rightarrow 0$, and the fixed-attacker fold is recovered; as $q \rightarrow 0$ (entry floods at the first sign of thinness), the safe region collapses.

Proof. The effective map is $\kappa = \Psi(\kappa, n(\kappa; a); a)$; differentiating its right-hand side in κ gives $\Gamma_{\text{tot}} = \partial_\kappa \Psi + \partial_n \Psi \cdot n'(\kappa)$, the first term $\Gamma_{\text{depth}} > 0$ and the second $\Gamma_{\text{entry}} = (\partial_n \Psi)(n') > 0$ as a product of two negatives. For the linear-supply instance $\Psi = (s - \ell_0 \bar{w} a n / \kappa) / c_s$ with free entry $n = (\alpha a^2 \bar{w}^2 / (F \kappa))^{1/q} = C a^{2/q} \kappa^{-1/q}$, one has $\partial_\kappa \Psi|_n = \ell_0 \bar{w} a n / (c_s \kappa^2) = \Gamma_{\text{depth}}$, $\partial_n \Psi = -\ell_0 \bar{w} a / (c_s \kappa)$, and $n'(\kappa) = -(1/q) n / \kappa$, so $\Gamma_{\text{entry}} = (\ell_0 \bar{w} a / (c_s \kappa)) (1/q) (n / \kappa) = (1/q) \Gamma_{\text{depth}}$; hence $\Gamma_{\text{tot}} = (1 + 1/q) \Gamma_{\text{depth}}$, unit at $\Gamma_{\text{depth}} = q/(q+1)$. Substituting n into the fixed point $c_s \kappa = s - \ell_0 \bar{w} a n / \kappa$ gives $c_s \kappa = s - \ell_0 \bar{w} C a^{1+2/q} \kappa^{-1-1/q}$; the saddle-node, where the right-hand side has unit κ -slope ($\Gamma_{\text{tot}} = 1$), satisfies $(1 + 1/q)(s/(c_s \kappa) - 1) = 1$, i.e. $\kappa_{\text{fold}} = s(q+1)/(c_s(2q+1))$, and for $q = 1$ the fixed point clears to the stated cubic. Because $\Gamma_{\text{tot}} > \Gamma_{\text{depth}}$ at every reliance, the fold is reached strictly earlier than in the fixed-attacker model, $a_{\text{tip}}^{\text{entry}} < a_{\text{tip}}$; the q -limits are immediate from $\Gamma_{\text{entry}} = \Gamma_{\text{depth}}/q$. $\square$

The consequence sharpens the over-trust catastrophe. A price-taking analyst who measures the depth-channel loop gain Γ_{depth} and reads the fold at one is wrong by the entry margin: the true fold is at $q/(q+1)$, so any reliance with $\Gamma_{\text{depth}} \in (q/(q+1), 1)$ has already passed the fold while still appearing safe. Theorem 10.4 and the reflexive results of Section 10 apply unchanged with $a_{\text{tip}}^{\text{entry}}$ in place of a_{tip} , because they hold for any microfoundation with a fold; the effect of endogenous entry is to relocate that fold inward, by an amount the congestion elasticity q governs. The remaining step, closing the loop on the dispersion $\text{Var}(w)$ and the welfare of the entrants themselves, rather than taking the per-attacker profit as the free-entry margin, is the joint attacker-provider welfare problem, which this decomposition isolates but does not solve.

10 Reflexive mark design

The robust mark of Section 6 and the entire prior literature on settlement and oracle manipulation share a hidden assumption: that the manipulability of the reference is a primitive, a number ρ the designer reads off the market and then optimizes against. Section 9 broke that assumption (depth, and hence ρ , responds to how much the mark relies on the reference), but treated the consequence as a correction to the optimal reliance. This section takes the consequence as the organizing principle. When manipulability is a fixed point of the design and the liquidity that supports the reference, mark design is a *reflexive* problem: the instrument changes the constraint it is chosen against. The price-taking analysis is the special case in which that feedback is switched off, and two results live only where it is on, one of which inverts the usual inference from a reference's observed depth.

The local sufficient statistic for the *strength* of the feedback at a given reliance is the *reflexivity gain*

$$\chi(a) := \frac{d \log \rho(a)}{d \log a} = -\frac{2a}{\kappa_0^+(a)} \frac{d\kappa_0^+}{da} \geq 0, \quad (10.1)$$

the elasticity of the reference's manipulability to the mark's reliance on it, with $\rho(a) = \text{Var}(w)/(\kappa_0^+(a)^2 \sigma^2)$ the manipulability at the honest depth of Proposition 9.2. The price-taking world is $\chi \equiv 0$; the manipulation trap is the regime in which χ grows without bound, diverging at the fold a_{tip} .

10.1 The reflexive optimum and the conservatism wedge

Proposition 10.1 (Reflexive responsiveness). *With endogenous depth, the welfare-optimal reliance solves the reflexive cubic*

$$2\rho(a)a^3 \left(1 + \frac{1}{4}\chi(a)\right) + Va - C = 0, \quad (10.2)$$

which is the cubic (6.2) of Theorem 6.2 with the marginal cost of reliance scaled up by the factor $1 + \chi(a)/4$. The reflexive optimum a_R is strictly below the price-taking optimum a_{PT} that an exchange obtains by reading depth off the undisturbed book and applying Theorem 6.2; the gap, the conservatism wedge, is increasing in the reflexivity gain and vanishes at $\chi = 0$.

Proof. The objective $L/\sigma^2 = 1 - 2Ca + Va^2 + \rho(a)a^4$ has first-order condition $-2C + 2Va + 4\rho(a)a^3 + \rho'(a)a^4 = 0$. Writing $\rho'(a)a = \chi(a)\rho(a)$ from (10.1) and dividing by two gives (10.2). The price-taking optimum a_{PT} solves (6.2) with ρ fixed at $\rho_0 = \text{Var}(w)/(\kappa_0^+(0)^2 \sigma^2)$, the manipulability at the undisturbed depth; since $\rho(a) \geq \rho_0$ and the reflexive condition carries the additional positive term $\frac{1}{4}\chi(a) \cdot 2\rho(a)a^3$, the left side of (10.2) exceeds that of (6.2) at every $a > 0$, so its root is smaller. The minimizer exists: the honest depth κ_0^+ is continuous on the compact $[0, a_{\text{tip}}]$ and finite and positive at the saddle-node merge, so ρ and hence $L/\sigma^2 = 1 - 2Ca + Va^2 + \rho(a)a^4$ are continuous there and a minimizer exists by Weierstrass; since $dL/da \rightarrow +\infty$ as $a \uparrow a_{\text{tip}}$, the marginal cost $\rho'(a)a^4 \rightarrow +\infty$ as $\chi \rightarrow \infty$, it is interior to $[0, a_{\text{tip}})$ and solves (10.2). Monotonicity in χ is immediate from the implicit function theorem applied to (10.2). $\square$

Proposition 10.1 characterizes the designer's reliance against an endogenous depth, Proposition 9.3 the depth providers supply at a given reliance, and Theorem 3.7 the adversary's best response at a given mark and depth. These are the three layers of the reflexive problem, established separately; we now define their joint object and prove it exists, so that the reflexive optimum a_R is an equilibrium reliance rather than the solution to a designer's program against an assumed depth.

Definition 10.2 (Reflexive equilibrium). A *reflexive equilibrium* is a triple $((a_R, c_R, \tau_R), \kappa_R, \delta_R(\cdot))$ of a committed mark, an equilibrium reference depth, and an adversary strategy such that:

- (i) (*Adversary.*) for every committed (a, c, τ) and every realized (w, v) , $\delta_R(w, v)$ maximizes the adversary's Stage-2 payoff at the prevailing depth, as in Definition 3.6(i);
- (ii) (*Providers.*) given the reliance a_R and the adverse selection $\Lambda(a_R \bar{w}/\kappa)$ that the adversary's best response inflicts, the reference depth is the honest fixed point $\kappa_R = \kappa_0^+(a_R)$, the largest stable root of $\kappa = \Psi(\kappa; a_R)$ (Proposition 9.3);
- (iii) (*Designer.*) (a_R, c_R, τ_R) minimizes the loss $L(a, c, \tau) = \mathbb{E}[(M(b) - v)^2]$ over $\mathcal{A}$, evaluated at the adversary's best response of (i) and at the endogenous depth $\kappa_0^+(a)$ of (ii), so the manipulability the designer faces, $\rho(a) = \text{Var}(w)/(\kappa_0^+(a)^2 \sigma^2)$, is the one its own reliance induces.

The equilibrium is the fixed point of the three layers: the depth at which providers rest is the depth the designer's reliance induces, and the manipulability the designer optimizes against is the manipulability that depth produces.

Theorem 10.3 (Existence of the reflexive equilibrium). *Under Assumptions 1–2, with depth supplied by increasing differentiable S, Λ as in Proposition 9.3 and the honest branch non-empty at zero reliance (the deep-reference regime $\Gamma(\cdot; 0) < 1$), a reflexive equilibrium exists. Its reliance is interior, $0 < a_R < a_{\text{tip}}$, and solves the reflexive cubic (10.2); its depth is $\kappa_R = \kappa_0^+(a_R) > 0$; its adversary strategy is the best response $\delta_R(\cdot)$ of Theorem 3.7 at depth κ_R ; and the intercept and window are $c_R = -\bar{w}a_R^2/\kappa_R$ and the interior τ_R of (7.2). The equilibrium is subgame-perfect: at every committed reliance the continuation, provider depth and adversary play, is itself an equilibrium.*

Proof. The three layers are established separately and then assembled as a consistent fixed point.

Stage 2 (adversary), at fixed reliance and depth. For any committed (a, c, τ) and any depth $\kappa > 0$ the adversary's Stage-2 payoff is strictly concave in δ (directional) or strictly concave on the feasible region $\mu a^2 < 1$ (saboteur), so the best response $\delta_R(w, v)$ exists, is unique, jointly measurable in (w, v) , and square-integrable; this is Theorem 3.7(i), applied at $\kappa = \kappa_R$.

Depth layer (providers), as a function of reliance. Fix reliance a . Providers face adverse selection $\Lambda(a\bar{w}/\kappa)$ from the aggregate of the Stage-2 best responses, $\delta^*(w) = w a / \kappa$ aggregated over the mean type $\bar{w}$, and supply depth solving $\kappa = \Psi(\kappa; a) = S(s - \Lambda(a\bar{w}/\kappa))$. By Proposition 9.3, in the deep-reference regime there is a largest root $\kappa_0^+(a)$, stable under the tâtonnement, which is the honest equilibrium. It is continuous and strictly positive on $[0, a_{\text{tip}}]$: on $[0, a_{\text{tip}})$ the honest root is stable, where $\Gamma = \partial_\kappa \Psi < 1$, so $\partial_\kappa(\kappa - \Psi) = 1 - \Gamma > 0$ and the implicit function theorem makes κ_0^+ a C^1 function of a ; as $a \uparrow a_{\text{tip}}$ the honest and unstable roots merge at the finite saddle-node value $\kappa_c > 0$, to which κ_0^+ extends continuously. Where $\Gamma > 1$ admits several roots the selection of the largest is unambiguous on $[0, a_M)$, the unique stochastically stable equilibrium there (Theorem 10.14), and $a_R < a_M$ below.

Designer layer, against the induced depth. The reduced loss $L(a) = \sigma^2(1 - 2Ca + Va^2 + \rho(a)a^4)$ with $\rho(a) = \text{Var}(w)/(\kappa_0^+(a)^2 \sigma^2)$ is, by the continuity and positivity of κ_0^+ on $[0, a_{\text{tip}}]$, continuous there; the intercept and window are concentrated out exactly as in Theorem 3.7(iii), giving $c^*(a) = -\bar{w}a^2/\kappa_0^+(a)$ and an interior τ^* under (7.2). By Proposition 10.1, L attains an interior minimizer $a_R \in (0, a_{\text{tip}})$ solving (10.2), Weierstrass on the compact $[0, a_{\text{tip}}]$, with $dL/da \rightarrow +\infty$ as $a \uparrow a_{\text{tip}}$ excluding the right endpoint and $L'(0) = -2C < 0$ excluding the left.

Assembly and consistency. Set a_R from the designer layer, $\kappa_R = \kappa_0^+(a_R)$ from the depth layer, and $\delta_R(\cdot) = \delta_R(\cdot; a_R, \kappa_R)$ from Stage 2. The triple satisfies Definition 10.2: (i) holds by Stage 2 at

depth κ_R ; (ii) holds because κ_R is by construction the honest fixed point at reliance a_R with adverse selection evaluated at the Stage-2 best response of (i); and (iii) holds because a_R minimizes L with ρ taken along the very map $a \mapsto \kappa_0^+(a)$ that providers realize, so the manipulability the designer optimized against, $\rho(a_R) = \text{Var}(w)/(\kappa_R^2 \sigma^2)$, is the one κ_R produces. No layer would revise given the others: the fixed point is consistent. Subgame perfection follows because for every committed a the depth $\kappa_0^+(a)$ and the adversary best response constitute the continuation equilibrium of (i)–(ii), against which a_R is the designer's optimum, reliance optimized against the book its own reliance produces, not against the undisturbed book, which is the content the price-taking program (6.2) omits. $\square$

The direction of this wedge was noted in Section 9; (10.2) is its general form, and χ is what an exchange would need to estimate to size it. The wedge, however, is only a quantitative correction. The qualitative result of this section is stronger: the price-taking optimum can be not merely too aggressive but *self-defeating*.

10.2 When the optimal mark destroys its own reference

Theorem 10.4 (Over-trust catastrophe). *Let reference depth be supplied by any increasing C^1 supply S and adverse selection Λ whose loop gain Γ exceeds one over a range of depths, so the depth map $\kappa_0^+(\cdot)$ has a saddle-node fold at some $a_{\text{tip}} < \infty$ (Proposition 9.3). Let $\rho_0 = \text{Var}(w)/(\kappa_0^+(0)^2 \sigma^2)$ be the manipulability at the undisturbed reference and a_{PT} the price-taking optimum of Theorem 6.2 computed at that depth. Then:*

- (i) *the reflexivity gain diverges at the fold, $\chi(a) \rightarrow \infty$ as $a \uparrow a_{\text{tip}}$, so the reflexive optimum is strictly interior, $a_R < a_{\text{tip}}$;*
- (ii) *the price-taking optimum overshoots the fold, $a_{\text{tip}} < a_{PT}$, if and only if*

$$\rho_0 < \bar{\rho}(a_{\text{tip}}) := \frac{C - V a_{\text{tip}}}{2 a_{\text{tip}}^3}, \quad (10.3)$$

a single inequality between the undisturbed manipulability and the fold location that does not depend on the supply primitives S, Λ ;

- (iii) *on the open set where (10.3) holds, $a_R < a_{\text{tip}} < a_{PT}$: under the price-taking mark realized depth passes the fold and the reference collapses to the manipulated floor $\underline{\kappa}$ (Remark 9.4), where the committed mark (still listening with reliance a_{PT} , since the designer does not revise it) realizes the floor loss $L_{\text{floor}}(a_{PT}) := L(a_{PT}; \rho_{\text{floor}})$ with $\rho_{\text{floor}} = \text{Var}(w)/(\underline{\kappa}^2 \sigma^2) > \rho_0$, while the reflexive mark preserves the honest equilibrium at $L(a_R)$. On any sufficiently thin manipulated floor, the collapse is strictly costlier than the honest branch, and $L_{\text{floor}}(a_{PT}) \geq \sigma^2$ once a mark that still listens imports more manipulation than marking to the anchor would. The price-taking collapse is therefore at least as costly as, and on a genuinely thin floor strictly costlier than, the σ^2 loss of an uninformative mark.*

The catastrophe is therefore a property of reflexive references as such (every microfoundation with a fold admits it), and it is triggered precisely when a reference that looks deep at rest (ρ_0 small) is fragile enough to fold early (a_{tip} small). The price-taking mark is optimal against the book it measures and destroys the book it needs.

Proof. (i) On the honest branch κ_0^+ solves $\kappa = \Psi(\kappa; a)$ with $\partial_\kappa \Psi = \Gamma < 1$, so implicit differentiation gives $d\kappa_0^+/da = \partial_a \Psi/(1-\Gamma)$. At the saddle-node $\Gamma \rightarrow 1^-$ while $\partial_a \Psi = -S' \Lambda' \bar{w}/\kappa < 0$ stays bounded

away from zero (since $S', \Lambda' > 0$), so $d\kappa_0^+/da \rightarrow -\infty$ and $\chi(a) = -2a(d\kappa_0^+/da)/\kappa_0^+ \rightarrow +\infty$. The reflexive marginal cost of reliance $4\rho(a)a^3 + \rho'(a)a^4$, with $\rho'(a) = \chi(a)\rho(a)/a$, therefore diverges as $a \uparrow a_{\text{tip}}$, while the marginal benefit $2C - 2Va$ stays bounded; hence $dL/da \rightarrow +\infty$ near the fold and the minimizer a_R lies strictly below a_{tip} .

(ii) The price-taking optimum is the unique positive root of the strictly increasing cubic $f(a) = 2\rho_0 a^3 + Va - C$ (Theorem 6.2). Since f is increasing, $a_{PT} > a_{\text{tip}} \iff f(a_{\text{tip}}) < 0 \iff 2\rho_0 a_{\text{tip}}^3 + Va_{\text{tip}} - C < 0 \iff \rho_0 < (C - Va_{\text{tip}})/(2a_{\text{tip}}^3)$. The threshold depends only on (C, V, a_{tip}) ; the supply primitives enter solely through the location of a_{tip} .

(iii) When (10.3) holds, the design a_{PT} sets reliance above a_{tip} , where by Proposition 9.3 the honest equilibrium has ceased to exist and depth collapses to the manipulated floor $\underline{\kappa}$ (Remark 9.4). A designer committed to a_{PT} does not revise it, so the realized loss is $L_{\text{floor}}(a_{PT}) = L(a_{PT}; \rho_{\text{floor}})$ with $\rho_{\text{floor}} = \text{Var}(w)/(\underline{\kappa}^2 \sigma^2)$; since $\underline{\kappa}$ is below the honest depth, $\rho_{\text{floor}} > \rho_0$ and $L(a; \rho_{\text{floor}}) > L(a; \rho_0)$ at every a . The design $a_R < a_{\text{tip}}$ preserves the honest branch at $L(a_R) = L(a_R; \rho_0)$. The collapse is costlier: $L_{\text{floor}}(a_{PT}) \geq \min_a L(a; \rho_{\text{floor}})$, and on a thin floor (ρ_{floor} large) the committed term $\rho_{\text{floor}} a_{PT}^4$ pushes $L_{\text{floor}}(a_{PT})$ above both σ^2 and $L(a_R)$. Marking to the anchor ($a = 0$) would cap the loss at $\sigma^2 = L(0)$, but that is a post-collapse revision the committed designer forgoes; σ^2 is therefore the re-optimized collapse loss and a lower bound on the committed one whenever listening to the floor is worse than ignoring it. The condition (10.3) is a strict inequality between continuous functions of the primitives, so the set on which it holds is open; it is non-empty by the instance of Figure 6. The numerical checks confirm (10.3) predicts the overshoot exactly under linear, convex, and concave-supply microfoundations alike. $\square$

Corollary 10.5 (A testable safety condition). *The over-trust condition (10.3) is estimable. The undisturbed manipulability ρ_0 is identified from the depth of the reference and the attack incentive (Section 12.1), and the fold a_{tip} is identified from the loop gain Γ , the elasticity of equilibrium depth to reliance (Propositions 9.3 and 12.2, which separate the form-free proximity to the fold from the form-dependent location of it). An exchange can therefore test, before committing to a price-taking mark, whether its reference lies in the over-trust region $\rho_0 < \bar{\rho}(a_{\text{tip}})$, and the reflexive margin $a_{\text{tip}} - a_R$ measures the buffer it must hold.*

Corollary 10.6 (The appearance trap). *The price-taking optimum a_{PT} is strictly decreasing in the measured manipulability ρ_0 (differentiate $2\rho_0 a^3 + Va - C = 0$). Hence any intervention that lowers measured manipulability without moving the fold (equivalently, that adds depth the exchange counts toward κ_0 but that does not raise the effective granularity or the robust supply that sets a_{tip}) moves the mark to rely more, toward and, once $\rho_0 < \bar{\rho}(a_{\text{tip}})$, past the fold. A reference made to look safer is for that reason trusted more, and if the added depth is fragile the added trust collapses it. Apparent depth that the robustness of the reference does not back is worse than none: the mark, calibrated to the appearance, leans on it past what the reality bears.*

Proof. $da_{PT}/d\rho_0 = -2a_{PT}^3/(6\rho_0 a_{PT}^2 + V) < 0$, so lowering ρ_0 raises a_{PT} ; by Theorem 10.4(ii) the catastrophe obtains once $a_{PT} > a_{\text{tip}}$, i.e. $\rho_0 < \bar{\rho}(a_{\text{tip}})$. “Fragile” depth is depth that enters the measured κ_0 (lowering ρ_0) but not the robust supply S or the effective granularity n_{eff} that locate a_{tip} (Proposition 9.3) and govern the run hazard (Proposition 10.11), so it lowers ρ_0 at fixed a_{tip} and shortens the expected time to a run. $\square$

Theorem 10.4 is unavailable to the price-taking framework by construction: with ρ a primitive there is no fold to cross, and the optimal mark is always safe. The catastrophe appears only once the reference’s depth is a fixed point of the reliance placed on it, and it has a concrete policy reading. An exchange that calibrates manipulability from its order book in normal times (the natural and

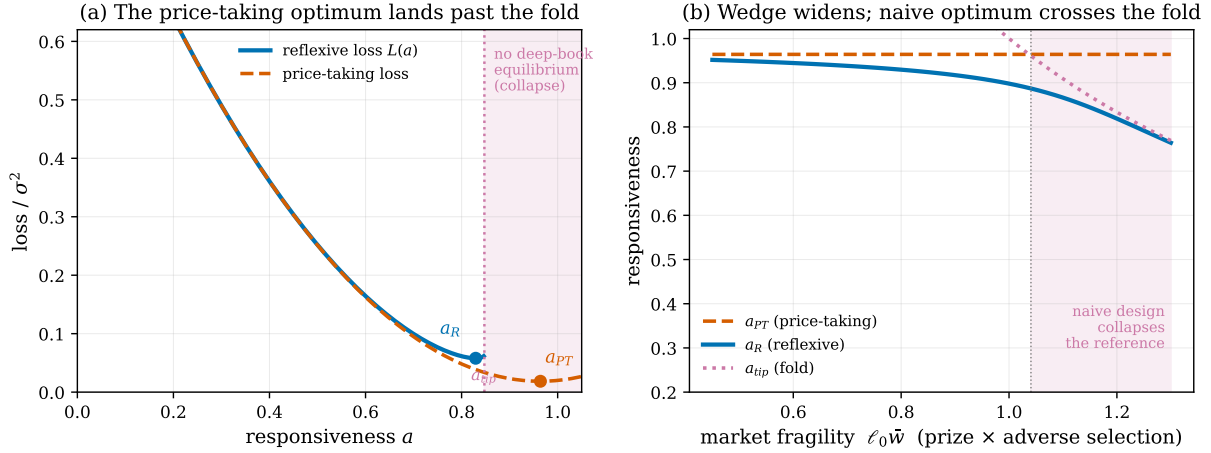


Figure 6: Reflexive mark design (Section 10). (a) For a fragile reference market, the reflexive loss $L(a)$ (solid) turns up before the fold a_{tip} and is minimized at a_R , safely inside the deep-book regime; the price-taking loss (dashed), computed at the undisturbed depth, is minimized at a_{PT} , which lies in the shaded region beyond the fold where no honest-depth equilibrium exists. The price-taking optimum is a design that destroys its own reference. (b) As the reference market grows more fragile (larger prize $\times$ adverse selection $\ell_0 \bar{w}$), the fold a_{tip} and the reflexive optimum a_R fall while the price-taking optimum a_{PT} , blind to the feedback, does not; past the crossing the naive design sits beyond the fold and collapses the reference.

the standard procedure), and then sets the mark by the robust rule of Theorem 6.2 will, on a fragile underlying, choose a reliance that looks optimal and safe on the measured book yet pushes the realized book past its tipping point. The deep, active spot market the regulator required is not a fixed endowment but a function of the very mark that leans on it, and a mark optimized against the endowment can spend it. The admissibility threshold of Section 8 must therefore be read reflexively: the depth that makes a fully responsive mark safe must be the depth that survives that responsiveness, not the depth observed before it.

10.3 Dynamic fragility: the time to a run

The catastrophe of Theorem 10.4 is static: beyond the fold the honest equilibrium is absent, below it the equilibrium is safe. But depth is not a deterministic equilibrium; it fluctuates as liquidity providers enter and exit, and a reference set safely below the fold can still be driven across the basin boundary by a run of adverse shocks. The static threshold then becomes a *hazard*, and the distance to the fold becomes a continuous fragility measure rather than a cliff. We model depth as a noisy tâtonnement around the honest branch, derived from the providers' entry and exit in Section 10.4,

$$d\kappa_t = \mu(\kappa_t; a) dt + \sigma_\kappa dW_t, \quad \mu(\kappa; a) = \Psi(\kappa; a) - \kappa, \quad (10.4)$$

whose drift μ vanishes at the honest depth κ_0^+ (stable, $\mu' < 0$) and at the unstable depth κ_0^- (the basin boundary, $\mu' > 0$), with σ_κ the volatility of liquidity supply. A *reference run* is the first passage of κ_t below κ_0^- , after which the drift carries depth to the manipulated floor. The object is economic rather than a borrowed escape problem: it is the benchmark-design analogue of a bank

run (Diamond and Dybvig, 1983; Goldstein and Pauzner, 2005) and of a market-liquidity crash (Brunnermeier and Pedersen, 2009; Cespa and Foucault, 2014), with the liquidity providers behind the reference in the role of depositors or dealers who withdraw when others do; the saddle-node and first-passage apparatus below is the formal language for that economic event, and the fragility it quantifies is the controllable, reference-price instance of the amplification surveyed by Goldstein et al. (2025).

Theorem 10.7 (Universal fragility law). *Let $U(\kappa) = -\int^\kappa \mu(\kappa'; a) d\kappa'$ be the potential of (10.4), with barrier $\Delta U(a) = U(\kappa_0^-) - U(\kappa_0^+)$. As $a \uparrow a_{\text{tip}}$ the barrier closes at the saddle-node rate*

$$\Delta U(a) = \Theta((a_{\text{tip}} - a)^{3/2}), \quad (10.5)$$

an exponent of 3/2 that is universal: it depends on neither the supply primitives S, Λ nor the parameters, only on the saddle-node geometry of the fold. The mean time to a run obeys the Kramers law

$$\mathbb{E}[T_{\text{run}}] \asymp \frac{2\pi}{\sqrt{U''(\kappa_0^+) |U''(\kappa_0^-)|}} \exp\left(\frac{2\Delta U(a)}{\sigma_\kappa^2}\right), \quad \log \mathbb{E}[T_{\text{run}}] \sim \frac{2\Delta U(a)}{\sigma_\kappa^2} \propto \frac{(a_{\text{tip}} - a)^{3/2}}{\sigma_\kappa^2}. \quad (10.6)$$

The static catastrophe is the $\Delta U = 0$ limit at $a = a_{\text{tip}}$, where the expected life is $O(1)$ (set by the closing exponent; the harmonic prefactor above is the deep-barrier form, which diverges as $a \uparrow a_{\text{tip}}$ and is superseded at the fold by the non-harmonic scaling of Remark 10.12); below the fold the run is exponentially rare in the distance to it, and the hazard rate $\mathbb{E}[T_{\text{run}}]^{-1}$ replaces the binary threshold with a smooth measure of fragility.

Proof. At the fold $a = a_{\text{tip}}$ the drift $\mu(\kappa; a) = \gamma(\Psi(\kappa; a) - \kappa)$ has a double zero at the merge point κ_c , where the honest and unstable equilibria coincide, so $\mu(\kappa_c; a_{\text{tip}}) = 0$ and $\partial_\kappa \mu(\kappa_c; a_{\text{tip}}) = 0$. Taylor-expanding in $(\kappa - \kappa_c, a_{\text{tip}} - a)$ and keeping leading terms, the vanishing $\partial_\kappa \mu$ removing the linear-in- κ term, with $\partial_a \mu > 0$ and $\partial_\kappa^2 \mu < 0$ at the fold (the nondegeneracy of a generic saddle-node, which holds here since $\partial_a \Psi < 0$ and Ψ is strictly concave in κ near κ_c where Γ crosses one), gives the saddle-node normal form $\mu(\kappa; a) \approx c(a_{\text{tip}} - a) - b(\kappa - \kappa_c)^2$ with $b, c > 0$, so the branches separate as $\kappa_0^\pm - \kappa_c = \pm \sqrt{c(a_{\text{tip}} - a)/b}$, the generic $\sqrt{\cdot}$ opening of a saddle-node. The potential is $U(y) = -c(a_{\text{tip}} - a)y + \frac{b}{3}y^3$ in $y = \kappa - \kappa_c$, and evaluating U at the two branch points gives $\Delta U = \frac{4}{3}c^{3/2}b^{-1/2}(a_{\text{tip}} - a)^{3/2}$, which is (10.5); the exponent 3/2 comes from the cubic potential of the normal form and is therefore independent of the microfoundation. With $U''(y) = 2by$, at the branch points $U'' = \pm 2\sqrt{bc(a_{\text{tip}} - a)}$. The mean first-passage time from the honest well over the saddle solves the backward equation $\frac{1}{2}\sigma_\kappa^2 T'' - U'(\kappa)T' = -1$, whose stationary-flux solution is the double integral $\mathbb{E}[T_{\text{run}}] = \frac{2}{\sigma_\kappa^2} \int e^{2U(y)/\sigma_\kappa^2} \left(\int^y e^{-2U(z)/\sigma_\kappa^2} dz \right) dy$. For small noise σ_κ , Laplace's method localizes the outer integrand at the barrier top and the inner at the well bottom, each contributing a Gaussian factor set by the curvature $|U''| = 2\sqrt{bc(a_{\text{tip}} - a)}$, giving $\mathbb{E}[T_{\text{run}}] \asymp \frac{\pi}{\sigma_\kappa^2 \sqrt{bc(a_{\text{tip}} - a)}} \exp(2\Delta U/\sigma_\kappa^2)$, the Kramers law (10.6) with the barrier (10.5). Direct simulation of (10.4) confirms the scaling: the measured barrier grows as $(a_{\text{tip}} - a)^{1.53}$ and $\log \mathbb{E}[T_{\text{run}}]$ is linear in $\Delta U/\sigma_\kappa^2$ with the Kramers slope, across the range of Figure 7. $\square$

Proposition 10.8 (The dynamic optimum and the value of a buffer). *Suppose the reference, once run, stays at the manipulated floor $\underline{\kappa}$, where the committed mark realizes the floor loss $L_{\text{floor}}(a) = L(a; \rho_{\text{floor}})$ with $\rho_{\text{floor}} = \text{Var}(w)/(\underline{\kappa}^2 \sigma^2) > \rho_0$ (Remark 9.4, not the re-optimized σ^2 , since the designer does not revise a post-collapse), and the designer discounts at rate r . The dynamic objective*

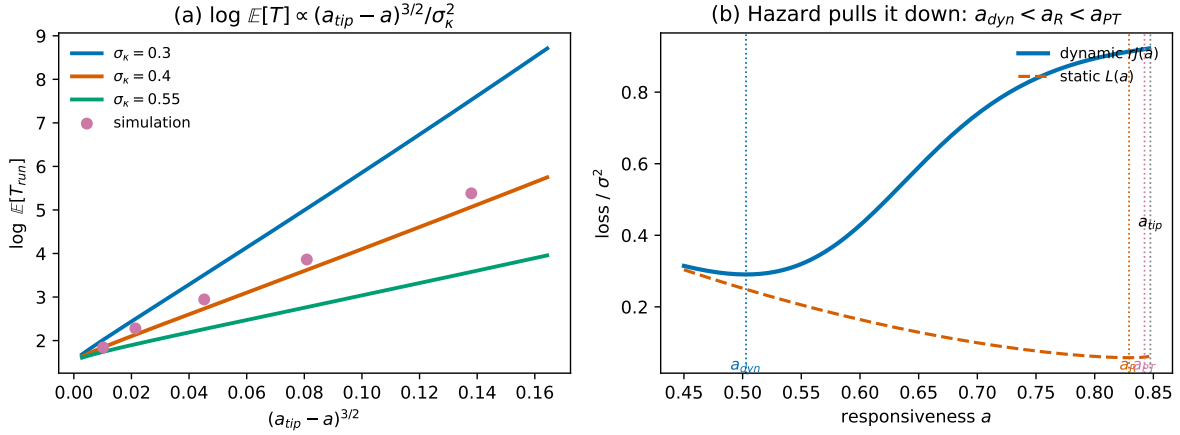


Figure 7: Dynamic reflexive fragility (Section 10.3). (a) The mean time to a reference run: $\log \mathbb{E}[T_{\text{run}}]$ is linear in $(a_{\text{tip}} - a)^{3/2}$ at every noise level (Kramers law (10.6)), with simulated first-passage times (points) on the predicted line; the slope steepens as liquidity-supply volatility σ_κ falls, so a reference held a little further from its fold survives exponentially longer. (b) The collapse hazard pulls the optimum down: the dynamic objective $rJ(a)$ is minimized at a_{dyn} , strictly below the static reflexive optimum a_R , which is below the price-taking optimum a_{PT} (here past the fold).

is $rJ(a) = [rL(a) + L_{\text{floor}}(a)h(a)]/(r + h(a))$, the discounted average of the honest-regime loss $L(a) = L(a; \rho_0)$ and the collapsed loss $L_{\text{floor}}(a)$ weighted by the discount and the hazard $h(a) = \mathbb{E}[T_{\text{run}}(a)]^{-1}$. Its minimizer a_{dyn} is strictly below the static reflexive optimum a_R of Proposition 10.1, so

$$a_{\text{dyn}} < a_R < a_{PT}.$$

The buffer $a_R - a_{\text{dyn}}$ also widens with the liquidity-supply volatility σ_κ , and near a_R a marginal reduction in reliance costs only second-order welfare (since $L'(a_R) = 0$) while lengthening the expected life at first order in the exponent: a small buffer buys an exponentially longer-lived reference.

Proof. L is minimized at a_R with $L'(a_R) = 0$, while $h(a)$ is strictly increasing in a (the barrier (10.5) shrinks and the hazard grows as $a \uparrow a_{\text{tip}}$). Differentiating rJ and evaluating at a_R , the L' term vanishes and the surviving terms are proportional to $h'(a_R)(L_{\text{floor}}(a_R) - L(a_R)) + h(a_R)L'_{\text{floor}}(a_R)$. The first factor is positive because $L_{\text{floor}}(a) - L(a) = (\rho_{\text{floor}} - \rho_0)a^4 > 0$, and $L'_{\text{floor}}(a_R) \geq 0$ over the relevant range (the floor loss rises in reliance once past its own small minimizer), so rJ is strictly increasing at a_R and its minimizer lies to the left, $a_{\text{dyn}} < a_R$; the price-taking ordering $a_R < a_{PT}$ is Proposition 10.1. The correction relative to the σ^2 accounting is signed: $L_{\text{floor}}(a) > L(a)$ for all a , so the committed collapse is strictly worse than the honest branch and the hazard term cannot vanish. The buffer comparative static in σ_κ follows from $h(a) = \Theta(\exp(-2\Delta U/\sigma_\kappa^2))$ increasing in σ_κ at fixed a , which raises the marginal hazard cost of reliance and pushes a_{dyn} down. The exponential-life statement is (10.6): a reduction $a_R \rightarrow a_R - \epsilon$ raises $(a_{\text{tip}} - a)^{3/2}$ by $\Theta((a_{\text{tip}} - a_R)^{1/2}\epsilon)$, hence multiplies $\mathbb{E}[T_{\text{run}}]$ by $\exp(\Theta(\epsilon))$, while the welfare cost is $O(\epsilon^2)$. $\square$

The dynamic theory completes the reflexive picture. The static framework gives a threshold and a discount; the dynamic framework turns the threshold into a fragility law with a universal exponent and shows that the optimal reliance is more conservative still, by a buffer whose value

is exponential and whose size grows with how unstable the reference market's liquidity is. The three reliances $a_{\text{dyn}} < a_R < a_{PT}$ are the same design question answered under three informational stances: price-taking in liquidity, reflexive in the static equilibrium, and reflexive in the dynamic equilibrium with the run hazard priced. Each step down is a refinement the step above cannot see.

10.4 The entry–exit foundation

The dynamics (10.4) were posited. We now derive them from the liquidity providers' problem, which is not a formality: the derivation pins down the noise amplitude σ_κ as an economic object and sharpens the fragility law into a statement about the granularity of liquidity provision.

Depth is supplied by a population of providers. Let there be n_0 potential providers per unit of depth (the *granularity* of liquidity provision, a measure of how many independent balance sheets stand behind the reference) of whom N_t are active, giving depth $\kappa_t = N_t/n_0$. Providers reassess at Poisson rate γ ; an inactive provider activates when the net return to provision $\pi(\kappa; a) = s - \Lambda(a\bar{w}/\kappa)$ is attractive and an active one exits when it is not, with π falling in reliance a and rising in depth κ exactly as in Section 9. The active count is then a birth–death process with entry rate $B(N) = n_0\gamma S(\pi(\kappa; a))$ and exit rate $D(N) = \gamma N$, where S is the supply share of Section 9.

Proposition 10.9 (Microfoundation of the depth dynamics). *As the granularity $n_0 \rightarrow \infty$, the depth process $\kappa_t = N_t/n_0$ converges to the solution of*

$$d\kappa_t = \gamma[\Psi(\kappa_t; a) - \kappa_t] dt + \sigma_\kappa(\kappa_t) dW_t, \quad \Psi = S(\pi), \quad \sigma_\kappa(\kappa)^2 = \frac{\gamma[\Psi(\kappa; a) + \kappa]}{n_0}, \quad (10.7)$$

which is exactly the posited tâtonnement (10.4): the drift is the net-return adjustment of Section 9, vanishing at the honest equilibrium $\Psi(\kappa_0^+) = \kappa_0^+$, and the noise is demographic, with an amplitude the microfoundation fixes, $\sigma_\kappa \propto n_0^{-1/2}$, so the volatility of liquidity supply falls as the square root of the number of independent providers.

Proof. The active count N_t is a density-dependent Markov jump process with up-rate $B(N) = n_0\gamma S(\pi(N/n_0; a))$ and down-rate $D(N) = \gamma N$, each a function of the density $\kappa = N/n_0$. Its generator on smooth f is

$$\mathcal{L}f(\kappa) = B[f(\kappa + \frac{1}{n_0}) - f(\kappa)] + D[f(\kappa - \frac{1}{n_0}) - f(\kappa)].$$

Taylor-expanding the increments in $1/n_0$,

$$\mathcal{L}f(\kappa) = \frac{B - D}{n_0} f'(\kappa) + \frac{B + D}{2n_0^2} f''(\kappa) + O(n_0^{-3}),$$

with scaled drift $(B - D)/n_0 = \gamma(S(\pi) - \kappa) = \gamma(\Psi - \kappa)$ and scaled infinitesimal variance $(B + D)/n_0^2 = \gamma(\Psi + \kappa)/n_0$. This is the generator of the diffusion (10.7); the convergence of $\kappa_t = N_t/n_0$ to its solution is Kurtz's functional law of large numbers and central-limit theorem for density-dependent chains, whose hypotheses hold here because S and π are Lipschitz on the relevant compact set of densities. The drift vanishes at $\Psi(\kappa_0^+) = \kappa_0^+$, the honest equilibrium, and the noise amplitude is $\sigma_\kappa(\kappa)^2 = \gamma(\Psi + \kappa)/n_0 \propto n_0^{-1}$. $\square$

Corollary 10.10 (Fragility is exponential in the granularity of liquidity). *Combining the demographic noise $\sigma_\kappa^2 = \Theta(1/n_0)$ with the fragility law (10.6), the expected time to a reference run is*

$$\mathbb{E}[T_{\text{run}}] \asymp \exp\left(\frac{2\Delta U(a)}{\sigma_\kappa^2}\right) = \exp\left(\Theta(n_0(a_{\text{tip}} - a)^{3/2})\right), \quad (10.8)$$

exponential in the granularity n_0 . A reference supported by more independent providers is exponentially more run-resistant: doubling the granularity of liquidity provision squares the expected life of the reference. The granularity that matters is the effective number of independent providers (the same quantity, net of cross-provider correlation, that governs joint-attack robustness), so a reference drawn from many venues that share liquidity is far more fragile than its nominal breadth suggests.

Proof. Equation (10.8) substitutes $\sigma_\kappa^2 = \Theta(1/n_0)$ from Proposition 10.9 and $\Delta U = \Theta((a_{\text{tip}} - a)^{3/2})$ from Theorem 10.7 into the Kramers law (10.6). Direct simulation of the birth–death chain confirms $\log \mathbb{E}[T_{\text{run}}]$ is linear in n_0 across a range of granularities, with the honest equilibrium and the fold at the values of Section 9. The effective-granularity statement follows because correlated providers reduce the number of independent demographic sources in the system-size expansion, replacing n_0 by its correlation-adjusted analogue, the substitution that turns nominal into effective breadth. $\square$

Proposition 10.11 (Effective granularity). *Let providers have heterogeneous sizes $\{w_i\}$ and reassess with pairwise correlation ρ in their active states. Then the demographic noise scales as $\sigma_\kappa^2 \propto 1/n_{\text{eff}}$ with the effective granularity*

$$n_{\text{eff}} = \frac{(\sum_i w_i)^2}{(1 - \rho) \sum_i w_i^2 + \rho (\sum_i w_i)^2}, \quad (10.9)$$

and the run-time law of Corollary 10.10 holds with n_0 replaced by n_{eff} . Three limits: with equal sizes and independent reassessment $n_{\text{eff}} = n_0$; with heterogeneous sizes and $\rho = 0$, $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2 = 1/\text{HHI}$, the inverse Herfindahl of provider sizes; with equal sizes and correlation $\rho > 0$, $n_{\text{eff}} = n_0 / [1 + (n_0 - 1)\rho] \rightarrow 1/\rho$. Both forces push n_{eff} strictly below the nominal count, so a reference is more fragile than its headline breadth: concentration of provision lowers run-resistance toward that of a handful of providers, and common shocks cap it at $1/\rho$, a ceiling no number of providers can breach.

Proof. Write $\kappa = \sum_i w_i X_i$ with X_i the active indicator, $\text{Var}(X_i) = v$, $\text{Cov}(X_i, X_j) = \rho v$. Then $\text{Var}(\kappa) = v [\sum_i w_i^2 + \rho \sum_{i \neq j} w_i w_j] = v [(1 - \rho) \sum_i w_i^2 + \rho (\sum_i w_i)^2]$. The independent equal-size benchmark with n_0 providers has $\text{Var}(\kappa) = v (\sum_i w_i)^2 / n_0$, so matching variances defines n_{eff} by (10.9); the demographic infinitesimal variance inherits the same factor, giving $\sigma_\kappa^2 \propto 1/n_{\text{eff}}$. Substituting into (10.8) replaces n_0 by n_{eff} . The three limits are computed directly from (10.9). With equal sizes $w_i \equiv w$ and $\rho = 0$: $n_{\text{eff}} = (n_0 w)^2 / (n_0 w^2) = n_0$. With heterogeneous sizes and $\rho = 0$: $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2 = 1/\text{HHI}$. With equal sizes and $\rho > 0$: the denominator is $(1 - \rho)n_0 w^2 + \rho n_0^2 w^2$, so $n_{\text{eff}} = n_0 / [1 + (n_0 - 1)\rho] \rightarrow 1/\rho$ as $n_0 \rightarrow \infty$. And $n_{\text{eff}} \leq n_0$ because $(\sum_i w_i)^2 \leq n_0 \sum_i w_i^2$ by Cauchy–Schwarz bounds the $\rho = 0$ value, while the term $\rho[(\sum_i w_i)^2 - \sum_i w_i^2] \geq 0$ added to the denominator for $\rho > 0$ only lowers it further. Simulation confirms both channels: heterogeneous sizes with squared coefficient of variation one ($n_{\text{eff}} = n_0/2$) reproduce the run time of a homogeneous reference of half the count, and equicorrelated reassessment saturates n_{eff} at $1/\rho$ as the nominal count grows. $\square$

The microfoundation closes the one posited element of the dynamic theory and, in doing so, gives σ_κ an economic reading and the fragility a primitive: the depth of a reference is not a number but a population, and a reference runs when that population thins past its basin boundary. The static catastrophe, the dynamic hazard, its exponential dependence on the granularity of liquidity provision, and the erosion of that granularity by concentration and common shocks are readings of the same fixed point, and together they make the policy lesson quantitative: the safety of a

reference is governed, exponentially, by the *effective* number of independent balance sheets behind it, which concentration and correlated withdrawal can drive far below the headline count.

Remark 10.12 (Scope of the fragility law: noise conventions, genericity, and the small- n_{eff} regime). Three qualifications fix the status of (10.6) and (10.8). *Noise conventions.* Theorem 10.7 reads the run as Kramers escape under additive noise of constant intensity σ_κ (the local diffusion approximation of the demographic birth–death dynamics about the honest well), while the stochastic selection of Theorem 10.14 uses the exact multiplicative quasi-potential. The two agree to leading exponential order: both place $\log \mathbb{E}[T_{\text{run}}]$ at twice the barrier between the wells, and the additive barrier ΔU and the quasi-potential coincide at leading order in the fluctuation size, differing only in the pre-exponential factor (the curvature prefactor of the additive form versus the large-deviation prefactor of the exact rate function). The *exponent* the paper uses is therefore convention-free; only the constant multiplying it is the local-diffusion value. The curvature prefactor is moreover the deep-barrier form: as $a \uparrow a_{\text{tip}}$ the well and barrier curvatures $U''(\kappa_0^\pm)$ vanish like $(a_{\text{tip}} - a)^{1/2}$, so the harmonic prefactor in (10.6) diverges and the formula ceases to hold uniformly up to the fold. The $O(1)$ catastrophic life at $a = a_{\text{tip}}$ is then the saddle-node limit of the vanishing *exponent*, $\log \mathbb{E}[T_{\text{run}}] \rightarrow 0$, not the harmonic prefactor evaluated at the merge; near the fold the pre-exponential factor is set by the non-harmonic (Airy-type) scaling of the coalescing wells. *Genericity.* The $3/2$ exponent is that of a codimension-one saddle-node, the generic fold; the proof verifies non-degeneracy ($\partial_a \mu > 0$ and $\partial_\kappa^2 \mu \neq 0$ at the merge κ_c). A higher-codimension degeneracy (a cusp, where $\partial_\kappa^2 \mu$ also vanishes) would replace $3/2$ by a different rational exponent, so the law is the generic-fold statement and is maintained as such. *The small- n_{eff} regime.* Equation (10.8) is a *scaling* law, asymptotic in large n_{eff} and in $a_{\text{tip}} - a$ bounded away from zero: it states how the log-run-time grows with granularity and proximity, not a numerical rate. Precisely where the economics bites ($n_{\text{eff}} \approx 1/\rho$, a handful of correlated balance sheets), the small-noise asymptotic is loosest and the neglected pre-exponential factors are not negligible, so the law there is qualitative. The comparative statics (more effective breadth and more distance buy exponentially more time; concentration and correlation buy exponentially less) are robust, but the absolute lifetime is not to be read off the leading exponential; a point estimate of the run probability at small n_{eff} requires the birth–death chain itself, which the simulations of Corollary 10.10 and Proposition 10.11 supply. The quantitative claims of this section are the scaling and its signs, not the bare exponential at granularities of order a few.

10.5 Equilibrium selection: the global basin

The multiplicity of Section 9 was resolved by selecting the honest branch on the ground that it is the stable rest point of the tâtonnement. That is a *local* argument, it says only that small perturbations of the honest depth decay. A reference, though, is chosen against large shocks and a noisy provider population, so the selection deserves a global footing: which equilibrium is reached from an arbitrary starting depth, and which one the noisy dynamics inhabit over the long run. Both questions have clean answers, and the second tightens the safety condition.

Proposition 10.13 (Global basin). *For the deterministic depth flow $\dot{\kappa} = \gamma(\Psi(\kappa; a) - \kappa)$ with three equilibria $\underline{\kappa} < \kappa_0^0 < \kappa_0^+$ (floor, unstable middle, honest), the basin of the honest equilibrium is exactly (κ_0^0, ∞) and that of the floor is $[0, \kappa_0^0)$: the unstable middle equilibrium is the global separatrix. Every reference whose depth ever falls below κ_0^0 collapses to the floor; every one that stays above it returns to the honest depth.*

Proof. The flow is one-dimensional and autonomous with $F := \gamma(\Psi - \kappa)$ continuous and vanishing only at the three equilibria. Stability signs (Proposition 9.3) give $F < 0$ on $(\underline{\kappa}, \kappa_0^0)$, $F > 0$ on

(κ_0^0, κ_0^+) , and $F < 0$ on (κ_0^+, ∞) . Hence from any $\kappa > \kappa_0^0$ the trajectory is monotone and converges to κ_0^+ , and from any $\kappa < \kappa_0^0$ it converges to $\underline{\kappa}$; κ_0^0 itself is the unstable separatrix. $\square$

Theorem 10.14 (Stochastic selection and the Maxwell reliance). *Under the demographic dynamics (10.7), the stationary law concentrates, as the granularity $n_0 \rightarrow \infty$, on the global minimum of the quasi-potential*

$$\Phi(\kappa; a) = -2 \int^{\kappa} \frac{\Psi(u; a) - u}{\Psi(u; a) + u} du, \quad (10.10)$$

the deeper well in the noise-weighted metric set by $\sigma_{\kappa}^2 \propto (\Psi + \kappa)/n_0$. There is a Maxwell reliance a_M , solving $\Phi(\kappa_0^+; a_M) = \Phi(\underline{\kappa}; a_M)$, such that the honest equilibrium is stochastically stable for $a < a_M$ and the manipulated floor for $a > a_M$. Because the honest well's barrier vanishes at the fold while the floor's does not, $\Phi(\kappa_0^0) - \Phi(\kappa_0^+) = \Theta((a_{\text{tip}} - a)^{3/2}) \rightarrow 0$ as $a \uparrow a_{\text{tip}}$ (Theorem 10.7), the floor is selected in a left-neighborhood of the spinodal, so $a_M < a_{\text{tip}}$ strictly.

Proof. The scalar diffusion (10.7) has stationary density $p(\kappa) \propto \sigma_{\kappa}^{-2}(\kappa) \exp(2 \int^{\kappa} \mu/\sigma_{\kappa}^2)$ with $\mu = \gamma(\Psi - \kappa)$ and $\sigma_{\kappa}^2 = \gamma(\Psi + \kappa)/n_0$, so $\mu/\sigma_{\kappa}^2 = n_0(\Psi - \kappa)/(\Psi + \kappa)$ and $p \propto \sigma_{\kappa}^{-2} \exp(-n_0\Phi)$ with Φ in (10.10). By Laplace's method the mass concentrates as $n_0 \rightarrow \infty$ at $\arg \min \Phi$, the stable equilibrium with the lower quasi-potential; the barrier separating a well from the saddle is $\frac{1}{2}[\Phi(\kappa_0^0) - \Phi(\text{well})]$, and the Kramers escape time $\asymp \exp(n_0[\Phi(\kappa_0^0) - \Phi(\text{well})])$ recovers Corollary 10.10. At small a the honest well is the deeper (its barrier $\Theta(1)$ while the floor sits just above $\underline{\kappa}$); as $a \uparrow a_{\text{tip}}$ the honest barrier vanishes as $(a_{\text{tip}} - a)^{3/2}$ by the saddle-node normal form of Theorem 10.7 while the floor barrier stays bounded below, so $\Phi(\kappa_0^+) - \Phi(\underline{\kappa})$ changes sign at some $a_M < a_{\text{tip}}$ by continuity, the Maxwell point. $\square$

Corollary 10.15 (The globally safe region is strictly smaller). *Local stability admits the honest reference up to the spinodal a_{tip} ; global selection admits it only up to the Maxwell reliance $a_M < a_{\text{tip}}$. On (a_M, a_{tip}) the honest equilibrium exists and is locally stable but is not stochastically stable: a reference operated there is metastable, certain to run given enough time and tipped by any shock clearing the now-lower barrier. The conservative design therefore targets $a < a_M$, and the over-trust condition of Theorem 10.4 tightens from the spinodal to the Maxwell point accordingly. In a representative calibration with saturating (Michaelis–Menten) supply and a thin-reference floor, solving the equal-quasi-potential condition $\Phi(\kappa_0^+; a_M) = \Phi(\underline{\kappa}; a_M)$ numerically gives $a_M \approx 0.58 a_{\text{tip}}$, so the globally safe band is roughly two-fifths narrower than local stability suggests; the ratio stays within $[0.52, 0.64]$ across nearby supply curvatures and floor depths, robustly and strictly inside the fold.²*

Remark 10.16 (The global-game reading). Depth provision is a coordination game with strategic complementarity: more active providers raise depth, which lowers the manipulator's bias and the adverse selection it inflicts, which raises the net return to provision, so each provider's incentive increases in the mass that provides. This is the regime-change structure of the bank-run and currency-attack literature (Diamond and Dybvig, 1983; Goldstein and Pauzner, 2005), and Section 10.6 develops it into a global-game selection theorem whose threshold coincides, in the vanishing-noise limit, with the deeper-well criterion of Theorem 10.14, demographic noise with a quasi-potential and dispersed information with a global game being two roads to the same selection of the honest branch over a region strictly inside the one local stability allows.

²Net return to provision $R = s - \ell_0 \bar{w} a/\kappa$, supply $S(R) = \kappa_{\max} R_+/(K + R_+)$, and depth floored at $\underline{\kappa}$, so $\Psi(\kappa; a) = \max\{\underline{\kappa}, S(R)\}$. The reported value uses $s = \kappa_{\max} = 1$, $K = 0.25$, $\ell_0 \bar{w} = 0.4$, $\underline{\kappa} = 0.05$, which places the fold at $a_{\text{tip}} \approx 0.95$ and the Maxwell point at $a_M \approx 0.55$. The ratio a_M/a_{tip} is invariant to $\ell_0 \bar{w}$, which only rescales the fold location, and rises with the floor thickness, from 0.52 at $\underline{\kappa} = 0.03$ to 0.64 at $\underline{\kappa} = 0.08$: a thicker floor is a less punishing collapse state, so the honest well keeps its quasi-potential advantage closer to the fold.

The selection thus sits on a global footing. The deterministic basin says the reference recovers from any shock that leaves its depth above the middle equilibrium and collapses from any that does not. The stochastic selection says that, even within the basin, the honest reference is the long-run resting state only below the Maxwell reliance, and on the metastable band above it the reference is locally fine but globally doomed. The practical reading sharpens Section 9: the safe reliance is bounded not by where the honest equilibrium ceases to exist but by where it ceases to be the one the market selects, and the gap between the two is a region of false comfort an exchange reading only local stability would mistake for safety.

Remark 10.17 (Two noise conventions and the range of the law). The fragility results use two noise models, and their relation and range bear stating. Theorem 10.7 forms the barrier ΔU from an additive-noise potential with σ_κ treated as constant; Theorem 10.14 uses the multiplicative-noise quasi-potential Φ with $\sigma_\kappa^2 \propto (\Psi + \kappa)/n_0$ state-dependent. These are the local and global descriptions of one process. Near a well or the saddle the diffusion coefficient is locally constant, so the additive potential delivers the barrier curvature and the 3/2 exponent correctly; the quasi-potential is the exact large-deviation rate function away from those points. The two agree to leading exponential order, both give $\log \mathbb{E}[T_{\text{run}}] = \Theta(n_{\text{eff}}(a_{\text{tip}} - a)^{3/2})$, and differ only in the pre-exponential factor, which the additive approximation does not claim. The 3/2 exponent assumes the fold is a generic, codimension-one saddle-node; a higher-codimension degeneracy would change it, but that is non-generic. Finally, the law is a *scaling* statement, sharpest as $n_{\text{eff}} \rightarrow \infty$. Where the interest is small n_{eff} (a handful of correlated balance sheets), the leading exponential is a qualitative guide rather than a precise rate, the neglected pre-exponential factors being of the same order; the statement to carry there is the comparative one (more granularity and more distance to the fold buy exponentially longer life), not the constant in front.

10.6 A global-game foundation for the selection

The quasi-potential of Theorem 10.14 selects through the dynamics; the same boundary can be reached through strategic uncertainty, which turns the selection from a stability argument into a theorem and places the provision stage squarely in the regime-change tradition of Diamond and Dybvig (1983) and Goldstein and Pauzner (2005). Model the provision stage as a global game (Carlsson and van Damme, 1993; Morris and Shin, 1998). A continuum of providers each chooses to provide or abstain; the mass ℓ that provides yields depth $\kappa(\ell)$, increasing, with a positive floor $\kappa(0) = \underline{\kappa} > 0$ and ceiling $\kappa(1) = \bar{\kappa}$. The fundamental θ is the net return to provision: it rises in the gross return s and falls in the manipulation scale $\bar{w}$. Each provider observes a private signal $x_i = \theta + \nu \xi_i$, the ξ_i i.i.d. with continuous density. The payoff to providing is the gross return net of the adverse selection the manipulator inflicts at the realized depth,

$$u(\theta, \ell) = \theta - \Lambda(a\bar{w}/\kappa(\ell)). \quad (10.11)$$

Theorem 10.18 (Global-game selection). *The provision game (10.11) has (state monotonicity) u strictly increasing in θ ; (strategic complementarity) u strictly increasing in ℓ , with $\partial_\ell u = \Lambda' a\bar{w} \kappa'(\ell)/\kappa(\ell)^2 > 0$; and (limit dominance) thresholds $\underline{\theta} < \bar{\theta}$ with abstention strictly dominant for $\theta < \underline{\theta}$ and provision strictly dominant for $\theta > \bar{\theta}$, the latter finite because $\underline{\kappa} > 0$. Hence as the signal noise $\nu \rightarrow 0$ the game has an essentially unique strategy surviving iterated deletion of dominated strategies, monotone in the signal (Carlsson and van Damme, 1993; Morris and Shin, 1998; Frankel et al., 2003): provide iff $x_i > x^*$, with the fundamental threshold θ^* the unique root of the Laplacian indifference condition*

$$\int_0^1 u(\theta^*, \ell) d\ell = 0, \quad \text{i.e.} \quad \theta^* = \int_0^1 \Lambda(a\bar{w}/\kappa(\ell)) d\ell. \quad (10.12)$$

The deep (honest) reference is selected for $\theta > \theta^*$ and the manipulated floor for $\theta < \theta^*$, with the complete-information multiplicity of Proposition 9.2 occupying exactly the interval the global game resolves.

Proof. The three properties are the displayed signs, using $\Lambda', \kappa' > 0$ throughout and $\kappa(0) = \underline{\kappa} > 0$ for a finite upper-dominance payoff $u(\theta, 0) = \theta - \Lambda(a\bar{w}/\underline{\kappa})$. A binary-action game with a continuum of players, strategic complementarity, state monotonicity, and two-sided dominance regions has, in the vanishing-noise limit, an essentially unique rationalizable strategy, monotone in the signal (Carlsson and van Damme, 1993; Morris and Shin, 1998), the supermodular generalization being Frankel et al. (2003). The threshold is fixed by the indifference of the critical type, who by the Laplacian property of the noise limit holds a belief over the aggregate ℓ uniform on $[0, 1]$, giving (10.12); monotonicity of u in θ makes the root unique. With linear $\Lambda(\delta) = \ell_0\delta$ and $\kappa(\ell) = \underline{\kappa} + (\bar{\kappa} - \underline{\kappa})\ell$, (10.12) integrates to $\theta^* = \ell_0 a\bar{w} \ln(\bar{\kappa}/\underline{\kappa})/(\bar{\kappa} - \underline{\kappa})$. Appendix A.8 reproves the selection from first principles for this game, without the general theorems, and shows that the additive separability of the payoff pins the threshold at θ^* exactly, at every noise level, not only as $\nu \rightarrow 0$. That exactness rests on the improper (flat, non-normalizable) prior, under which the critical type's posterior over the aggregate is uniform at every noise level; it is the clean idealization of the small-noise limit, and with a proper prior the threshold returns to coinciding with θ^* as $\nu \rightarrow 0$ rather than identically. $\square$

Remark 10.19 (Two routes, one selection). The threshold (10.12) is the risk-dominant selection of the provision game, and for the symmetric coordination structure here it coincides, in the respective small-noise limits, with the stochastically stable selection of Theorem 10.14. The global game perturbs common knowledge with private signals; the quasi-potential perturbs the dynamics with demographic noise; both place the boundary where the provision return balances the adverse-selection loss integrated over aggregate states: the Laplacian condition (10.12) and the equal-quasi-potential Maxwell condition being two forms of that balance. What is proven is that each route, in its own small-noise limit, selects a threshold strictly inside the spinodal (the global game in Appendix A.8, the dynamics in Theorem 10.14), and that the two limiting thresholds are the same balance condition for this symmetric structure. The map from the global-game threshold θ^* to the Maxwell reliance a_M runs through that shared balance and is exact in the limits; a finite-noise equivalence of the static and dynamic selections is not claimed, and is left open. The threshold θ^* rises in reliance a and in the manipulation scale $\bar{w}$: a more relied-upon, more contested reference needs better fundamentals to stay honest, and the selected boundary lies strictly inside the spinodal, recovering the conservative safe region of Corollary 10.15. The earlier remark's informal threshold argument is thus discharged, in the small-noise limit, as a theorem, at the cost only of the standard global-game regularity the provision game has been shown to satisfy.

10.7 The framework and its limits

Proposition 10.20 (Nesting). *The reflexive program nests the paper's two regimes. When the loop gain $\Gamma < 1$ everywhere (weak feedback), χ is bounded, the depth map has no fold, the over-trust catastrophe cannot occur, and as $\chi \rightarrow 0$ the reflexive cubic (10.2) collapses to the robust mark of Theorem 6.2: the price-taking baseline and the prior literature. When $\Gamma > 1$ over a range (strong feedback), the fold of Section 9 appears, χ diverges at a_{tip} , and the conservatism wedge and the over-trust catastrophe are the design-side consequences of the same feedback that produces the trap.*

Proof. Immediate from Proposition 9.3: $\Gamma < 1$ gives a unique, fold-free honest equilibrium with κ_0^+ smooth and χ bounded, so (10.2) $\rightarrow$ (6.2) as the feedback vanishes; $\Gamma > 1$ on an interval produces the saddle-node, at which $\chi \rightarrow \infty$, delivering Proposition 10.1 and Theorem 10.4. $\square$

The reframing is the contribution of this section. The robust mark that names the paper is one point, the $\chi = 0$ point, of a design problem in which manipulability is endogenous to the design, and the move from a number ρ to a fixed point $\rho(a)$ is what separates this framework from the price-taking treatments it nests. The reflexivity gain χ is the *local* order parameter for the conservatism wedge: at zero it returns the existing theory, and as it grows it produces the quantitative discount. The qualitative catastrophe is a *global* property, it turns on the whole honest branch $\kappa_0^+(\cdot)$, through a_{tip} relative to a_{PT} (Theorem 10.4), not on χ at a single point, and χ diverging at the fold is the local signature of that global object rather than a substitute for it. With that distinction kept, everything downstream (the liquidation convention, the funding clamp, the multi-venue index, the identification) can be read at either $\chi = 0$ or $\chi > 0$, and the gap between the two readings is the value of taking reflexivity seriously.

11 The liquidation convention and localization

The mark governs *which* positions are liquidated, but the price at which a liquidation is executed and marked is a separate design choice. This section compares two conventions using the cross-section of Section 4.3, now with an attacker who profits from triggering liquidations, and shows that the convention which is more accurate in quiet markets is the one that most spreads a manipulation in attacked ones.

11.1 The cross-section, conventions, and the attacker

Parametrize each marginal position by its *distance to liquidation* $s \geq 0$, the price move required to cross its boundary, with continuous cross-sectional density $h(\cdot)$ and $h(0) = h_0 > 0$ (the boundary density g_0 of Section 4.3). Two settlement conventions are available.

Definition 11.1 (Conventions). Under *mark liquidation*, every position is liquidated and settled at the common mark M ; the idiosyncratic execution error in the common mark has standard deviation σ_M . Under *own-price liquidation*, each position is liquidated and settled at its own execution price; the idiosyncratic error in a position's own price has standard deviation σ_e . The common mark averages idiosyncratic prints, so $\sigma_M \leq \sigma_e$.

An attacker profits $\pi > 0$ from each marginal position it causes to be liquidated, for instance through a position that gains in the ensuing cascade or through an auto-deleveraging claim. The conventions differ in how a manipulation reaches positions.

Definition 11.2 (Manipulation technology under each convention). Under *mark* liquidation, a single bias δ at cost $\frac{\kappa}{2}\delta^2$ moves the common mark by $a\delta$ and so liquidates every position with $s < a\delta$. Under *own-price* liquidation, settlements are idiosyncratic: moving an individual position's own price across its boundary s costs $\frac{\kappa}{2}s^2$ and affects only that position.

This is the formal content of localization: under a common mark, one manipulation is felt by the whole boundary; under own prices, the attacker must pay position by position, with no spillover.

Remark 11.3 (Both conventions inherit the same primitives). The two technologies in Definition 11.2 are the common-reference and idiosyncratic readings of the impact cost of Section 3.4. Under a common mark the attacker pays once, $\frac{\kappa}{2}\delta^2$ with $\kappa = \kappa(\tau)$ of Lemma 3.1, to move the shared reference; under own prices it pays the same impact technology position by position, $\frac{\kappa}{2}s^2$ to push a single execution price across its own boundary s . The boundary density h is the population of Section 4.1 in the distance-to-liquidation coordinate, with $h_0 = g_0$. The noise ordering $\sigma_M \leq \sigma_e$ of

Definition 11.1 is the same averaging at work: a common mark pooled from n idiosyncratic prints has execution-error variance $\sigma_M^2 = \sigma_e^2/n$. The localization result is thus one population and one cost technology read through two settlement rules, not a fresh set of primitives. The own-price technology's self-limiting property (moving one position's execution price affects only that position, at cost $\frac{\kappa}{2}s^2$ each) is a *separability* assumption, and it is the engine of the result. It holds when own-price executions draw on idiosyncratic liquidity: a position-specific auction, or a sufficiently fragmented book, so that an attacker contesting many positions pays the convex cost once per position with no shared displacement. If instead the executions share one book, moving many own-prices enjoys increasing returns from a single displacement, the own-price cost ceases to be separable, and the linear-versus-square-root advantage of Theorem 11.5 erodes toward the common-mark case. The result is therefore a statement about settlement designs that execute positions against their own liquidity, and its force is exactly the degree of that separability; a shared-book own-price rule sits between the idiosyncratic ideal and the common mark, nearer the latter the more the book is shared.

11.2 The attacker's reach under each convention

Lemma 11.4 (Manipulation mass). *Let h_0 be the boundary density. To leading order in the attack size,*

- (a) *under mark liquidation the attacker chooses $\delta^* = \pi h_0 a / \kappa$ and wrongly liquidates mass $m_{\text{mark}} = \pi h_0^2 a^2 / \kappa$, increasing linearly in manipulation cheapness $1/\kappa$ and proportional to the entire boundary density h_0 ;*
- (b) *under own-price liquidation the attacker liquidates exactly the positions with $s < \sqrt{2\pi/\kappa}$, a mass $m_{\text{own}} = h_0 \sqrt{2\pi/\kappa}$, increasing as $\sqrt{1/\kappa}$, with no spillover across positions.*

Proof. (a) Liquidating all positions within $a\delta$ of the boundary yields mass $\int_0^{a\delta} h(s) ds = h_0 a\delta + o(\delta)$, so the attacker solves $\max_\delta \pi h_0 a\delta - \frac{\kappa}{2}\delta^2$, with maximizer $\delta^* = \pi h_0 a / \kappa$ and mass $h_0 a\delta^* = \pi h_0^2 a^2 / \kappa$. (b) Position s is worth attacking iff $\pi \geq \frac{\kappa}{2}s^2$, i.e. $s \leq \sqrt{2\pi/\kappa}$, and each is attacked independently, giving mass $\int_0^{\sqrt{2\pi/\kappa}} h(s) ds = h_0 \sqrt{2\pi/\kappa} + o(1)$. $\square$

The asymmetry in the exponents, linear in $1/\kappa$ under the common mark versus square-root under own prices, is the engine of the result: cheap manipulation hurts a common mark disproportionately because one cheap move reaches the whole density, whereas an own-price attacker must keep paying for each additional victim.

11.3 The welfare comparison

Wrong liquidations are costly at rate $c_L > 0$ per unit mass. Each convention loses mass to manipulation (Lemma 11.4) and to idiosyncratic noise: a position is wrongly liquidated by noise when its own settlement error exceeds its distance to the boundary, a mass $\int_0^\infty h(s) \mathbb{P}(|\text{error}| > s) ds = h_0 \mathbb{E}|\text{error}| + o(1) = h_0 \sigma \sqrt{2/\pi} + o(1)$ for a mean-zero Gaussian error of standard deviation σ . Hence the welfare losses are

$$W_{\text{mark}} = c_L \left[h_0 \sigma_M \sqrt{2/\pi} + \pi h_0^2 a^2 / \kappa \right], \quad (11.1)$$

$$W_{\text{own}} = c_L \left[h_0 \sigma_e \sqrt{2/\pi} + h_0 \sqrt{2\pi/\kappa} \right]. \quad (11.2)$$

Theorem 11.5 (Localization). *Compare the conventions as manipulation severity, measured by $1/\kappa$ (equivalently by the manipulability index $\rho \propto 1/\kappa^2$), varies.*

- (a) Absent manipulation ($1/\kappa \rightarrow 0$), mark liquidation is strictly preferred, $W_{\text{mark}} < W_{\text{own}}$, because the common mark averages away idiosyncratic execution noise ($\sigma_M < \sigma_e$).
- (b) There is a unique severity threshold κ^* (equivalently ρ^*) at which the conventions are equally good. For more robust references ($\kappa > \kappa^*$, mild manipulation) mark liquidation is preferred; for less robust references ($\kappa < \kappa^*$, severe manipulation) own-price liquidation is preferred.
- (c) At the switch, the marginal manipulation externality is equated across conventions: the spillover that a common mark imposes on the whole boundary exactly offsets its noise-averaging advantage.

Proof. Write $y := 1/\kappa \geq 0$ and consider $\Delta(y) := W_{\text{mark}} - W_{\text{own}} = c_L h_0 [A + B y - D \sqrt{y}]$ from (11.1)-(11.2), with

$$A = \sqrt{2/\pi} (\sigma_M - \sigma_e) < 0, \quad B = \pi h_0 a^2 > 0, \quad D = \sqrt{2\pi} > 0.$$

Then $\Delta(0) = c_L h_0 A < 0$, giving (a). The function $g(y) := A + B y - D \sqrt{y}$ has $g'(y) = B - \frac{D}{2\sqrt{y}}$ and $g''(y) = \frac{D}{4} y^{-3/2} > 0$, so g is strictly convex with $g(0) = A < 0$ and $g(y) \rightarrow +\infty$. A strictly convex function that is negative at the left endpoint and diverges to $+\infty$ has exactly one positive root: it decreases to a unique interior minimum (where $g' = 0$, at $y_{\min} = (D/2B)^2$) and then increases, crossing zero once on the increasing branch. Let y^* be that root and $\kappa^* = 1/y^*$; for $y < y^*$ (i.e. $\kappa > \kappa^*$) $\Delta < 0$ and mark liquidation wins, for $y > y^*$ own-price wins, which is (b). Since $\rho \propto 1/\kappa^2$ is strictly increasing in y , the threshold maps to a unique ρ^* . At y^* , $\Delta = 0$ means the manipulation gap $B y^* - D \sqrt{y^*}$ exactly cancels the noise gap A ; the derivative interpretation of (c) is that the common mark's spillover, which grows linearly, has caught up with its fixed averaging advantage. $\square$

Theorem 11.5 is the perpetual-market analogue of the contrast between uniform-price and pay-as-bid allocation. A common mark is a uniform price: efficient when noise dominates, because it pools idiosyncratic errors, but exposed when manipulation dominates, because a single displacement resolves every marginal position at once. Own-price settlement is pay-as-bid: noisier in quiet markets but self-limiting under attack, since the manipulator pays separately for each position it picks off. The empirically dominant convention is the common mark, which Theorem 11.5 rationalizes precisely for underlyings deep enough that $\kappa > \kappa^*$ (Figure 8); the result also delineates when, for a sufficiently manipulable reference, a pay-as-bid liquidation would do less harm.

12 Identifying the fragility condition

The fragility condition of Theorem 10.4, $\rho_0 < \bar{\rho}(a_{\text{tip}})$, is useful only if its two arguments can be measured, and an honest account must separate what a single cross-section of execution-cost and order-flow data delivers from what it does not. The static objects (the value dynamics, the reference's instantaneous depth and resilience, and the *level* of its measured manipulability ρ_0) are point-identified from one cross-section. The objects that locate the *fold* are harder, for a reason intrinsic to the mechanism: a_{tip} is a property of the depth *response* to reliance, and a single cross-section reveals depth at the prevailing reliance, not its slope. Concretely, the loop gain $\Gamma = S' \Lambda' a \bar{w} / \kappa_0^2$ turns on the elasticities of liquidity supply (S') and adverse selection (Λ'), which are identified only from variation in reliance or in providers' net return (a panel of venues, or the time-series response of depth to past returns and dislocations), and the incentive scale $\text{Var}(w)$ is set-identified absent manipulation episodes. The over-trust test is therefore sharp in its level inputs

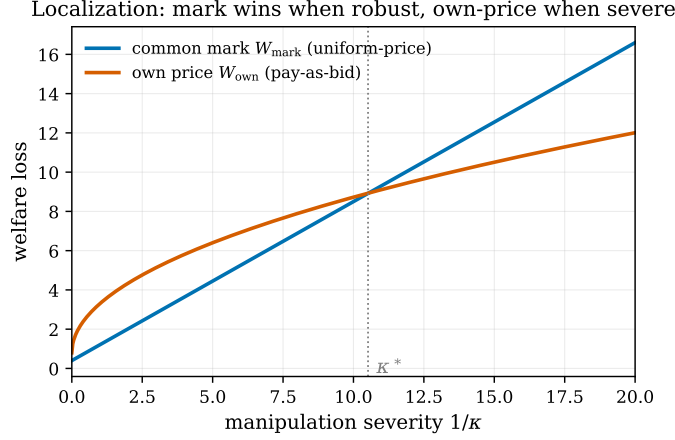


Figure 8: The localization threshold of Theorem 11.5: welfare losses (11.1)–(11.2) under the two conventions as manipulation severity $1/\kappa$ rises. The common mark (uniform-price) is preferred when references are robust, because it pools idiosyncratic execution noise; own-price settlement (pay-as-bid) is preferred under severe manipulation, because the attacker must pay position by position. The curves cross once, at the severity κ^* where the common mark’s spillover catches up with its noise-averaging advantage.

and banded in its fold inputs, and the boundary between the two is exactly the boundary between observing a market and observing its response to being relied upon.

A prior question frames all of this, and the apparatus does not prejudge it: whether a given reference is fold-bearing at all. A reference with weak feedback, $\Gamma < 1$ throughout, has a unique and globally safe equilibrium with no fold (Proposition 9.3); only $\Gamma > 1$ over a range of depths produces the multiplicity, the tipping point, and the entire fragility apparatus. Which regime a particular reference inhabits is thus the first thing the measurement decides, not a maintained hypothesis, and it is decided form-free, by the sign of $\Gamma - 1$ at the operating point, read from how depth responds to a single change in reliance (Proposition 12.2(i)), before any supply curve is fit or any fold is located. The fragility results bind for the references the data place in the $\Gamma > 1$ regime and are silent, by construction, on those they do not; the empirical content of the theory is this sorting, and the claim that any specific real reference is near its fold is an assertion about measured feedback, not a consequence of the model.

12.1 Identification

Proposition 12.1 (Identification). *Suppose the econometrician observes the index path, venue-level order flow and prices, and the cross-section of positions and liquidations. Then:*

- (i) *the value dynamics (ϕ, ς^2) are point-identified from the index autocovariance, $\text{Cov}(v_t, v_{t-h}) = \sigma^2 e^{-\phi h}$ with $\varsigma^2 = 2\phi\sigma^2$;*
- (ii) *the instantaneous robustness $\kappa_0 = 1/\lambda$ is point-identified from the contemporaneous price-impact coefficient λ in $\Delta p_t = \lambda \text{OF}_t + u_t$;*
- (iii) *the resilience (β, τ_0) is point-identified from the impact-decay response $G(\ell) = \kappa_0^{-1}(1 + \ell/\tau_0)^{-\beta}$ fit to the lagged price response to order flow;*

- (iv) the cross-venue cost matrix K , hence the joint-attack robustness $\kappa_{\text{eff}} = \mathbf{1}'K\mathbf{1}$ and the depth level entering the loop gain, are point-identified from the cross-venue impact matrix, the price response of each venue to order flow at each other;
- (v) the supply and adverse-selection elasticities (S', Λ') , hence the loop gain Γ of Proposition 9.3 and the fold a_{tip} , are point-identified given variation in reliance or in providers' net return (the depth response $d\kappa_0/dR$ and the dislocation-sensitivity of provider losses, observed across venues or over time), and are only set-identified from a single cross-section, which fixes the level but not the slope of the supply schedule;
- (vi) the manipulation-incentive dispersion $\text{Var}(w)$ is only set-identified absent manipulation episodes: it is bounded below by the largest sustained reference dislocation times κ , and above by the prize implied by the position mass within reach of a boundary.

The optimal window τ^* and responsiveness a^* are therefore point-identified up to the $\text{Var}(w)$ band; the static admissibility threshold of Corollary 8.2 is point-identified up to that same band; and the distance to the fold $a_{\text{tip}} - a$ is point-identified given panel or time-series variation in reliance (which delivers S', Λ') and set-identified without it. The over-trust test is thus sharp where it reads depth levels and banded where it reads the depth response: the very response that constitutes the trap, so the one quantity hardest to measure is the one the mechanism is about.

Proof. Each claim identifies its parameter from a population moment of observable data. (i) is the OU autocovariance, whose decay rate identifies ϕ and whose level identifies σ^2 , with $\zeta^2 = 2\phi\sigma^2$. (ii) is the Kyle price-impact regression. (iii) is propagator estimation: the lagged response of price to a unit of order flow traces G , whose power-law fit identifies (β, τ_0) and whose intercept identifies κ_0 . (iv) follows because the cross-venue impact matrix is K^{-1} up to scale, so inverting it recovers K and the scalar $\mathbf{1}'K\mathbf{1}$, which is the depth level. (v): the loop gain $\Gamma = S'\Lambda'a\bar{w}/\kappa_0^2$ contains the supply slope $S' = d\kappa_0/dR$ and the adverse-selection slope Λ' ; a single cross-section reports κ_0 at the prevailing net return R but not $d\kappa_0/dR$, so S' (and likewise Λ' , the marginal provider loss per unit dislocation) require either a panel of venues spanning different reliance or net returns, or the time-series regression of depth on lagged net return and of provider P&L on lagged dislocation; with that variation the slopes are ordinary regression coefficients and Γ , hence a_{tip} , is point-identified, while without it only the level is pinned and the fold is bounded. (vi): the directional bias is $\delta^* = wa/\kappa$, so an observed sustained dislocation δ implies $w = \kappa\delta/a$, bounding $\text{Var}(w)$ below by the dispersion of observed dislocations and above by the largest prize the position mass can offer. The policy claims follow because the cubic (6.2) and the window condition (7.4) are functions of (C, V, ρ) with $\rho = \text{Var}(w)/(\kappa^2\sigma^2)$: given $(\phi, \zeta^2, \kappa_0, \beta, \tau_0)$ the pair (C, V) and the κ -dependence of ρ are pinned down, leaving the $\text{Var}(w)$ scale and, for the fold alone, the supply/adverse-selection slopes as the objects that a single cross-section does not deliver. $\square$

12.2 Sharp identification of the fold from reliance variation

The set-identification of the fold in Proposition 12.1 is a property of a *single* cross-section, which fixes depth at the prevailing reliance but not its slope. Variation in reliance removes it. A panel of venues at different mark responsiveness, or a venue that changes its mark methodology over time, traces the honest branch $\kappa_0^+(\cdot)$ itself: its local slope reads the loop gain (and hence proximity to the fold) form-free, while its extrapolation to the vertical tangent locates the fold conditional on a supply specification, without any observation near a collapse.

Proposition 12.2 (Sharp identification of the fold). *Suppose reliance varies, within venue or instrumented, so the honest depth branch $\kappa_0^+(a)$ is traced on a sub-interval of $[0, a_{\text{tip}})$. Then:*

- (i) (Form-free.) *The loop gain is $\Gamma(a) = \eta(a)/(\eta(a) - 1)$, where $\eta(a) = d \ln \kappa_0^+ / d \ln a \in (-\infty, 0)$ is the reliance-elasticity of honest depth; the fold a_{tip} is the reliance at which $\eta \rightarrow -\infty$, equivalently $\Gamma \rightarrow 1$ and the branch turns vertical. The loop gain is thus read directly off the branch slope, with no knowledge of the supply or adverse-selection forms.*
- (ii) (Linear model.) *The branch is $\kappa_0^+(a) = \frac{1}{2}[\theta_1 + \sqrt{\theta_1^2 - 4\theta_2 a}]$ with $\theta_1 = s/c_s$ and $\theta_2 = \ell_0 \bar{w}/c_s$, and $a_{\text{tip}} = \theta_1^2/(4\theta_2)$. The pair (θ_1, θ_2) is point-identified by least squares of measured depth on reliance, and the fold depends only on these two ratios: neither the supply scale c_s nor the incentive scale $\bar{w}$ need be identified separately.*
- (iii) (Extrapolation is form-dependent.) *Under the linear branch of (ii), $d\kappa_0^+/da \rightarrow -\infty$ as $a \uparrow a_{\text{tip}}$ in a known square-root shape, so its parameters, hence the vertical tangent, are pinned by the branch on any sub-interval and the fold is recovered by extrapolation from the safe region. This step, unlike (i), is form-dependent: Proposition 9.3 allows any increasing S and Λ , under which the branch need not be the square root, and extrapolating a fitted parametric shape to its vertical tangent can be biased under supply misspecification. The robust content therefore splits. Proximity to the fold is form-free: the loop gain $\Gamma(a) = \eta(a)/(\eta(a) - 1)$ of (i) is read off the local branch slope with no functional-form knowledge. Locating the fold by extrapolation inherits the parametric supply form of (ii); Figure 9 demonstrates the recovery in-model, fitting the family the simulation generates, and so speaks to estimation efficiency rather than to misspecification. In practice one reports the form-free Γ at the operating point and treats the extrapolated a_{tip} as conditional on a flexible supply specification, with a misspecification band.*

Proof. At the honest equilibrium $\kappa_0 = \Psi(\kappa_0; a)$, implicit differentiation gives $d\kappa_0^+/da = \partial_a \Psi / (1 - \Gamma)$ with $\Gamma = \partial_{\kappa_0} \Psi$ the loop gain of Proposition 9.3. Since $\partial_a \Psi = -S' \Lambda' \bar{w} / \kappa_0$ and $\Gamma = S' \Lambda' a \bar{w} / \kappa_0^2$, one has $\partial_a \Psi = -\Gamma \kappa_0 / a$, so $d\kappa_0^+/da = -\Gamma \kappa_0 / (a(1 - \Gamma))$ and $\eta = (a/\kappa_0) d\kappa_0^+/da = -\Gamma / (1 - \Gamma) = \Gamma / (\Gamma - 1)$, which inverts to $\Gamma = \eta / (\eta - 1)$; as $\Gamma \uparrow 1$, $\eta \rightarrow -\infty$, the vertical tangent. This is (i). For (ii), the honest root of $c_s \kappa_0^2 - s \kappa_0 + \ell_0 \bar{w} a = 0$ is $\kappa_0^+ = \frac{1}{2}[\theta_1 + \sqrt{\theta_1^2 - 4\theta_2 a}]$ after dividing by c_s , with vanishing discriminant at $a_{\text{tip}} = \theta_1^2/(4\theta_2)$; reliance a is a committed design parameter, so regressing measured depth (a noisy outcome) on a identifies (θ_1, θ_2) without the errors-in-variables that a depth regressor would carry, and a_{tip} is their stated function. For (iii), $d\kappa_0^+/da = -\theta_2 / \sqrt{\theta_1^2 - 4\theta_2 a} \rightarrow -\infty$ as $a \rightarrow a_{\text{tip}}$; the branch is real-analytic on $[0, a_{\text{tip}})$, so its values on any sub-interval determine (θ_1, θ_2) and hence the fold, consistently under measurement error in depth. $\square$

The identifying variation is in reliance itself, which is observed: a is the responsiveness of a venue's mark to its reference (the weight the published mark formula places on the reference and the length of its averaging window), and as a committed Stage-0 choice it is predetermined relative to the contemporaneous depth shocks the providers' demographic noise generates. The threat to identification is cross-venue heterogeneity in liquidity-market structure $(s, c_s, \ell_0, \bar{w})$ correlated with the reliance a venue chooses; within-venue variation, a venue's own mark-methodology changes, or reliance instrumented by exogenous redesigns or regulatory mandates removes it, tracing each venue's own branch with its structure held fixed. Under that variation the over-trust gap $a - a_{\text{tip}}$ becomes a point estimate with a delta-method standard error rather than the set of Proposition 12.1: the test is sharp. In simulation the estimator recovers a fold of $a_{\text{tip}} = 0.25$ essentially without bias from data confined to $a \leq 0.2$ (never within a fifth of the fold) with a standard error that falls in the number of venues and the span of reliance, and is robust to ten-percent multiplicative noise in measured depth. The economic content is that one need not witness a reference run to locate the cliff: the cliff announces itself in how steeply depth falls as the mark is made to listen, well inside the region where the reference is still safe.

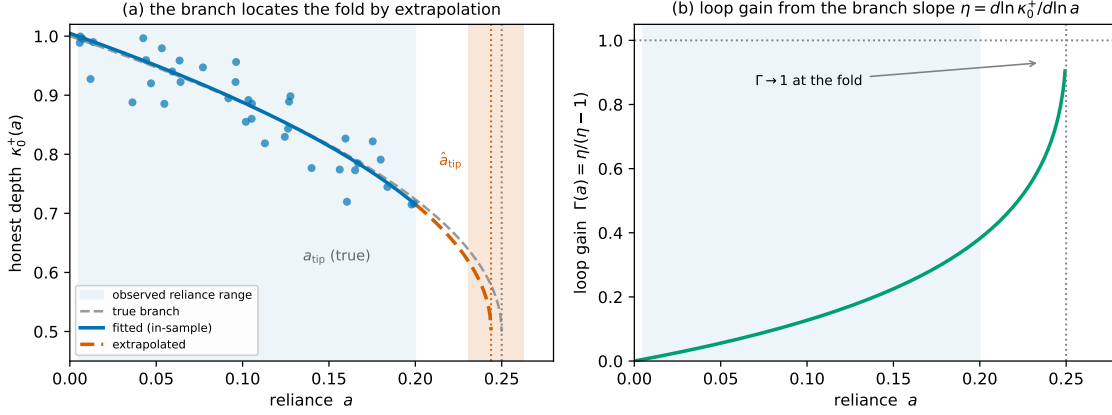


Figure 9: Sharp identification of the fold from reliance variation (Proposition 12.2). (a) A panel of venues observed only in the safe region (shaded, $a \leq 0.8 \hat{a}_{\text{tip}}$) traces the honest depth branch; least squares on the structural form (solid) extrapolates (dashed) to a vertical tangent, recovering the fold $\hat{a}_{\text{tip}}$ within a tight band of the truth without any near-collapse observation. (b) The loop gain $\Gamma(a) = \eta/(\eta - 1)$, read off the branch’s log-slope $\eta = d \ln \kappa_0^+ / d \ln a$, rising to one at the fold.

12.3 Identification under venue heterogeneity

The branch of Proposition 12.2 belongs to a venue, and venues differ. If structure $(s_j, c_{s,j}, \ell_{0,j}, \bar{w}_j)$ varies and (as the reflexive optimum implies, each venue relying below and comoving with its own fold), the chosen reliance is a function of structure, then a single point per venue cannot be pooled into a branch: the cross-venue comparison confounds movement *along* a branch with movement *across* branches. The identification must run within the venue.

Proposition 12.3 (Identification under heterogeneity and selection). *Let each venue j have its own branch $\kappa_{0,j}^+(\cdot)$ and fold $a_{\text{tip},j}$, with its operating reliance a_j a function of its structure. Then:*

- (i) (Pooling is inconsistent.) *The cross-venue locus $\{(a_j, \kappa_{0,j}^+(a_j))\}_j$ is the selection rule composed with the venue-indexed branch, not a structural branch; a common branch fit to it estimates a functional of the selection rule, and its estimand changes when the selection rule changes at fixed structure.*
- (ii) (Within-venue identification.) *Observed at two or more reliance settings of its own mark-methodology history, venue j ’s parameters $(\theta_{1,j}, \theta_{2,j})$ and fold $a_{\text{tip},j} = \theta_{1,j}^2 / (4\theta_{2,j})$ are identified from its own data (the venue is its own control) consistently as its reliance span widens.*
- (iii) (Form-free proximity.) *The local loop gain at the operating point, $\Gamma_j(a_j) = \eta_j / (\eta_j - 1)$ with $\eta_j = d \ln \kappa_{0,j}^+ / d \ln a$ the local within-venue elasticity, is a sufficient statistic for over-trust proximity ($\Gamma_j < 1$ safe, $\Gamma_j \rightarrow 1$ at the fold) invariant to the venue’s structural level and to the supply and adverse-selection forms, and recoverable from a single reliance nudge.*

Proof. (i) Each venue contributes the single point $(a_j, \kappa_{0,j}^+(a_j))$ on its own branch; with a_j a function of $(\theta_{1,j}, \theta_{2,j})$, the map $j \mapsto (a_j, \kappa_{0,j}^+(a_j))$ is the selection rule pushed through the branch, and holding the distribution of structure fixed while changing the selection rule changes this locus, hence any common-branch estimand. (ii) Within a venue $(\theta_{1,j}, \theta_{2,j})$ are constant, so Proposition 12.2

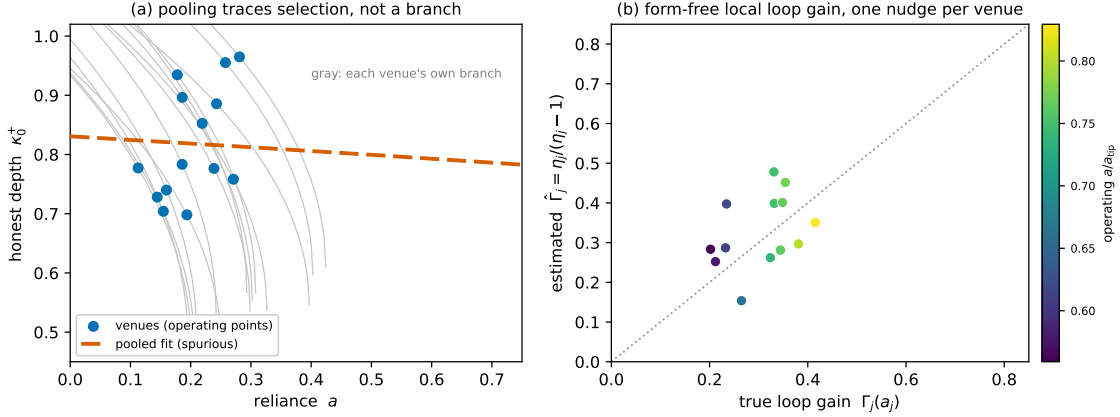


Figure 10: Identification under venue heterogeneity (Proposition 12.3). (a) Heterogeneous venues, each on its own branch (gray), sit at operating points (dots) that a pooled fit (dashed) threads into a spurious near-flat “branch”: the graph of the selection rule, not structure. (b) The form-free local loop gain $\hat{\Gamma}_j = \eta_j/(\eta_j - 1)$, from one within-venue reliance nudge, against the truth; color is the operating $a_j/a_{tip,j}$. The estimator tracks the loop gain and orders venues by proximity to their own fold.

applies to its own time series, consistent by analyticity as the observed interval widens. (iii) By Proposition 12.2(i), $\Gamma_j(a) = \eta_j/(\eta_j - 1)$ within the venue, and Γ is strictly increasing in a with $\Gamma = 1$ at a_{tip} , so the operating-point value orders venues by proximity to their own folds using only the local slope. $\square$

The three readings are confirmed numerically on eighty heterogeneous venues whose chosen reliance is a structure-dependent fraction of their own fold (Figure 10). A single pooled branch fit returns a fold outside the entire range of true folds and shifts when only the selection rule is changed, inconsistent, as (i) predicts. The within-venue estimator recovers each venue’s own fold with a median error that falls below five percent as the methodology history widens, its residual finite-sample bias conservative: it understates the fold, erring toward caution. And the form-free local loop gain orders the venues by proximity to their folds with a rank correlation of one, recovering Γ to within 0.08 from a small methodology nudge under realistic measurement noise. The lesson for the empirical program is a hierarchy of data demands. A venue’s *proximity* to its own cliff is read, model-free and heterogeneity-proof, from a single change in how much its mark listens; *locating* the cliff asks for a wider methodology history; and only the naive cross-venue comparison (which the selection of reliance to structure contaminates) is to be avoided. When within-venue variation is scarce, reliance instrumented by exogenous redesigns or regulatory mandates, orthogonal to contemporaneous structure, restores the consistency of the pooled estimator.

13 Welfare, the social planner, and optimal policy

The loss $L = \mathbb{E}[(M - v)^2]$ minimized throughout is, by Proposition 4.9, the social deadweight a mark error inflicts: the liquidation distortion plus the funding distortion. The designer who minimizes it is a social planner, and the characterizations of Sections 6–10.5 are its constrained-efficient policy.

This section reads them as such: it isolates the externality that makes a decentralized reference inefficient, states the constrained-efficient reliance, and gives the instruments that decentralize it.

13.1 The externality and the inefficiency of laissez-faire

A mark is chosen by an exchange that is small relative to the reference market: it reads the reference's depth off the book and optimizes responsiveness against it, taking that depth as given. Section 9 identified this as the omission the reflexive feedback punishes, reliance thins the depth every user of the reference shares, and the price-taker prices none of it. The gap is an externality with two layers. *Within the venue*, the planner internalizes that its own reliance lowers depth: it optimizes against $\rho(a)$, not the frozen ρ_0 , and so chooses the reflexive optimum a_R of Proposition 10.1 rather than the price-taking a_{PT} . *Across venues*, when several contracts settle on one reference (the empirically dominant case, a few spot composites underlying most perpetuals), each venue's reliance degrades the common depth, a textbook common-pool problem in which the reference is a shared resource every venue over-grazes, and the social cost of a venue's reliance is summed over all the contracts that lean on it.

Proposition 13.1 (Inefficiency of laissez-faire reliance). *Whenever the reflexivity gain is positive, the decentralized price-taking reliance exceeds the constrained-efficient one, $a_{PT} > a_R$, by the conservatism wedge of Proposition 10.1; the gap is strictly increasing in χ , and with J identical contracts sharing the reference it widens further, the marginal external cost being J times the single-venue term. If the price-taking optimum overshoots the fold, $a_{PT} > a_{tip}$, the decentralized reference is not merely inefficient but self-destroying (Theorem 10.4); and when prudence against runs binds, the constrained-efficient reliance is capped below the spinodal at the Maxwell reliance, $a^* = a_R \wedge a_M$ with $a_M < a_{tip}$ (Corollary 10.15).*

Proof. $a_{PT} > a_R$ and monotonicity in χ are Proposition 10.1. With J identical contracts of reliance a on one reference, the common depth responds to the aggregate reliance, and each venue ignoring its effect on the shared depth omits a marginal external cost that, by symmetry, the planner counts J times: the planner's first-order condition carries $J\rho'(a)a^4$ where the private one carries none, so its root falls further below a_{PT} as J grows. The over-trust and Maxwell statements are Theorem 10.4 and Corollary 10.15. $\square$

13.2 Decentralizing the efficient reference

The efficient mark can be implemented either by constraining the instrument or by pricing the externality.

Proposition 13.2 (Decentralization). *The constrained-efficient reliance $a^* = a_R \wedge a_M$ is decentralized by either of two equivalent instruments:*

- (i) *a responsiveness cap $a \leq a^*$, binding the venue directly; or*
- (ii) *a Pigouvian charge on responsiveness whose marginal rate equals the depth externality an atomistic venue omits,*

$$t'(a) = J\rho'(a)a^4 > 0,$$

which raises the private marginal cost of reliance to its social level and makes a depth-taking venue's first-order condition coincide with the planner's.

Two further instruments enlarge the region over which any reliance is safe rather than choosing reliance within it: a window floor $\tau \geq \tau_{\min}$, which raises the manipulation cost $\kappa(\tau)$ and pushes the fold a_{tip} and the Maxwell point a_M outward (Theorem 7.2); and an effective-granularity floor $n_{\text{eff}} \geq \underline{n}$, which, since run-resistance is exponential in n_{eff} (Corollary 10.10, Proposition 10.11), buys exponential safety per independent balance sheet and caps the damage of correlated withdrawal.

Proof. A venue that takes the prevailing equilibrium depth $\rho(a)$ as given minimizes $1 - 2Ca + Va^2 + \rho(a)a^4$ holding ρ fixed, with first-order condition $-2C + 2Va + 4\rho(a)a^3 = 0$; the planner internalizes that depth responds to reliance, adding the term $\rho'(a)a^4$ to give $-2C + 2Va + 4\rho(a)a^3 + \rho'(a)a^4 = 0$. The difference $\rho'(a)a^4$ (the marginal fall in shared depth a venue ignores) is the externality, counted J times when J contracts share the reference (Proposition 13.1); a charge with marginal rate $J\rho'(a)a^4$ reproduces the planner's condition, and the cap $a \leq a^*$ implements the same allocation directly. The naive price-taker a_{PT} that reads depth off the undisturbed book (using ρ_0 in place of $\rho(a)$) additionally mis-measures depth: a separate information error the cap corrects but the charge does not. The window and granularity floors shift $a_{\text{tip}} = s^2/(4c_s\ell_0\bar{w})$ through κ and the run barrier through n_{eff} , both monotone, by Theorem 7.2 and Corollary 10.10. $\square$

The convention and the index are policy levers too. Under the conditions of Theorem 11.5, own-price liquidation welfare-dominates common-mark liquidation by confining each attack to the position it targets, so a planner mandates the localized convention; and the clean-index normalization (Proposition 4.7, Remark 4.8) is its funding-side analogue, an index built broad enough that the same trades cannot move it together with the mark.

13.3 The planner is not the exchange

The foregoing identifies the welfare-optimal design and the policy that decentralizes it among *price-taking* venues. A profit-maximizing exchange is a different agent: it values fee revenue and volume, and to the extent that a livelier, more manipulable reference draws volume it may under-provide robustness relative to even the price-taking benchmark, so the case for the responsiveness cap and the granularity floor is stronger than the externality alone implies. Competition for order flow can sharpen rather than soften this. A venue that loosens its mark to attract flow exports run risk to every contract sharing its reference, a race to the bottom in reference robustness that the common-pool externality already names and that a floor on robustness, a ceiling on responsiveness, a minimum window, a minimum effective breadth, is the natural instrument to arrest. Characterizing the profit-maximizing exchange's chosen mark, and the competitive equilibrium among exchanges, is the positive counterpart this normative section leaves open (Section 16).

13.4 Policy instruments and a supervisory test

The framework speaks to a live regulatory question. The cash-settled perpetual has recently moved onshore under a case-by-case listing review, and its settlement reference is a regulated benchmark; the operative standard is the long-standing one that a contract's reference be reliable and not readily susceptible to manipulation, applied as a static verification of the reference's depth and integrity at listing. The results of this paper say the object so verified is not fixed: depth, and with it manipulability, is a fixed point of the reliance the contract places on the reference (Section 9), so a reference that passes the test at rest can be pushed past its fold by the very mark the test approves (Theorem 10.4). The efficient response is to supervise *reliance* against an estimated fold, continuously, rather than to certify depth once.

Instruments. Proposition 13.2 gives the efficient design as a constrained responsiveness; four implementable instruments deliver it.

- (i) A *responsiveness ceiling*, equivalently a *window floor*, holding reliance below the Maxwell margin a_M of Corollary 10.15, rather than the spinodal a_{tip} , since a reference between the two exists but is not the selected equilibrium. The floor is reference-specific and lengthens with the attack incentive (Theorem 7.2).
- (ii) An *effective-granularity floor* $n_{\text{eff}} \geq \underline{n}$ on the liquidity behind the reference, accounting for concentration and correlated withdrawal, since the expected time to a run is exponential in n_{eff} (Corollary 10.10, Proposition 10.11): an anti-concentration requirement on the reference’s providers, not on the contract.
- (iii) A *liquidation-convention rule* keyed to the severity threshold κ^* of Theorem 11.5: localized own-price settlement for fragile references, common-mark settlement only where the reference is robust enough that noise-pooling dominates spillover.
- (iv) An *index-breadth requirement* (Proposition 4.7, Remark 4.8): a contemporaneous composite wide enough that the trades that move the mark cannot move the index with it.

A supervisory test. The instruments are only as good as the fold they target, and the identification results make the target estimable. The efficient test is the reliance-adjusted admissibility condition

$$\Gamma_j(a_j) < \underline{\Gamma} \quad \text{equivalently} \quad a_j < a_{M,j}, \quad \underline{\Gamma} := \Gamma_j(a_{M,j}) < 1,$$

evaluated reference by reference, where $\Gamma_j(a_j) = \eta_j/(\eta_j - 1)$ is the local loop gain at the venue’s operating reliance and $\eta_j = d \ln \kappa_0^+ / d \ln a$ its reliance-elasticity of depth; the equivalence is the strict monotonicity of Γ in a (with $\Gamma = 1$ at the fold). By Propositions 12.2–12.3 this gauge is identified from within-venue methodology changes, model-free and robust to cross-venue heterogeneity, and it rank-orders references by proximity to their own folds; the naive cross-venue comparison is contaminated by the selection of reliance to structure and must be avoided. The data the test requires are mark-methodology histories (the responsiveness and window each reference has used, and when it changed them) which a benchmark administrator already holds and a supervisor can require. This is the content the case-by-case review lacks and could adopt: not “is the reference deep,” but “how close is this reference to its fold at the reliance the mark places on it, and how fast does its depth fall as that reliance rises.”

Why quantity floors, and at the reference. Three features of the problem select the instrument. The loss is discontinuous at the fold, so a quantity limit dominates a price instrument when the damage is sharply convex near a threshold. The appearance trap (Corollary 10.6) makes disclosed depth a perverse signal, since fragile depth lowers measured manipulability and *induces* more reliance, so disclosure alone can move a reference toward its fold; and the exchange’s private objective (Section 13.3) means a profit-maximizing venue may absorb a charge and compete for flow by under-protecting the mark, against which a binding floor on robustness is the surer instrument. Finally, because a shared reference is a common pool (Proposition 13.1), the floor must be set at the *reference* level and coordinated across every contract that settles on it: the benchmark-administrator level at which manipulation-resistance is already regulated. The contribution of the model to that architecture is the reflexivity: a benchmark’s robustness is not a property to certify but a budget reliance spends, so the supervised quantity is the reliance-adjusted distance to the fold, not the resting depth.

14 Robustness of the modeling choices

The framework rests on parametric choices made for tractability: Gaussian primitives, a quadratic manipulation cost, an affine mark, a linear-quadratic loss, an iso-elastic robustness, a specific depth supply. A theory whose conclusions turned on these would be a theory of its functional forms. They do not. The central results are already form-free where it matters, and the two parametric choices a reader of a cryptocurrency application presses hardest (the Gaussian value and the quadratic cost) leave the conclusions intact.

What is already form-free. The reflexive fold holds for *any* increasing depth supply and *any* increasing adverse selection (Proposition 9.3), so the tipping point, the multiplicity, and the hysteresis are properties of the feedback’s strength rather than of a functional form. The 3/2 saddle-node exponent of the fragility law comes from the normal form and is, as its proof notes, independent of the microfoundation (Theorem 10.7). The linear-quadratic welfare loss is the leading term of an exact loss whose remainder is bounded by the relative mark error (Proposition A.6), exact in the basis-point regime and controlled outside it. The optimal-window characterization holds for the exact, non-Brownian loss (Propositions A.2, A.3), and the first-passage law is the exact Ornstein–Uhlenbeck one for any mean reversion (Proposition A.7). The affine form of the mark rests not on Gaussianity but on a worst-case optimality we prove directly (Theorem 14.2): against an adversary whose type law is known only through its variance, the affine mark minimizes the worst-case loss, the Gaussian least-favorable and Proposition 3.5 its best response: the mark’s analogue of the robust-contracts principle of Carroll (2015). What remains is the dependence of the *coefficients*, the discount and the window, on the value distribution and the cost, which the next two results settle.

Proposition 14.1 (Second-moment robustness). *The robust mark of Theorem 6.2 (its responsiveness a^* , intercept c^* , and loss) depends on the joint law of (v, ε, w) only through the second moments $\sigma^2 = \text{Var}(v)$, the staleness pair (C, V) , and $\rho = \text{Var}(w)/(\kappa^2 \sigma^2)$. Any finite-variance primitives sharing these moments therefore yield the identical optimal mark: the Gaussian specification is a representative of its second-moment class, and a heavy-tailed but finite-variance value (the empirically relevant case for cryptocurrency) leaves the characterization unchanged. When $\text{Var}(v) = \infty$ the quadratic criterion is the small-error leading term of an exact welfare loss that is bounded by the finite equity at risk, so the design problem remains well posed and the discount is governed by the bulk second moments with a bounded tail correction.*

Proof. At the directional best response $b = v - \varepsilon + \delta^*(w)$ with $\delta^*(w) = wa/\kappa$ independent of (v, ε) (Remark 5.3), so $\mathbb{E}[(ab + c - v)^2]$ is a quadratic form in (a, c) whose coefficients are the second moments $\mathbb{E}[v^2] = \sigma^2$, $\mathbb{E}[v\bar{v}_\tau]$, $\mathbb{E}[\bar{v}_\tau^2]$, and $\mathbb{E}[\delta^{*2}] = \text{Var}(w)a^2/\kappa^2$, the cross terms in v and δ^* vanishing by independence. Collected, $L/\sigma^2 = 1 - 2Ca + Va^2 + \rho a^4$ (Theorem 6.2), a function of (σ^2, C, V, ρ) alone, so its minimizer is invariant to any reshaping of the laws preserving these moments; the staleness pair (C, V) are themselves normalized second moments of the averaged path, closing the statement within second moments. For $\text{Var}(v) = \infty$, $\mathbb{E}[(M - v)^2]$ diverges, but the exact liquidation loss $\mathbb{E}[D_L] = \lambda_L \mathbb{E} \int_0^e s g(v + s) ds$ is bounded: g has finite mass and a finite first moment, so $\int_0^e s g(v + s) ds \leq \int_{\mathbb{R}} |u| g(u) du < \infty$ uniformly in e (a position cannot lose more than its bounded equity), and the welfare criterion is finite for any value law, with the quadratic its leading term where the error is small (Proposition A.6). $\square$

The second-moment robustness has a sharper consequence than invariance: it makes the affine mark *minimax*-optimal, which is the worst-case foundation the related literature reaches for by

citing [Carroll \(2015\)](#). We prove it here directly.

Theorem 14.2 (Worst-case optimality of the affine mark). *Suppose the designer knows the directional adversary's type law only through its variance, $\text{Var}(w) \leq \bar{V}$ (its mean absorbed by the intercept), and minimizes the worst-case loss $\sup_{G \in \mathcal{G}} L(M, G)$ over that class $\mathcal{G}$. Then the affine mark of Theorem 6.2 is minimax-optimal,*

$$\min_{M \text{ measurable}} \sup_{G \in \mathcal{G}} L(M, G) = \sup_{G \in \mathcal{G}} L(a^*, G) = L(a^*; \bar{V}),$$

and the Gaussian type of variance $\bar{V}$ is a least-favorable distribution, so the affine mark and the Gaussian type form a saddle point. The optimal rule is therefore robust to the shape of the type law, rather than to its second moment alone.

Proof. The affine loss is detail-free. By Proposition 14.1 the affine mark's loss against any G depends on G only through $\text{Var}(w)$: $L(a^*, G) = 1 - 2Ca^* + Va^{*2} + \rho a^{*4}$ with $\rho = \text{Var}(w)/(\kappa^2 \sigma^2)$, increasing in $\text{Var}(w)$. Hence $\sup_{G \in \mathcal{G}} L(a^*, G) = L(a^*; \bar{V})$, attained at every G of variance $\bar{V}$. The Gaussian is least-favorable and the affine mark is its best response. Let G_N be the Gaussian type of variance $\bar{V}$. With (v, ε) Gaussian and w independent, (v, ε, w) is jointly Gaussian, so Proposition 3.5 makes the loss-minimizing mark over *all* measurable functions against G_N the affine a^* : $L(M, G_N) \geq L(a^*, G_N) = L(a^*; \bar{V})$ for every measurable M . *Minimax.* Combining, for any measurable M ,

$$\sup_{G \in \mathcal{G}} L(M, G) \geq L(M, G_N) \geq L(a^*; \bar{V}) = \sup_{G \in \mathcal{G}} L(a^*, G).$$

Taking the infimum over M yields $\min_M \sup_G L = \sup_G L(a^*, \cdot) = L(a^*; \bar{V})$, attained by the affine mark; and $L(a^*, G_N) = \min_M L(M, G_N) = \max_{G \in \mathcal{G}} L(a^*, G)$ is the saddle-point identity, so (a^*, G_N) is a saddle point. The nonlinear mark that would beat a^* against a particular type law is exposed against the variance-matched worst case, while the affine mark sits at that worst case uniformly; this is the mark's analogue of the robust-contracts principle of [Carroll \(2015\)](#), here obtained directly. $\square$

Proposition 14.3 (General convex manipulation cost). *Replace the quadratic establishment cost $\frac{\kappa}{2}\delta^2$ by any strictly convex $\mathcal{C}(\delta)$ with $\mathcal{C}'(0) = 0$ and $\mathcal{C}'' > 0$. Then (i) the directional best response is $\delta^*(w) = (\mathcal{C}')^{-1}(wa)$, increasing in the prize wa and equal to wa/κ for the quadratic; (ii) the reflexive fold of Proposition 9.3 holds unchanged, the cost entering only through an adverse selection still increasing in a and decreasing in depth; (iii) the optimal mark still under-weights, the discount governed by the spread of the attack through the convex penalty $a^2 \text{Var}(\delta^*)$, with the quadratic the representative that makes the penalty exactly quartic; (iv) the saboteur's payoff is convex, unbounded absent a budget, iff $2wa^2 > \mathcal{C}''(\delta)$, the general form of $\mu a^2 > 1$, and the capital bound of Proposition 8.3 regularizes it for any convex cost.*

Proof. (i) The directional payoff $wa\delta - \mathcal{C}(\delta)$ is strictly concave with first-order condition $wa = \mathcal{C}'(\delta^*)$; $\mathcal{C}'$ is increasing and onto $[0, \infty)$, so $\delta^* = (\mathcal{C}')^{-1}(wa)$ is unique and increasing in wa , reducing to wa/κ when $\mathcal{C}'(\delta) = \kappa\delta$. (ii) Equilibrium depth solves $\kappa_0 = S(s - \Lambda(\delta^*(\bar{w}; \kappa_0)))$ with δ^* increasing in a and decreasing in κ_0 ; this is the hypothesis of Proposition 9.3 with composite adverse selection $\tilde{\Lambda}(a, \kappa_0) = \Lambda(\delta^*)$, so the fold, multiplicity, and hysteresis persist. (iii) After the intercept removes the mean, the manipulation contributes $a^2 \text{Var}(\delta^*(w))$ to $\mathbb{E}[(M - v)^2]$, a strictly increasing convex penalty in a , so $1 - 2Ca + Va^2 + a^2 \text{Var}(\delta^*)/\sigma^2$ is minimized at $a^* < 1$; with $\delta^* = wa/\kappa$ the penalty is $\rho a^4 \sigma^2$, the quartic. (iv) The saboteur payoff $w e(\delta)^2 - \mathcal{C}(\delta)$ has second derivative $2wa^2 - \mathcal{C}''(\delta)$,

positive (convex, unbounded above) exactly when $2wa^2 > \mathcal{C}''(\delta)$, which is $\mu a^2 > 1$ for $\mathcal{C}'' = \kappa$; Proposition 8.3's budget caps the dislocation wherever convexity makes the unbudgeted problem unbounded. $\square$

Both are confirmed numerically: a Student- t value with the variance of the Gaussian yields the identical optimal mark despite a hundredfold larger kurtosis, and the discount $a^* < 1$ persists across power costs $\mathcal{C}(\delta) \propto |\delta|^{1+\gamma}$ for every $\gamma > 0$.

The affine restriction and the reflexive result. Finally, the over-trust catastrophe is not an artifact of the affine mark. The depth feedback runs through the *effective reliance* the mark places on the reference (the pass-through that multi-venue weighting, trimming, and clamping move below one) not through the coefficient a as a literal multiplier. Any mark, affine or not, has an effective reliance, and the fold is crossed when that reliance thins the book past its tipping point; replacing the affine rule by the optimal nonlinear mark changes the map from primitives to reliance but not the existence of the fold or the over-trust region around it (Section 10). The reflexive collapse is a statement about how heavily a reference is leaned on, robust to the form in which the leaning is expressed.

15 The economics of listening less

The responsiveness discount is a trade, and a complete account weighs both sides in welfare terms: what robustness buys, what staleness costs, who gains and who loses, and when the design is worth running. The loss of Section 4 already holds the calculus (the manipulation term ρa^4 against the staleness terms $1 - 2Ca + Va^2$), but its terms are abstract; this section reads them as economics.

What robustness buys. The direct benefit is the manipulation rent the mark ceases to pay. Extraction scales with ρa^4 (Theorem 6.2), so lowering reliance from full tracking to the robust a^* cuts it quartically: the funding transfers and liquidation-trigger profits a manipulator would capture are wealth taken from honest counterparties on the wrong side of a manufactured move, and a mark that listens less shrinks the prize. The Mango Markets episode of Section 1 (a reference ramped to drain a lending pool) is the canonical realization; a sufficient discount is what renders such an attack unprofitable, since extraction falls to zero as reliance does. Beyond the transfer lies a deadweight. A manipulation that pushes the mark across a cluster of maintenance boundaries triggers liquidations the true value would not (Lemma 4.6, Theorem 11.5), destroying value through forced unwinds, gap risk, and cascade contagion on bystanders; these are pure social loss, and a mark harder to push averts them. The funding channel gains in parallel: funding is the perpetual's tether to spot, a manipulated mark mis-charges it and turns the mechanism into a conduit for rents, and the robust mark holds the transfer near the fundamental basis so that funding does its work rather than the manipulator's.

The largest benefit is insurance against the tail. Because reference depth is endogenous, leaning on a market thins the liquidity that keeps it honest (Proposition 9.3), a mark that listens too much can carry reliance past the fold a_{tip} , where the honest equilibrium vanishes and the reference collapses to a manipulated state that is costly and slow to reverse (Theorem 10.4). This is not a marginal loss but a regime change, the difference between a functioning reference and a collapsed one. A mark that listens less stays below the fold and preserves the infrastructure that reads the reference, and the value of doing so is greatest precisely where the reference is most fragile, near its tipping point, which is where the loop-gain measurement of Proposition 12.2 locates the cliff.

Finally, the design economizes on information the designer lacks: the robust mark is worst-case optimal over the attack distribution given only its variance (Theorem 14.2), so a designer who commits before the attack and cannot observe it still chooses well.

What staleness costs. The mirror of robustness is lag. A mark that listens less tracks fundamentals with delay (the $1 - 2Ca + Va^2$ terms), so when value genuinely moves (real news, not manipulation), the mark is wrong and everything keyed to it is mis-set. Staleness is basis risk levied on every honest user to deter a manipulating few, and its incidence is the cost side of the ledger.

The sharpest cost, and one the quadratic loss understates, runs through solvency. Liquidation exists to protect the protocol: a position must close before its loss exceeds its collateral. A mark that listens less liquidates late in a genuine crash: true value has fallen through the maintenance boundary while the anchored mark has not caught up, so the position runs underwater. The resulting bad debt is socialized, drawn from an insurance fund, auto-deleveraged onto solvent traders, or left as protocol loss. The anchoring that defeats a fake move delays the liquidation a real one requires, converting manipulation risk into solvency risk; the direction-dependent liquidation cost of Section 4 is where the model registers it. The floor on a is therefore set not by price discovery alone but by the protocol's tolerance for bad debt: a reference too slow to track a real crash is as dangerous as one too fast to yield to a fake one, and the responsiveness optimal against manipulation must be checked against the insolvency it would tolerate. A second cost is informational. The mark is itself a price others read. A less responsive mark contributes less to discovery and hands hedgers a worse instrument, raising hedging error and muddying the funding signal as a reading of the basis: listening less to the reference makes the market listen less to the perpetual. A stale mark also manufactures its own arbitrage: a trader who sees value move while the mark has not can trade the predictable update, a latency rent the discount creates. This is more benign than manipulation (it accrues to informed traders and pulls the mark toward value), but it is a rent the lag pays. And the design is not free to run: committing (a, τ) through governance, calibrating to a venue's manipulability and depth, and monitoring the fold consume resources, and the identification that locates the fold (Proposition 12.2) is informative only where reliance has varied. A mis-calibrated design is worse than none: too much robustness pays the staleness cost without deterring a commensurate attack, too little leaves the rents in place and courts the fold.

Net welfare, distribution, and the private–social wedge. The two sides net in favor of robustness when manipulability is high and the reference fragile, and against it when the reference is deep and manipulation already dear; the interior a^* of Theorem 6.2 is the balance, and its comparative statics (robustness rising with manipulability ρ , falling with the genuine volatility the mark must track) are the first-order guidance. The distribution matters as much as the level. The benefits accrue to honest traders spared the transfers and the cascades, to hedgers and to protocol solvency spared the collapse, and to the reference market's liquidity providers spared the thinning. The costs fall on users exposed to genuine moves through mis-priced funding and late liquidation, and through the solvency channel on the insurance fund and the traders who backstop it. The design trades a rare, large, adversarial loss for a steady, small, frictional one, buying down the tail risk of manipulation with the ongoing carry of staleness.

A last wedge separates private from social value. A venue choosing its own reliance does not internalize that leaning on a commonly read reference thins it for all who read it, so private reliance exceeds the social optimum and the privately chosen mark listens too much (Propositions 13.1 and 13.2). A Pigouvian correction restores the optimum; absent it, the decentralized outcome over-trusts, making the social benefit of listening less larger than any single venue's private benefit.

The over-trust catastrophe (Theorem 10.4) is the tail of this wedge: the regime in which privately optimal reliance collapses a shared reference. Because the benefit is convex in manipulability and discontinuous at the fold while the cost is smooth in staleness, the design’s net value is modest, even negative, for deep and hard-to-manipulate references, and large and rising as a reference nears its fold. Robustness should be spent where the loop gain says the cliff is close and withheld where the reference can carry the weight.

16 Discussion and conclusion

A perpetual future is tethered by a mark price (a manipulable public signal that allocates transfers and forces liquidations), and the question this paper answers is how much the mark should listen to it. The baseline answer is the familiar one: listen less, discount the contaminated reference, and average it over a window long enough to make manipulation costly and short enough to stay fresh. The contribution is to show this answer is incomplete in a way that inverts it. The reference’s depth is not a primitive; it is supplied by the traders who lose to the manipulator, so it thins as the mark relies on it more. Manipulability is therefore a fixed point of reliance. The mark that is optimal taking depth as given, the answer the literature gives, can, on a fragile reference, choose a reliance that thins the book past its tipping point and destroys the reference it reads. A reference made to look deeper is for that reason trusted more, so fragile depth the reference’s robustness does not back is worse than none. This reflexive collapse, its estimable trigger, and its dynamic form (a universal law making the expected time to a reference run exponential in the effective number of independent balance sheets behind it) are what the price-taking treatment of manipulation cannot see. They are the paper’s reason for being. The design that surrounds the result is complete on its own terms: a welfare loss microfounded from the liquidation and funding channels, the listen-less mark and the optimal window, the sharp depth-admissibility threshold, and the common-versus-own-price liquidation trade-off that is the perpetual analogue of uniform-price versus pay-as-bid allocation. Most consequentially for an empirical program, the fragility condition $\rho_0 < \bar{\rho}(a_{\text{tip}})$ is identified from execution-cost and order-flow data (Section 12), so the central prediction is testable rather than merely qualitative.

A theory of a designed object earns its sharpness by choosing what to hold fixed, and the choices here are load-bearing. They are worth stating plainly, not as apologies but because in each case the result survives relaxing the assumption, and the way it survives is itself a finding.

The cross-section of leveraged positions is taken exogenous to the attack. The welfare coefficient and the boundary density g_0 (and through the liquidation prize $w_L = \pi_{A0}$, the location of the fold) come from a population (Section 4.1) that does not anticipate manipulation; letting it anticipate makes g_0 a fixed point of the design. This is a normalization rather than a restriction.

Proposition 16.1 (Rational-expectations invariance). *Let traders anticipate manipulation when choosing buffers, making the boundary density $g(\cdot)$ (and through the prize $w_L = \pi_{A0}$, the manipulability ρ and the fold a_{tip}) a fixed point of the design. Then to leading order in the mark error:*

- (i) *the welfare loss keeps the form $\frac{1}{2}(\lambda_L g_0 + \lambda_F) \mathbb{E}[(M - v)^2]$ with $g_0 = g_{\text{RE}}(v)$, since the reduction of Lemma 4.6 uses only continuity of g at v , not its exogeneity;*
- (ii) *the robust-mark cubic (6.2) and the window condition (7.4) hold verbatim, read at the rational-expectations cross-section: the functional form of the optimum is unchanged, and anticipation enters only by setting the level of the prize w_L , hence of ρ and a_{tip} ;*

(iii) *that feedback is self-stabilizing. Anticipating manipulation adds a liquidation hazard decreasing in the buffer, so traders hold wider buffers; by (4.6) this thins g_0 near the mark, lowering the prize w_L , which both lowers the manipulability $\rho \propto \text{Var}(w)$ (so the robust mark relies more) and pushes the fold $a_{\text{tip}} = s^2/(4c_s\ell_0\bar{w})$ out: the reference becomes at once less manipulable and further from collapse.*

Proof. (i) Lemma 4.6 expands $D_L(e) = \frac{1}{2}\lambda_L g(v)e^2 + o(e^2)$ using only that g is continuous at v with $g(v) = g_0 > 0$; with $g = g_{\text{RE}}$ the same expansion holds at $g_0 = g_{\text{RE}}(v)$, and the funding term is unchanged. (ii) After the normalization of Proposition 4.9 the objective is $\mathbb{E}[(M - v)^2]$, whose minimizer is characterized by (6.2) and (7.4) as identities in (C, V, ρ) ; these hold with ρ evaluated at the rational-expectations prize, the only object anticipation moves. (iii) A trader anticipating manipulation faces, beyond the noise first-passage $p_\ell(\ell)$, a manipulation-liquidation hazard $p_m(\ell) = \Pr(a\delta > \ell)$, strictly decreasing in ℓ ; the buffer program $\min_\ell r\ell + \varpi(p_\ell + p_m)$ has a larger marginal benefit of ℓ than without anticipation, so ℓ^* rises. By (4.6) a wider buffer slides boundaries away from v , and since f_P decreases away from its mode, $g_0 = \mathbb{E}_{\hat{\ell}}[f_P(v + \hat{\ell})]$ falls; the prize $w_L = \pi_A g_0$ falls with it, so $\bar{w}$ and $\text{Var}(w)$ fall, lowering $\rho = \text{Var}(w)/(\kappa_0^2 \sigma^2)$ and, through $\bar{w}$, raising $a_{\text{tip}} = s^2/(4c_s\ell_0\bar{w})$. All steps are monotone. $\square$

The destabilizing direction is the one a full treatment would need: forced deleveraging in a stressed market concentrates positions *toward* boundaries, raising the prize exactly as depth thins. Proposition 16.1 prices the voluntary-anticipation channel, which stabilizes; the stress channel, which destabilizes, the model holds fixed, and closing it is part of the agenda below.

The attacker is drawn rather than entered in the baseline, which separates the reliance–depth loop from the entry loop. The entry extension restores a single law G by drawing each entrant from it, so the dispersion that drives the discount and the mean that drives the fold are two moments of one population (Proposition 9.6). Endogenizing the *number* of attackers does not relax the fold but compounds it: free entry adds a positive term to the loop gain and relocates the tipping point inward by an amount the congestion elasticity governs. What entry leaves open (the entrants’ own welfare, solved jointly rather than through the per-attacker zero-profit margin) is the joint attacker–provider equilibrium, an agenda item rather than a caveat.

The rule is a constant commitment, the conservative benchmark. The designer commits at Stage 0 to a responsiveness it does not later revise; the dynamic analysis (Proposition 10.8) shows a state-dependent rule that cuts reliance as depth falls dominates any constant one. The committed a^* is therefore a welfare lower bound and the right object precisely when depth is not contractible in real time or the exchange cannot credibly commit to cut reliance mid-attack; the full depth-contingent design is the companion’s impulse-control problem.

The funding channel’s welfare content is allocative, and enters as a normalization. The funding loss (4.9) is the distortion to a position chosen against a strictly concave private payoff. For a pure directional speculator, whose payoff is effectively linear, mistimed funding is closer to a transfer than a deadweight, so λ_F loads on the hedging mass alone. The liquidation channel carries no such caveat (a wrong liquidation destroys equity whoever holds the position), and it is the channel that drives the fold.

The objective is social welfare, and the private objective only sharpens the policy case. A profit-maximizing exchange values fee revenue and volume and may tolerate more manipulability than is efficient. Section 13 shows this widens rather than closes the gap the Pigouvian instruments address, so the normative design is, if anything, a conservative floor on what a regulator would require. Characterizing the profit-maximizing mark and the competitive equilibrium among venues is the positive counterpart this normative analysis leaves open.

Underlying these, the analysis is local: the welfare loss, the adversary’s benefit, the saboteur’s explosion at $\mu a^2 = 1$, and the saddle-node exponent are leading-order in the mark error and the distance to the fold, with remainder bounds in Appendix A.6 and the exact first passage in Appendix A.7. This is the right theory in the empirically relevant regime, where mark errors are basis points against the price scale on which the leverage density varies. The saboteur’s unbounded exploitation at $\mu a^2 \geq 1$ is the local model’s signal that the design must stay in the concave region (Corollary 8.2), with the finite, capital-bounded behavior in that region given by Proposition 8.3. The affine, Gaussian benchmark is itself a deliberate choice (optimal against a linear adversary under the Gaussian primitives) with the optimal nonlinear mark for non-Gaussian types developed in the companion paper.

These scope choices are not a list of things undone; each names a question the reflexive framework is now equipped to pose, and three are the natural next steps. The first is empirical: whether a given real reference has a fold at all (whether its loop gain exceeds one) is a question about venue-level liquidity data. The identification of $\rho_0 < \bar{\rho}(a_{\text{tip}})$ from execution-cost and reliance variation (Section 12) is the instrument that would settle it, turning the central prediction into a measurement. The second is the joint attacker–provider equilibrium: Proposition 9.6 endogenizes the count of attackers and shows entry compounds the fold. But closing the loop on the type *dispersion* and the entrants’ welfare (so that the prize, the dispersion, and the depth are determined together) is the equilibrium of which the present results are the fixed-dispersion slice. The third is the joint design of the mark and the funding rate through the clamp. Its robustness benefit is already in the model, since the clamp truncates the funding loss below the quadratic. Its cost is not: the weakened anchoring of the perpetual to spot under a capped funding rate requires a model of perpetual price formation this paper does not carry. So the anchoring-versus-robustness frontier is the one that genuinely needs the companion’s pricing apparatus joined to this paper’s manipulation apparatus.

Beyond perpetuals, the object of the paper is a benchmark that is at once a public good and an attack surface, and whose informativeness decays because it is used. That the decay is *reflexive* (that reliance on a measure manufactures the manipulability the measure must then be discounted for) is a force the price-taking treatment of manipulation cannot represent. It is the benchmark-design counterpart of the feedback by which trading on a price corrupts the information it carries. The practical reading is a warning an exchange or a regulator can act on: depth measured in calm is not depth available under reliance, the deep and active reference a sound mark requires is partly manufactured by the mark itself, and the variable to monitor is not the reference’s depth today but the gain that measures how close that depth sits to its fold. A reference can be made to look safer and thereby be made more dangerous; the contribution of this paper is the framework that says when, and the collapse it reveals.

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Appendix A Supplementary results and the exact-loss window

This appendix records properties of the staleness functions used implicitly in the main text and states the optimal-window condition for the exact loss (6.1), showing that the existence and uniqueness arguments do not rely on the locally-Brownian approximation.

A.1 Derivatives of the staleness functions

Lemma A.1 (Monotonicity of C and V). *For $x > 0$,*

$$C'(x) = \frac{(1+x)e^{-x} - 1}{x^2} < 0, \quad V'(x) = -\frac{2\psi(x)}{x^3} < 0, \quad \psi(x) = x - 2 + (x+2)e^{-x} > 0,$$

so both C and V are strictly decreasing, with $C(0^+) = V(0^+) = 1$ and $C, V \rightarrow 0$ as $x \rightarrow \infty$. Their second derivatives

$$C''(x) = \frac{2 - (x^2 + 2x + 2)e^{-x}}{x^3} > 0, \quad V''(x) = \frac{2[2x - 6 + (x^2 + 4x + 6)e^{-x}]}{x^4} > 0$$

are bounded on $[0, \infty)$, with $C''(0^+) = \frac{1}{3}$ and $V''(0^+) = \frac{1}{6}$. We also have $-2C'(0^+) = 1$ and $V'(0^+) = -\frac{1}{3}$, recovering the locally-Brownian slopes.

Proof. Differentiate $C(x) = (1 - e^{-x})/x$ by the quotient rule: $C'(x) = [xe^{-x} - (1 - e^{-x})]/x^2 = [(1+x)e^{-x} - 1]/x^2$, negative because $h(x) := (1+x)e^{-x}$ satisfies $h(0) = 1$ and $h'(x) = -xe^{-x} < 0$, so $h(x) < 1$ for $x > 0$. Differentiate $V(x) = 2(x - 1 + e^{-x})/x^2$: $V'(x) = 2[(1 - e^{-x})x^2 - 2x(x - 1 + e^{-x})]/x^4 = 2[(1 - e^{-x})x - 2(x - 1 + e^{-x})]/x^3 = -2[x - 2 + (x+2)e^{-x}]/x^3 = -2\psi(x)/x^3$. The numerator $\psi(x) = x - 2 + (x+2)e^{-x}$ has $\psi(0) = 0$, $\psi'(x) = 1 - (x+1)e^{-x}$ with $\psi'(0) = 0$, and $\psi''(x) = xe^{-x} > 0$; hence $\psi' > 0$ and $\psi > 0$ on $(0, \infty)$, so $V' < 0$. Both C and V decrease from $C(0^+) = V(0^+) = 1$ (l'Hôpital) to 0 as $x \rightarrow \infty$. Differentiating once more, $C''(x) = [2 - (x^2 + 2x + 2)e^{-x}]/x^3$, whose numerator $g(x) = 2 - (x^2 + 2x + 2)e^{-x}$ has $g(0) = 0$ and $g'(x) = x^2e^{-x} > 0$, so $C'' > 0$; and $V''(x) = 2[2x - 6 + (x^2 + 4x + 6)e^{-x}]/x^4$, whose numerator $G(x)$ has $G(0) = G'(0) = G''(0) = G'''(0) = 0$ and $G^{(4)}(x) = x^2e^{-x} > 0$, so $G > 0$ and $V'' > 0$. The slopes and curvatures at the origin follow from $e^{-x} = \sum_{k \geq 0} (-x)^k/k!$: substituting gives $C'(x) = -\frac{1}{2} + \frac{x}{3} - \frac{x^2}{8} + O(x^3)$, $V'(x) = -\frac{1}{3} + \frac{x}{6} - \frac{x^2}{20} + O(x^3)$, $C''(x) = \frac{1}{3} - \frac{x}{4} + O(x^2)$, and $V''(x) = \frac{1}{6} - \frac{x}{10} + O(x^2)$, so $-2C'(0^+) = 1$, $V'(0^+) = -\frac{1}{3}$, $C''(0^+) = \frac{1}{3}$, $V''(0^+) = \frac{1}{6}$, the locally-Brownian values, and C'', V'' are continuous and bounded on $[0, \infty)$ (finite at 0 by these limits, decaying like $2/x^3$ and $4/x^3$ as $x \rightarrow \infty$). $\square$

A.2 The optimal window for the exact loss

Proposition A.2 (Exact-loss window condition). *For the exact loss (6.1), with $a^*(\tau)$ the responsiveness of Theorem 6.2 and $x = \phi\tau$, the optimized loss $L^*(\tau) = \min_a L(a, \tau)$ has*

$$\frac{1}{\sigma^2} \frac{dL^*}{d\tau} = \phi a^* (-2C'(x) + V'(x) a^*) + \rho'(\tau) a^{*4}. \quad (\text{A.1})$$

The first term, the staleness marginal, is strictly positive for all $\tau > 0$; the second, the manipulation marginal, is strictly negative. Hence an interior optimal window exists whenever the manipulation marginal dominates at $\tau = 0$, exactly as in Proposition 7.1, and an interior stationary point is a local minimum wherever the staleness marginal is increasing in τ relative to the manipulation marginal, the exact-loss analogue of (7.6).

Proof. By the envelope theorem, $\frac{1}{\sigma^2} dL^*/d\tau = \partial_\tau(L/\sigma^2)|_{a^*}$. From (6.1), $\partial_\tau(L/\sigma^2) = -2C'(x)\phi a + V'(x)\phi a^2 + \rho'(\tau)a^4$, which at $a = a^*$ is (A.1). For the staleness marginal, since $a^* \in (0, 1)$ and $V' < 0$ we have $V'a^* > V'$, so $-2C' + V'a^* > -2C' + V'$, and it suffices that $-2C'(x) + V'(x) > 0$. Using Lemma A.1,

$$-2C'(x) + V'(x) = \frac{2}{x^3} [2 - (x^2 + 2x + 2)e^{-x}],$$

and $g(x) := 2 - (x^2 + 2x + 2)e^{-x}$ has $g(0) = 0$ and $g'(x) = x^2 e^{-x} > 0$, hence $g > 0$ and the staleness marginal is strictly positive. The manipulation marginal $\rho'a^{*4} < 0$ since $\rho' < 0$. The existence and local-minimum claims then follow by the sign argument of Propositions 7.1 and 7.3, with $(-2C'(x), V'(x))$ in place of the leading slopes $(1, -\frac{1}{3})$. $\square$

The exact condition differs from the locally-Brownian one only in replacing the constant slopes by the functions $-2C'(x)$ and $V'(x)$ of Lemma A.1, which converge to those constants as $x \rightarrow 0$. Since the calibrated optima sit at $x = \phi\tau^* \ll 1$, the leading-order analysis of Section 7 characterizes them, and (A.1) shows the structure persists for any window.

A.3 Uniqueness of the optimal window for the exact loss

Proposition A.2 gives the exact first-order condition; we now close the uniqueness gap noted in the conclusion, showing that the strict-local-minimum property of Proposition 7.3 survives in the exact loss (6.1) under the same slow-mean-reversion condition.

Proposition A.3 (Exact-loss uniqueness). *For the exact loss (6.1), write $\Xi_e(\tau) := \phi(-2C'(x) + V'(x)a^*) + \rho'(\tau)a^{*3}$ with $x = \phi\tau$, so that $\frac{1}{\sigma^2} L^* = a^* \Xi_e$. At any interior stationary point ($\Xi_e = 0$),*

$$\frac{1}{\sigma^2} \frac{d^2 L^*}{d\tau^2} = a^* \phi \left[\frac{(2\beta + 1)(-2C'(x) + V'(x)a^*)}{\tau_0 u} - \phi R(x, a^*) \right],$$

where $u = 1 + \tau/\tau_0$ and

$$R(x, a^*) := 2C''(x) - V''(x)a^* - \frac{1}{\phi^2} \frac{da^*}{d\tau} (\phi V'(x) + 3\rho'(\tau)a^{*2})$$

is bounded, with $C''(x) = [2 - (x^2 + 2x + 2)e^{-x}]/x^3$ and $V''(x) = 2[2x - 6 + (x^2 + 4x + 6)e^{-x}]/x^4$ the bounded second derivatives of Lemma A.1 and $da^*/d\tau, \rho'(\tau)$ each $O(\phi)$ (so the last group is $O(1)$). The leading bracket term is $O(1)$ and positive by Lemma A.1; the correction is $O(\phi)$. Hence every interior stationary point is a strict local minimum, and the optimal window is unique, whenever $\phi < \phi_e^* := \inf_\tau (2\beta + 1)(-2C'(x) + V'(x)a^*)/(\tau_0 u R(x, a^*)) > 0$.

Proof. At a stationary point $\frac{1}{\sigma^2}L^{**} = a^* d\Xi_e/d\tau$. Differentiating Ξ_e totally,

$$\frac{d\Xi_e}{d\tau} = \phi^2(-2C'' + V''a^*) + \rho''a^{*3} + \frac{da^*}{d\tau}(\phi V' + 3\rho'a^{*2}),$$

with $da^*/d\tau = -[\phi(-C' + V'a^*) + 2\rho'a^{*3}]/(V + 6\rho a^{*2})$ from the inner first-order condition. At the stationary point $\rho'a^{*3} = -\phi(-2C' + V'a^*) := -\Pi_e$, which is $O(\phi)$ since $-2C' + V'a^* = O(1)$; hence $\rho = O(\phi)$ there, $da^*/d\tau = O(\phi)$, and the term $\frac{da^*}{d\tau}(\phi V' + 3\rho'a^{*2}) = O(\phi^2)$, as is $\phi^2(-2C'' + V''a^*)$. By Assumption 4, $\rho''a^{*3} = (2\beta + 1)\Pi_e/(\tau_0 u) = (2\beta + 1)\phi(-2C' + V'a^*)/(\tau_0 u)$, the unique $O(\phi)$ term (using $\rho'' = 2\beta(2\beta + 1)\rho/(\tau_0 u)^2$ and $\rho a^{*3} = \Pi_e \tau_0 u/(2\beta)$ from the stationary condition). Collecting this as the leading bracket term and the two $O(\phi^2)$ groups, $\phi^2(-2C'' + V''a^*)$ and $\frac{da^*}{d\tau}(\phi V' + 3\rho'a^{*2})$, as $-a^*\phi^2 R$ gives exactly the displayed $R(x, a^*) = 2C'' - V''a^* - \phi^{-2}\frac{da^*}{d\tau}(\phi V' + 3\rho'a^{*2})$. It is bounded: C'', V'', V' are bounded on $[0, \infty)$ (Lemma A.1), and $\phi^{-2}\frac{da^*}{d\tau}(\phi V' + 3\rho'a^{*2})$ is $O(1)$ because each of $da^*/d\tau$ and ρ' is $O(\phi)$, so the product of the two $O(\phi)$ factors divided by ϕ^2 is $O(1)$. Both $-2C' + V'a^* > 0$ (proof of Proposition A.2) and $C'' > 0$ (Lemma A.1) hold, so for $\phi < \phi_e^*$ the bracket is positive and $L^{**} > 0$. As in Proposition 7.3, every interior stationary point being a strict local minimum precludes two minima, giving uniqueness. $\square$

The exact threshold ϕ_e^* differs from the locally-Brownian ϕ^* of (7.6) only through the bounded staleness derivatives, which converge to the leading slopes as $x \rightarrow 0$; at the calibrated optima $x = \phi\tau^* \ll 1$ and the two coincide to leading order, so the numerically verified uniqueness is now a theorem for the exact model as well.

A.4 Existence of equilibrium and well-posedness of the designer's program

This appendix proves Theorem 3.7: the pointwise best responses of Section 5 assemble into a measurable map, the loss is a finite continuous functional of the instruments, and its minimization has a solution, so a subgame-perfect equilibrium exists.

Proof of Theorem 3.7. (i) Best response. For the directional adversary the Stage-2 payoff (5.2) is strictly concave in δ with maximizer $\delta^*(w) = wa/\kappa(\tau)$ (Lemma 5.2), linear hence Borel-measurable in w and, being independent of v , jointly measurable in (w, v) ; it is square-integrable because $\mathbb{E}[w^2] < \infty$. For the saboteur on $\{\mu(\tau)a^2 < 1\}$ the payoff (5.3) is strictly concave with maximizer $\delta^*(w, v) = \frac{\mu a}{1 - \mu a^2}[(a - 1)v + c]$ (Lemma 5.4), measurable in (w, v) , rational in $\mu = 2w/\kappa(\tau)$ and affine in v , and square-integrable because v is Gaussian and w has finite variance on the feasible region. Strict concavity gives uniqueness.

(ii) *Finiteness and continuity.* Write the error at the best response as $e(w, v, \varepsilon) = M(b) - v = (a\bar{v}_\tau - v) + a\delta^* + c$. In the directional case $e = (a - 1)v - a\varepsilon + a^2w/\kappa(\tau) + c$ is a linear combination of the jointly Gaussian (v, ε) and the independent, finite-variance w , so $\mathbb{E}[e^2] < \infty$. By Lemma 6.1,

$$L(a, c, \tau) = \sigma^2(1 - 2Ca + Va^2) + \rho(\tau)a^4\sigma^2 + (c + \bar{w}a^2/\kappa(\tau))^2,$$

manifestly finite, and jointly continuous on $\mathcal{A}$ because $C(\phi\tau)$, $V(\phi\tau)$ (Lemma 5.5) and $\kappa(\tau)$ (Lemma 3.1) are continuous in τ and L is a continuous function of these moments and of (a, c) . The saboteur case is identical on its feasible region, with $L(a) = \sigma^2(1 - a)^2/(1 - \mu a^2)^2$ (Proposition 8.1), continuous where $\mu a^2 < 1$.

(iii) *Concentrating out c and a , and existence of τ^* .* For fixed (a, τ) , L is strictly convex in c (unit coefficient on c^2) with unique minimizer $c^*(a, \tau) = -\bar{w}a^2/\kappa(\tau)$, which annihilates the mean term and leaves the variance objective (6.1). For fixed τ , $\tilde{L}(a, \tau) := \min_c L$ is strictly convex in a on all of $\mathbb{R}$ (second derivative $2V + 12\rho a^2 > 0$), and its unconstrained minimizer is the unique real root

$a^*(\tau)$ of the cubic (6.2), lying in $(0, 1)$ (Theorem 6.2); hence the constraint $a \in [0, 1]$ does not bind. Finally $L^*(\tau) := \min_a \tilde{L}(a, \tau)$ is continuous on $[0, \infty)$ by Berge's maximum theorem (continuous objective, compact-valued constant correspondence $[0, 1]$), with $L^*(0) < \sigma^2$ (at any small $a > 0$, $(1 - a)^2 + \rho_0 a^4 < 1$) and $L^*(\tau) \rightarrow \sigma^2$ as $\tau \rightarrow \infty$ (Corollary 6.3). Since $L^*(\tau) \rightarrow \sigma^2 > L^*(0)$, there is $\bar{\tau}$ with $L^*(\tau) > L^*(0)$ for all $\tau > \bar{\tau}$; thus $\inf_{[0, \infty)} L^* = \min_{[0, \bar{\tau}]} L^*$ is attained at some finite τ^* by continuity on the compact interval, interior under (7.2) and equal to 0 otherwise (Proposition 7.1).

(iv) *Equilibrium.* The pair $((a^*, c^*, \tau^*), \delta^*(\cdot))$, with δ^* from (i) and (a^*, c^*, τ^*) the minimizer from (iii), satisfies Definition 3.6: the adversary best-responds in every subgame by (i), and the designer minimizes the well-defined loss by (iii). The equilibrium loss is $L^* = \min_{\mathcal{A}} L$. $\square$

For the saboteur the same argument applies on the feasible region, where Proposition 8.1 gives the closed-form minimizer directly: a corner $a^* = 1$ with zero loss when $\mu \leq 1$, and the interior $a^* = 1 - \sqrt{1 - 1/\mu}$ when $\mu > 1$. Continuity of L^* across the threshold fails, the jump of Proposition 8.1(c), but the program remains well posed on each side, and the optimal window is then chosen so as to keep $\mu(\tau) \leq 1$ where admissibility requires it (Corollary 8.2).

Remark A.4 (The mixed adversary). The general type of Section 5.1 carries both a directional component w_{dir} and a saboteur component w_{sab} , with Stage-2 payoff $w_{\text{dir}}(a\delta) + w_{\text{sab}} e(\delta)^2 - \frac{\kappa}{2} \delta^2$ and $e(\delta) = (a - 1)v + c + a\delta$. This is strictly concave in δ iff $2w_{\text{sab}}a^2 < \kappa$, the same feasibility as the pure saboteur with w_{sab} in the role of w . On the admissible region where it holds for the largest saboteur type in the support of G (Corollary 8.2), the best response is the unique interior root of a linear first-order condition, jointly measurable and square-integrable as in part (i), and the loss is the sum of the directional and saboteur quadratics, finite and continuous as in part (ii), so the minimization of parts (iii)–(iv) goes through unchanged. Theorem 3.7 thus extends to the general adversary on the admissible region, the directional mean absorbed by the intercept c^* and the saboteur curvature governing feasibility.

A.5 The cross-section under multivariate types

Lemma 4.1 stated the boundary density in the scalar case and invoked the coarea formula for $k > 1$. We record the supplement.

Lemma A.5 (Coarea density). *Let $\zeta \in Z \subseteq \mathbb{R}^k$ have continuous density f_ζ and let $B \in C^1(Z)$ be a submersion ($\nabla B \neq 0$). Then $b = B(\zeta)$ has density*

$$g(b) = \int_{\{B=b\}} \frac{f_\zeta(\zeta)}{|\nabla B(\zeta)|} d\mathcal{H}^{k-1}(\zeta),$$

and $g_0 = g(v) > 0$ iff the level set $\{B = v\}$ carries positive f_ζ -mass. If in addition f_ζ has bounded support (or, on unbounded Z , is uniformly integrable along the level sets near b) then g is continuous at b ; without such tightness g may jump, since a continuous density can have a discontinuous marginal.

Proof. By the coarea formula, for any test function φ ,

$$\mathbb{E}[\varphi(B(\zeta))] = \int_Z \varphi(B(\zeta)) f_\zeta(\zeta) d\zeta = \int_{\mathbb{R}} \varphi(b) \left(\int_{\{B=b\}} \frac{f_\zeta}{|\nabla B|} d\mathcal{H}^{k-1} \right) db,$$

identifying the inner integral as the density $g(b)$. The local structure of the level sets follows from the implicit function theorem: where $\nabla B \neq 0$ the level sets $\{B = b\}$ form a C^1 family of $(k - 1)$ -manifolds depending continuously on b (locally graphs over a fixed hyperplane), so

the surface measure $d\mathcal{H}^{k-1}$ and the integrand $f_\zeta/|\nabla B|$ vary continuously with b . When the type support is bounded (collateral capacities and valuations lie in a compact set, the empirically relevant case), each level set $\{B = b\}$ is compact, the integrand is continuous on a continuously varying compact family, and g is continuous in b . Boundedness is not dispensable. With unbounded support the pushforward density can jump where mass escapes to infinity along the level sets: a continuous density on $\mathbb{R}^2$ can have a discontinuous marginal (a unit-mass bump whose center recedes to infinity as the parameter approaches a point leaves a marginal that is one nearby and zero at the point), with the compactly truncated densities converging to it only pointwise, not uniformly. For unbounded Z the conclusion therefore requires the additional condition that f_ζ be uniformly integrable along the level sets near b , no escaping mass, a tightness the bounded-support hypothesis supplies automatically. Positivity at $b = v$ is positivity of the surface integral over $\{B = v\}$. The scalar case $k = 1$ collapses the surface integral to the single preimage point, recovering $g(b) = f_\zeta(B^{-1}(b))/|B'(B^{-1}(b))|$. $\square$

The hypotheses are mild and hold for the model's population: $\nabla B \neq 0$ asks only that no fat set of types share an exact boundary, continuity of f_ζ that the population of leverage choices have no atoms at the marginal boundary, and bounded support that collateral capacities and valuations lie in a compact set, which also delivers the tightness the continuity of g requires. Under them the welfare coefficient $\frac{1}{2}(\lambda_L g_0 + \lambda_F)$ of Proposition 4.9 is a continuous, strictly positive functional of the population, and the leading-order objective (4.10) is the exact welfare loss to the order of Definition 4.12.

A.6 The leading-order reduction and its remainder

The objective (4.10) is the leading term of the welfare loss in the small-error regime of Definition 4.12. We bound the remainder, so that “to leading order” carries a quantitative rate.

Proposition A.6 (Cubic remainder). *Suppose the boundary density g is Lipschitz with constant L_g near v and the private payoff $\pi(\cdot; \nu)$ is C^3 with $|\pi_{qqq}|$ bounded. Then the exact welfare loss obeys*

$$\mathbb{E}[D_L] + \mathbb{E}[D_F] = \left(\frac{1}{2}\lambda_L g_0 + \frac{1}{2}\lambda_F\right) \mathbb{E}[(M - v)^2] (1 + O(\bar{e}/\underline{s})),$$

with $\bar{e} = (\mathbb{E}[(M - v)^2])^{1/2}$ and $\underline{s}$ as in Definition 4.12. After the normalization of Proposition 4.9, $L = \mathbb{E}[(M - v)^2] (1 + O(\bar{e}/\underline{s}))$, so the affine optimum of Theorem 6.2 is welfare-optimal up to a relative $O(\bar{e}/\underline{s})$ correction.

Proof. In Lemma 4.6, $D_L(e) = \lambda_L \int_0^e s g(v + s) ds = \frac{1}{2}\lambda_L g_0 e^2 + \lambda_L \int_0^e s [g(v + s) - g_0] ds$; the Lipschitz bound gives $|g(v + s) - g_0| \leq L_g |s|$, so the second term is at most $\lambda_L L_g \int_0^{|e|} s^2 ds = \frac{1}{3}\lambda_L L_g |e|^3 = O(|e|^3)$. Taking expectations, the liquidation remainder is $O(\mathbb{E}|e|^3)$, a relative factor $O(\bar{e}/\underline{s})$ against the leading $\frac{1}{2}\lambda_L g_0 \mathbb{E}[e^2]$, with $\underline{s} = g_0/L_g$ the scale on which g varies. For funding, the third-order Taylor remainder of π about q^* is $O(f^3)$ with $f = \lambda_I(M - v)$, so the funding remainder is $O(\mathbb{E}|M - v|^3)$, again relative $O(\bar{e}/\underline{s})$ with $\underline{s} = |\pi_{qq}/\pi_{qqq}|$ the curvature scale. Summing the two channels gives the stated bound, and the normalization rescales the leading coefficient to one. $\square$

The reduction is therefore exact in the limit $\bar{e}/\underline{s} \rightarrow 0$ and accurate to a controlled relative error away from it. Empirically $\bar{e}$ is a few basis points while $\underline{s}$ is the price scale over which the leverage cross-section thins, orders of magnitude larger, so the affine, quadratic problem of Sections 6–11 is the right object on the relevant range, with nonlinear corrections the first term beyond it.

A.7 Exact Ornstein–Uhlenbeck first passage

Lemma 4.3 used the Brownian leading order. We record the exact first-passage law of the mean-reverting marking price, valid for any ϕT , in terms of parabolic cylinder functions, and reconcile it with the reflection value in the regime $\phi T \ll 1$ where the calibrated optima sit.

Let X solve $dX = -\phi X du + \varsigma dB$ with $X_0 = 0$ and stationary variance $\sigma^2 = \varsigma^2/(2\phi)$, and let $\tau_{-\ell} = \inf\{u : X_u = -\ell\}$, $\ell > 0$, be the liquidation time of a long with buffer ℓ .

Proposition A.7 (Exact OU first passage). *Write $\nu = \lambda/\phi$ and let D_p denote the parabolic cylinder function. The λ -harmonic functions of the OU generator $A = \frac{\varsigma^2}{2}\partial_x^2 - \phi x\partial_x$ are $H_\lambda(x) = e^{x^2/(4\sigma^2)}D_{-\nu}(x/\sigma)$, and the Laplace transform of the first-passage time is*

$$\mathbb{E}_0[e^{-\lambda\tau_{-\ell}}] = \frac{H_\lambda(0)}{H_\lambda(-\ell)} = e^{-\ell^2/(4\sigma^2)} \frac{D_{-\nu}(0)}{D_{-\nu}(-\ell/\sigma)}, \quad D_{-\nu}(0) = \frac{\sqrt{\pi} 2^{-\nu/2}}{\Gamma(\frac{1+\nu}{2})}.$$

Inverting, the survival probability has the spectral expansion

$$\mathbb{P}(\tau_{-\ell} > T) = \sum_{n \geq 1} a_n(\ell) e^{-\phi \rho_n(\ell) T},$$

where $\{\rho_n(\ell)\}$ are the ordered positive zeros of $p \mapsto D_p(-\ell/\sigma)$ (the Dirichlet eigenvalues of the OU generator killed at $-\ell$), and the coefficients $\{a_n\}$ are given in closed form in Proposition A.8; the principal eigenvalue $\phi \rho_1(\ell) > 0$ governs the long-horizon decay. The mean first-passage time is the Siebert integral

$$\mathbb{E}_0[\tau_{-\ell}] = \frac{\sqrt{2\pi}}{\phi} \int_{-\ell/\sigma}^0 e^{u^2/2} \Phi(-u) du.$$

As $\phi T \rightarrow 0$ the first-passage probability $1 - \mathbb{P}(\tau_{-\ell} > T)$ converges to the reflection value $2\Phi(-\ell/(\varsigma\sqrt{T}))$ of Lemma 4.3, the leading correction negative of order ϕT and given in closed form in Corollary A.9, so the optimal buffer (4.5) contracts by $O(\phi T)$ relative to its Brownian value.

Proof. In standardized coordinates $z = x/\sigma$, $\nu = \lambda/\phi$, the eigenequation $Au = \lambda u$ becomes $u'' - zu' - \nu u = 0$; substituting $u = e^{z^2/4}w$ gives the parabolic cylinder equation $w'' + (\frac{1}{2} - \nu - \frac{z^2}{4})w = 0$, solved by $w = D_{-\nu}(z)$, so $H_\lambda(x) = e^{x^2/(4\sigma^2)}D_{-\nu}(x/\sigma)$ is the λ -harmonic function decaying as $x \rightarrow +\infty$. From the asymptotics $D_{-\nu}(z) \sim z^{-\nu}e^{-z^2/4}$ as $z \rightarrow +\infty$ one has $H_\lambda(x) \sim (x/\sigma)^{-\nu} \rightarrow 0$, and H_λ is positive and strictly decreasing on $[-\ell, \infty)$ (it is the decreasing harmonic function; the second, increasing solution $e^{x^2/(4\sigma^2)}D_{-\nu}(-x/\sigma)$ blows up at $+\infty$ and is discarded). Because H_λ is λ -harmonic, $N_u := e^{-\lambda u}H_\lambda(X_u)$ is a local martingale; on $\{u \leq \tau_{-\ell}\}$ the path satisfies $X_u \geq -\ell$, where $0 < H_\lambda(X_u) \leq H_\lambda(-\ell)$, so $0 < N_u \leq H_\lambda(-\ell)$ and N is a bounded martingale on the stopped interval. Optional stopping at $\tau_{-\ell} \wedge t$ gives $H_\lambda(0) = N_0 = \mathbb{E}_0[N_{\tau_{-\ell} \wedge t}]$; letting $t \rightarrow \infty$, bounded convergence applies because $N_{\tau_{-\ell} \wedge t} \rightarrow e^{-\lambda\tau_{-\ell}}H_\lambda(-\ell)$ on $\{\tau_{-\ell} < \infty\}$ and $\rightarrow 0$ on $\{\tau_{-\ell} = \infty\}$ (there $e^{-\lambda u} \rightarrow 0$ for $\lambda > 0$ while $H_\lambda(X_u)$ stays bounded), so $H_\lambda(0) = H_\lambda(-\ell) \mathbb{E}_0[e^{-\lambda\tau_{-\ell}} \mathbf{1}\{\tau_{-\ell} < \infty\}]$, i.e. $\mathbb{E}_0[e^{-\lambda\tau_{-\ell}}] = H_\lambda(0)/H_\lambda(-\ell)$ with the convention $e^{-\lambda\cdot\infty} = 0$; evaluating with the tabulated $D_{-\nu}(0) = \sqrt{\pi} 2^{-\nu/2}/\Gamma(\frac{1+\nu}{2})$ yields the ratio. The poles of $\lambda \mapsto 1/D_{-\lambda/\phi}(-\ell/\sigma)$ are at $\lambda = -\phi\rho_n(\ell)$, where the order $-\lambda/\phi = \rho_n$ solves $D_{\rho_n}(-\ell/\sigma) = 0$; these are the (negative) eigenvalues of A with a Dirichlet condition at $-\ell$, and the residue expansion of the inverse Laplace transform is the spectral sum (Proposition A.8). For the mean first-passage time, write $u(x) = \mathbb{E}_x[\tau_{-\ell}]$ in scale-speed form $u(x) = \int_{-\ell}^x s'(y) (\int_y^\infty m(z) dz) dy$ with scale density $s'(y) = e^{y^2/(2\sigma^2)}$ and speed density $m(z) = \frac{2}{\varsigma^2} e^{-z^2/(2\sigma^2)}$. This solves the MFPT equation directly: $u'(x)/s'(x) = \int_x^\infty m$, so $\frac{d}{dx}(u'/s') = -m(x)$; since $s''/s' = x/\sigma^2$, expanding the left side gives $u'' - (x/\sigma^2)u' = -m(x)s'(x) =$

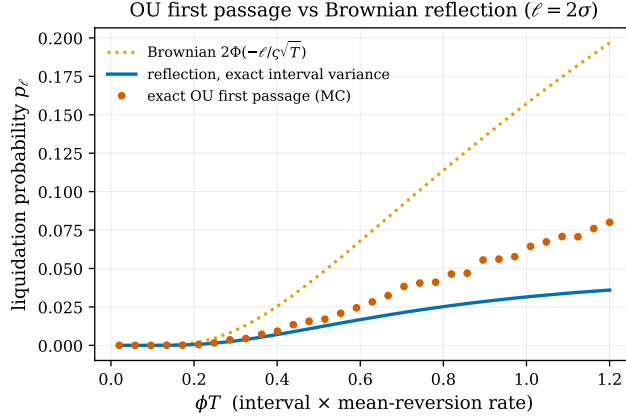


Figure 11: Exact Ornstein–Uhlenbeck first-passage probability (Monte Carlo) against the Brownian reflection value of Lemma 4.3, for a 2σ buffer. The naive Brownian (constant variance rate ς^2) overstates hitting and the reflection with the exact interval variance understates it; the exact OU lies between, and all three coincide in the slow-reversion regime $\phi T \ll 1$ where the calibrated mark windows sit, confirming the leading order of Lemma 4.3 and Proposition A.7.

$-2/\varsigma^2$, and multiplying by $\varsigma^2/2$ with $\varsigma^2/(2\sigma^2) = \phi$ yields $\frac{\varsigma^2}{2}u'' - \phi x u' = -1$, with $u(-\ell) = 0$ by construction. The inner integral is $\int_y^\infty m = \frac{\sqrt{2\pi}}{\phi\sigma} \Phi(-y/\sigma)$, so $u(0) = \frac{\sqrt{2\pi}}{\phi\sigma} \int_{-\ell}^0 e^{y^2/(2\sigma^2)} \Phi(-y/\sigma) dy$, and the substitution $u = y/\sigma$ gives the Siegert form. The $\phi T \rightarrow 0$ limit is the convergence of the killed-OU transition density to the killed-Brownian one, recovering the reflection value; the explicit $O(\phi T)$ correction, negative, is computed in closed form in Corollary A.9. $\square$

Proposition A.8 (Closed-form spectral coefficients). *The coefficients in the survival expansion of Proposition A.7 are*

$$a_n(\ell) = -e^{-\ell^2/(4\sigma^2)} \frac{D_{\rho_n}(0)}{\rho_n \partial_p D_p(-\ell/\sigma)|_{p=\rho_n}}, \quad D_{\rho_n}(0) = \frac{\sqrt{\pi} 2^{\rho_n/2}}{\Gamma(\frac{1-\rho_n}{2})},$$

where $\partial_p D_p$ is the derivative of the parabolic cylinder function in its order, evaluated at the eigenvalue ρ_n . They satisfy $\sum_{n \geq 1} a_n = 1$, understood as the boundary value $\lim_{T \downarrow 0} \sum_n a_n e^{-\phi \rho_n T} = \mathbb{P}(\tau_{-\ell} > 0) = 1$: the coefficients decay only as $|a_n| \sim \rho_n^{-1/2}$ (the initial datum does not vanish at the absorbing boundary), so the series is conditionally, not absolutely, summable: an Abel sum, not an absolutely convergent one. The long-horizon survival is $\mathbb{P}(\tau_{-\ell} > T) \sim a_1 e^{-\phi \rho_1 T}$ (Figure 12), and the mean first-passage time of Proposition A.7 is recovered as $\mathbb{E}_0[\tau_{-\ell}] = \sum_{n \geq 1} a_n / (\phi \rho_n)$.

Proof. Write $\psi(\lambda) = \mathbb{E}_0[e^{-\lambda \tau_{-\ell}}] = e^{-\ell^2/(4\sigma^2)} D_{-\lambda/\phi}(0) / D_{-\lambda/\phi}(-\ell/\sigma)$ from Proposition A.7. The survival probability has Laplace transform $\int_0^\infty e^{-\lambda T} \mathbb{P}(\tau > T) dT = (1 - \psi(\lambda)) / \lambda$, whose singularities are the poles of ψ at $\lambda = -\phi \rho_n$; the apparent pole at $\lambda = 0$ is removable because $\psi(0) = 1$. Inverting by residues, $\mathbb{P}(\tau > T) = \sum_n a_n e^{-\phi \rho_n T}$ with $a_n = R_n / (\phi \rho_n)$ and $R_n = \text{Res}_{\lambda = -\phi \rho_n} \psi(\lambda) = e^{-\ell^2/(4\sigma^2)} D_{\rho_n}(0) / \partial_\lambda D_{-\lambda/\phi}(-\ell/\sigma)|_{-\phi \rho_n}$. Since $\partial_\lambda D_{-\lambda/\phi}(-\ell/\sigma) = -\phi^{-1} \partial_p D_p(-\ell/\sigma)|_{\rho_n}$, the residue is $-\phi e^{-\ell^2/(4\sigma^2)} D_{\rho_n}(0) / \partial_p D_{\rho_n}(-\ell/\sigma)$, giving the stated a_n . Letting $T \downarrow 0$, $\sum_n a_n e^{-\phi \rho_n T} \rightarrow \mathbb{P}(\tau > 0) = 1$; the bare series $\sum_n a_n$ converges only conditionally ($|a_n| \sim \rho_n^{-1/2}$, the boundary datum not

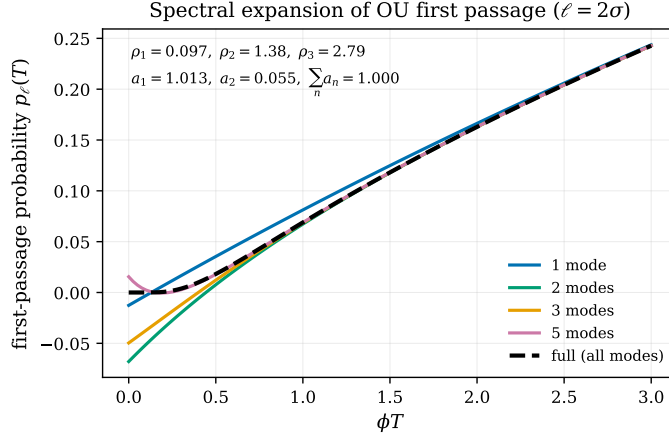


Figure 12: The spectral expansion of Propositions A.7–A.8 for a 2σ buffer, reconstructed from the closed-form coefficients with increasing numbers of modes against the full first-passage probability. The principal mode $a_1 e^{-\phi \rho_1 T}$ captures the long-horizon decay; the higher eigenvalues $\rho_2, \rho_3, \dots$ supply the short-horizon correction. The coefficients sum to one, $\sum_n a_n = 1$, as required, and the displayed values were verified against a direct discretization of the killed generator.

vanishing at the absorbing point), so the equality is this Abel limit. Integrating the expansion, legitimate term by term because it converges absolutely and uniformly on $[\epsilon, \infty)$ for each $\epsilon > 0$ while the $T < \epsilon$ contribution is $O(\epsilon)$ uniformly, gives $\mathbb{E}_0[\tau] = \int_0^\infty \sum_n a_n e^{-\phi \rho_n T} dT = \sum_n a_n / (\phi \rho_n)$, consistent with the Siegert integral. The value $D_{\rho_n}(0) = \sqrt{\pi} 2^{\rho_n/2} / \Gamma(\frac{1-\rho_n}{2})$ is the parabolic cylinder function at the origin. $\square$

Corollary A.9 (Closed-form mean-reversion correction). *Let $\zeta = \ell/s$ with $s = \varsigma\sqrt{T}$, write φ for the standard normal density and Φ for its distribution function. In the regime $\phi T \ll 1$ the first-passage probability of Lemma 4.3 has the explicit expansion*

$$p_\ell(\ell; T) = 2\Phi(-\zeta) + \phi T [\zeta \varphi(\zeta) - 2\zeta^2 \Phi(-\zeta)] + O((\phi T)^2),$$

the bracket negative for $\zeta \gtrsim 0.61$, so mean reversion lowers the liquidation probability at any buffer beyond roughly six-tenths of the interval scale; the time-changed Brownian clock produces a small positive correction only for very small buffers below that scale. The induced optimal buffer (4.5) is, to the same order,

$$\ell^* = \ell_0^* - \phi T \frac{s}{2} \frac{g'(\zeta_0)}{\zeta_0 \varphi(\zeta_0)}, \quad \zeta_0 = \frac{\ell_0^*}{s}, \quad g'(\zeta) = \varphi(\zeta)(1 + \zeta^2) - 4\zeta \Phi(-\zeta),$$

with ℓ_0^ the Brownian buffer of (4.5). Since $g'(\zeta_0) > 0$ for $\zeta_0 \gtrsim 1.5$, the optimal buffer contracts by $O(\phi T)$: a mean-reverting marking price is less likely to breach a distant boundary, so the trader rationally posts less collateral than the Brownian heuristic prescribes.*

Proof. The path is a scaled, time-changed Brownian motion, $X_t = e^{-\phi t} \tilde{B}(\tau(t))$ with $\tau(t) = \sigma^2(e^{2\phi t} - 1)$, so X hits $-\ell$ exactly when the standard Brownian motion $\tilde{B}$ hits the curved boundary $-\ell\sqrt{1 + \tau/\sigma^2}$ at clock τ . Over $[0, \tau(T)]$ with $\tau(T) = \sigma^2(e^{2\phi T} - 1) = s^2(1 + \phi T + O((\phi T)^2))$, the

boundary is linear up to $O((\phi T)^2)$, namely $-\ell - m\tau$ with $m = \ell/2\sigma^2$. The first passage of $\tilde{B}$ to this line has a closed form, obtained by reduction to a flat boundary: hitting $\tilde{B}_\tau = -\ell - m\tau$ is the event $\{W_\tau = -\ell \text{ for some } \tau \leq \tau(T)\}$ for $W_\tau := \tilde{B}_\tau + m\tau$, a Brownian motion with drift m . Removing the drift by the Girsanov change of measure $dQ/dP = \exp(-m\tilde{B}_{\tau(T)} - \frac{1}{2}m^2\tau(T))$ (under Q , W is a standard Brownian motion) and applying the reflection principle to the Q -Brownian motion at the level $-\ell$, with the Radon–Nikodym factor evaluated along the reflected path contributing $e^{2m(-\ell)} = e^{-2m\ell}$, gives

$$\mathbb{P}\left(\min_{\tau \leq \tau(T)} W_\tau \leq -\ell\right) = \Phi\left(\frac{-\ell - m\tau(T)}{\sqrt{\tau(T)}}\right) + e^{-2m\ell} \Phi\left(\frac{-\ell + m\tau(T)}{\sqrt{\tau(T)}}\right),$$

the standard drifted first-passage law. Substituting $m\tau(T) = \ell\phi T + O((\phi T)^2)$, $2m\ell = \ell^2/\sigma^2 = 2\zeta^2\phi T$, and $\sqrt{\tau(T)} = s(1 + \frac{\phi T}{2})$, and expanding both terms to first order in ϕT , the $O(\phi T)$ contributions combine to $\zeta\varphi(\zeta) - 2\zeta^2\Phi(-\zeta)$, giving the displayed p_ℓ . The buffer follows by perturbing the first-order condition $r = (2\varpi/s)\varphi(\ell/s)$ of (4.5): with $p_\ell = 2\Phi(-\ell/s) + \phi T g(\ell/s)$, $g(\zeta) = \zeta\varphi(\zeta) - 2\zeta^2\Phi(-\zeta)$, the condition $r + \varpi \partial_\ell p_\ell = 0$ gives $\ell^* = \ell_0^* - \phi T \partial_\ell [\varpi g(\ell/s)] / \partial_{\ell\ell} [2\varpi\Phi(-\ell/s)]|_{\ell_0^*}$, and $\partial_\ell g = g'(\zeta)/s$, $\partial_{\ell\ell} [2\Phi(-\ell/s)] = (2/s^2)\zeta\varphi(\zeta)$ yield the stated ℓ^* . Both expansions were verified against Monte Carlo (continuity-corrected) and against direct minimization of the corrected objective, with residuals of the predicted $O((\phi T)^2)$. $\square$

A.8 Self-contained proof of the global-game selection

Theorem 10.18 was stated by invoking the general global-game selection results. Here we prove it from first principles for the provision game, and find that its additive structure delivers more than the general theorems: the selected threshold is the Laplacian value *exactly*, at every noise level, with no small-noise limit required.

Fix the model of Section 10.6: a continuum of providers, signals $x_i = \theta + \nu\xi_i$ with ξ_i i.i.d. of continuous symmetric density φ and c.d.f. Φ , an improper uniform prior on θ (so the posterior of θ given $x_i = x$ has density $\nu^{-1}\varphi((x - \theta)/\nu)$), abstention normalized to payoff 0, and provision payoff $u(\theta, \ell) = \theta - \Lambda(a\bar{w}/\kappa(\ell))$ with κ increasing, $\kappa(0) = \underline{\kappa} > 0$, and Λ increasing. Write $\Lambda_*(\ell) := \Lambda(a\bar{w}/\kappa(\ell))$, decreasing in ℓ .

Lemma A.10 (Cutoff best responses and complementarity). *Against any profile in which the aggregate provision at fundamental θ is $\ell(\theta)$, a provider's expected payoff to providing, $U(x) = \int [\theta - \Lambda_*(\ell(\theta))] \nu^{-1}\varphi((x - \theta)/\nu) d\theta$, is strictly increasing in the signal x ; hence every best response is a cutoff strategy “provide iff $x_i \geq c$.” If all others use the common cutoff $\hat{x}$, the aggregate is $\ell(\theta; \hat{x}) = 1 - \Phi((\hat{x} - \theta)/\nu)$, decreasing in $\hat{x}$, so the best-response cutoff $T(\hat{x})$ is nondecreasing in $\hat{x}$: the game is supermodular.*

Proof. Since the posterior of θ given x is a location family shifting right in x , it increases in x in the monotone-likelihood-ratio order; $u(\theta, \ell(\theta))$ is increasing in θ (unit coefficient, and $\ell(\theta)$ increasing while u increases in ℓ), so its posterior mean $U(x)$ is strictly increasing in x . A best response provides exactly where $U(x) \geq 0$, an upper interval, i.e. a cutoff. Raising the common cutoff $\hat{x}$ lowers $\ell(\theta; \hat{x})$ pointwise, hence raises Λ_* , hence lowers U at every x , so the indifference point $T(\hat{x})$ rises. $\square$

Lemma A.11 (The symmetric indifference is exact and affine). *Under a common cutoff $\hat{x}$, the marginal type $x_i = \hat{x}$ has expected provision payoff*

$$G(\hat{x}) := U(\hat{x})|_{\ell(\cdot; \hat{x})} = \hat{x} - \int_0^1 \Lambda_*(\ell) d\ell = \hat{x} - \theta^*,$$

for every noise level $\nu > 0$, where $\theta^* = \int_0^1 \Lambda(a\bar{w}/\kappa(\ell)) d\ell$.

Proof. Change variables $\theta = \hat{x} + \nu z$, $z = (\theta - \hat{x})/\nu$, in the marginal type's expectation. The posterior weight $\nu^{-1}\varphi((\hat{x} - \theta)/\nu) d\theta$ becomes $\varphi(z) dz$ (using symmetry of φ), and the induced aggregate is $\ell(\theta; \hat{x}) = 1 - \Phi((\hat{x} - \theta)/\nu) = 1 - \Phi(-z) = \Phi(z)$. Hence

$$G(\hat{x}) = \int_{\mathbb{R}} [(\hat{x} + \nu z) - \Lambda_*(\Phi(z))] \varphi(z) dz = \hat{x} + \nu \int_{\mathbb{R}} z \varphi(z) dz - \int_{\mathbb{R}} \Lambda_*(\Phi(z)) \varphi(z) dz.$$

The middle term vanishes by symmetry ($\int z \varphi = 0$), and the last term, under the substitution $\ell = \Phi(z)$, $d\ell = \varphi(z) dz$, equals $\int_0^1 \Lambda_*(\ell) d\ell = \theta^*$, with no residual ν -dependence. The marginal type's belief over the aggregate $\ell = \Phi(z)$ is thus uniform on $[0, 1]$, the Laplacian property, and here it holds exactly, not merely in the limit, because θ enters u additively. So $G(\hat{x}) = \hat{x} - \theta^*$, affine with unit slope. $\square$

Proposition A.12 (Exact selection at every noise level). *For every $\nu > 0$ the provision game has a unique strategy surviving iterated deletion of dominated strategies: provide iff $x_i \geq \theta^*$. The deep equilibrium is realized for $\theta > \theta^*$ and the manipulated floor for $\theta < \theta^*$.*

Proof. Upper dominance: $u(\theta, 0) = \theta - \Lambda(a\bar{w}/\kappa)$ is positive for θ above some $\bar{\theta}$ (finite since $\kappa > 0$), so for signals high enough that the posterior puts sufficient mass on $\{\theta > \bar{\theta}\}$, providing is dominant; symmetrically abstaining is dominant for signals below some bound. Every rationalizable cutoff therefore lies in a bounded interval $[\underline{c}, \bar{c}]$, and the best-response map T of Lemma A.10 maps $[\underline{c}, \bar{c}]$ into itself, is nondecreasing, and is continuous. A symmetric cutoff $\hat{x}$ is an equilibrium iff the marginal type is indifferent, $G(\hat{x}) = 0$; by Lemma A.11 the unique such $\hat{x}$ is θ^* , so T has the unique fixed point θ^* . Iterating T from $\underline{c}$ gives a nondecreasing sequence and from $\bar{c}$ a nonincreasing one, each converging to a fixed point of T (monotone bounded convergence, T continuous), hence both to θ^* . By the supermodular structure (Lemma A.10) these two sequences bound every rationalizable strategy, so the rationalizable set collapses to the single cutoff θ^* . Finally, with all using cutoff θ^* , the realized regime is provision when the mass providing is positive in the relevant range, i.e. for $\theta > \theta^*$, and collapse for $\theta < \theta^*$. $\square$

The argument reproves Theorem 10.18 without invoking the general selection theorems, and sharpens it: where those theorems deliver a unique threshold only as $\nu \rightarrow 0$ and leave its value as a limit, the additive separability of the provision payoff makes the marginal-type indifference $G(\hat{x}) = \hat{x} - \theta^*$ exactly affine, so the threshold is the Laplacian value $\theta^* = \int_0^1 \Lambda(a\bar{w}/\kappa(\ell)) d\ell$ at every noise level and the rationalizable set is the single threshold strategy for all $\nu > 0$. The general machinery is what licenses the step from “unique symmetric cutoff equilibrium” to “unique rationalizable outcome” in games where G is not monotone; here G is affine, so that step is immediate and the selection is exact. The exactness should be read for what it is: an artifact of the improper uniform prior on θ . No proper prior is uniform on $\mathbb{R}$, and with a proper prior the marginal type's belief over the aggregate is only approximately uniform, so G acquires an $O(\nu)$ curvature and the threshold is the Laplacian value only in the small-noise limit, exactly the limit the general theorems already deliver. The improper-prior calculation is therefore a clean closed-form benchmark, exact in its idealized information structure and the leading term for a diffuse proper prior, not a strengthening that escapes the small-noise qualification in substance; its value is that it identifies the selected threshold in closed form and confirms it lies strictly inside the spinodal.